

Assignment 2.1 - Cure Garbage In, Garbage Out

Goal : To predict home credit defaulters (TARGET). 1 = defaulter and 0 = non-defaulter

2.1 - Minimal Feature Engineering

1.1 - Install binaries

Used Python version - 3.9.18, because dataprep is compatible with 3.8, 3.9 and 3.10 python versions.

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In [ ]: ## Select kernal python 3.9.18
        %pip install --user dataprep
        !pip install imblearn
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Requirement already satisfied: dataprep in c:\users\jagdi\appdata\roaming\python\python39\site-packages (0.4.5)

Requirement already satisfied: aiohttp<4.0,>=3.6 in d:\tech\ai\envs\aa1-501-m5\lib\site-packages (from dataprep) (3.11.10)

Requirement already satisfied: bokeh<3,>=2 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from dataprep) (2.4.3)

Requirement already satisfied: dask<2022.3.0 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from dask[array,dataframe,delayed]>=2022.3.0->dataprep) (2024.2.1)

Requirement already satisfied: flask<3,>=2 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from dataprep) (2.2.5)

Requirement already satisfied: flask_cors<4.0.0,>=3.0.10 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from dataprep) (3.0.10)

Requirement already satisfied: ipywidgets<8.0,>=7.5 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from dataprep) (7.8.5)

Requirement already satisfied: jinja2<3.1,>=3.0 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from dataprep) (3.0.3)

Requirement already satisfied: jsonpath-ng<2.0,>=1.5 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from dataprep) (1.8.0)

Requirement already satisfied: metaphone<0.7,>=0.6 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from dataprep) (0.6)

Requirement already satisfied: nltk<4.0.0,>=3.6.7 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from dataprep) (3.9.2)

Requirement already satisfied: numpy<2.0,>=1.21 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from dataprep) (1.26.4)

Requirement already satisfied: pandas<2.0,>=1.1 in d:\tech\ai\envs\aa1-501-m5\lib\site-packages (from dataprep) (1.5.3)

Requirement already satisfied: pydantic<2.0,>=1.6 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from dataprep) (1.10.26)

Requirement already satisfied: pydot<2.0.0,>=1.4.2 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from dataprep) (1.4.2)

Requirement already satisfied: python-crfsuite==0.9.8 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from dataprep) (0.9.8)

Requirement already satisfied: python-stdnum<2.0,>=1.16 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from dataprep) (1.20)

Requirement already satisfied: rapidfuzz<3.0.0,>=2.1.2 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from dataprep) (2.15.2)

Requirement already satisfied: regex<2022.0.0,>=2021.8.3 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from dataprep) (2021.11.10)

Requirement already satisfied: scipy<2.0,>=1.8 in d:\tech\ai\envs\aa1-501-m5\lib\site-packages (from dataprep) (1.10.1)

Requirement already satisfied: sqlalchemy==1.3.24 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from dataprep) (1.3.24)

Requirement already satisfied: tqdm<5.0,>=4.48 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from

dataprep) (4.67.3)
Requirement already satisfied: varname<0.9.0,>=0.8.1 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from dataprep) (0.8.3)
Requirement already satisfied: wordcloud<2.0,>=1.8 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from dataprep) (1.9.6)
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Requirement already satisfied: frozenlist>=1.1.1 in d:\tech\ai\envs\aaai-501-m5\lib\site-packages (from aiohttp<4.0,>=3.6->dataprep) (1.5.0)
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Requirement already satisfied: packaging>=16.8 in d:\tech\ai\envs\aaai-501-m5\lib\site-packages (from bokeh<3,>=2->dataprep) (24.2)
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Requirement already satisfied: itsdangerous>=2.0 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from flask<3,>=2->dataprep) (2.2.0)
Requirement already satisfied: click>=8.0 in d:\tech\ai\envs\aaai-501-m5\lib\site-packages (from flask<3,>=2->dataprep) (8.1.8)
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Requirement already satisfied: comm>=0.1.3 in d:\tech\ai\envs\aaai-501-m5\lib\site-packages (from ipywidgets<8.0,>=7.5

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->dataprep) (0.2.1)
Requirement already satisfied: ipython-genutils~=0.2.0 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from ipywidgets<8.0,>=7.5->dataprep) (0.2.0)
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Requirement already satisfied: jupyterlab-widgets<3,>=1.0.0 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from ipywidgets<8.0,>=7.5->dataprep) (1.1.11)
Requirement already satisfied: MarkupSafe>=2.0 in d:\tech\ai\envs\aa1-501-m5\lib\site-packages (from jinja2<3.1,>=3.0->dataprep) (3.0.2)
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Requirement already satisfied: python-dateutil>=2.8.1 in d:\tech\ai\envs\aa1-501-m5\lib\site-packages (from pandas<2.0,>=1.1->dataprep) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in d:\tech\ai\envs\aa1-501-m5\lib\site-packages (from pandas<2.0,>=1.1->dataprep) (2025.2)
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Requirement already satisfied: colorama in d:\tech\ai\envs\aa1-501-m5\lib\site-packages (from tqdm<5.0,>=4.48->dataprep) (0.4.6)
Requirement already satisfied: asttokens<3.0.0,>=2.0.0 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from varname<0.9.0,>=0.8.1->dataprep) (2.4.1)
Requirement already satisfied: executing<0.9.0,>=0.8.3 in d:\tech\ai\envs\aa1-501-m5\lib\site-packages (from varname<0.9.0,>=0.8.1->dataprep) (0.8.3)
Requirement already satisfied: pure_eval<1.0.0 in d:\tech\ai\envs\aa1-501-m5\lib\site-packages (from varname<0.9.0,>=0.8.1->dataprep) (0.2.2)
Requirement already satisfied: notebook>=4.4.1 in d:\tech\ai\envs\aa1-501-m5\lib\site-packages (from widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (7.4.4)
Requirement already satisfied: matplotlib in d:\tech\ai\envs\aa1-501-m5\lib\site-packages (from wordcloud<2.0,>=1.8->dataprep) (3.7.1)
Requirement already satisfied: idna>=2.0 in d:\tech\ai\envs\aa1-501-m5\lib\site-packages (from yarl<2.0,>=1.17.0->aiohttp<4.0,>=3.6->dataprep) (3.7)
Requirement already satisfied: cloudpickle>=1.5.0 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from dask>=2022.3.0->dask[array,dataframe,delayed]>=2022.3.0->dataprep) (3.1.2)
Requirement already satisfied: fsspec>=2021.09.0 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from dask>=2022.3.0->dask[array,dataframe,delayed]>=2022.3.0->dataprep) (2025.10.0)
Requirement already satisfied: partd>=1.2.0 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from dask>=2022.3.0->dask[array,dataframe,delayed]>=2022.3.0->dataprep) (1.4.2)
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Requirement already satisfied: zipp>=3.20 in d:\tech\ai\envs\aa-501-m5\lib\site-packages (from importlib-metadata>=
3.6.0->flask<3,>=2->dataprep) (3.21.0)
Requirement already satisfied: backcall in d:\tech\ai\envs\aa-501-m5\lib\site-packages (from ipython>=4.0.0->ipywidg
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Requirement already satisfied: decorator in d:\tech\ai\envs\aa-501-m5\lib\site-packages (from ipython>=4.0.0->ipywid
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->ipywidgets<8.0,>=7.5->dataprep) (0.1.6)
Requirement already satisfied: pickleshare in d:\tech\ai\envs\aa-501-m5\lib\site-packages (from ipython>=4.0.0->ipyw
idgets<8.0,>=7.5->dataprep) (0.7.5)
Requirement already satisfied: prompt-toolkit!=3.0.37,<3.1.0,>=3.0.30 in d:\tech\ai\envs\aa-501-m5\lib\site-packages
(from ipython>=4.0.0->ipywidgets<8.0,>=7.5->dataprep) (3.0.43)
Requirement already satisfied: pygments>=2.4.0 in d:\tech\ai\envs\aa-501-m5\lib\site-packages (from ipython>=4.0.0->
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Requirement already satisfied: stack-data in d:\tech\ai\envs\aa-501-m5\lib\site-packages (from ipython>=4.0.0->ipywi
dgets<8.0,>=7.5->dataprep) (0.2.0)
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3.1.0,>=3.0.30->ipython>=4.0.0->ipywidgets<8.0,>=7.5->dataprep) (0.2.13)
Requirement already satisfied: parso<0.9.0,>=0.8.4 in d:\tech\ai\envs\aa-501-m5\lib\site-packages (from jedi>=0.16->
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Requirement already satisfied: jupyter-server<3,>=2.4.0 in d:\tech\ai\envs\aa-501-m5\lib\site-packages (from noteboo
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Requirement already satisfied: anyio>=3.1.0 in d:\tech\ai\envs\aa-501-m5\lib\site-packages (from jupyter-server<3,>=
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Requirement already satisfied: jupyter-client>=7.4.4 in d:\tech\ai\envs\aa-501-m5\lib\site-packages (from jupyter-se
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Requirement already satisfied: jupyter-core!=5.0.*,>=4.12 in d:\tech\ai\envs\aa-501-m5\lib\site-packages (from jupyt
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Requirement already satisfied: jupyter-events>=0.11.0 in d:\tech\ai\envs\aa-501-m5\lib\site-packages (from jupyter-s

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erver<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (0.12.0)
Requirement already satisfied: jupyter-server-terminals>=0.4.4 in d:\tech\ai\envs\aa-501-m5\lib\site-packages (from
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Requirement already satisfied: nbconvert>=6.4.4 in d:\tech\ai\envs\aa-501-m5\lib\site-packages (from jupyter-server<
3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (7.16.6)
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3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (5.10.4)
Requirement already satisfied: overrides>=5.0 in d:\tech\ai\envs\aa-501-m5\lib\site-packages (from jupyter-server<3,
>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (7.4.0)
Requirement already satisfied: prometheus-client>=0.9 in d:\tech\ai\envs\aa-501-m5\lib\site-packages (from jupyter-s
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Requirement already satisfied: pywinpty>=2.0.1 in d:\tech\ai\envs\aa-501-m5\lib\site-packages (from jupyter-server<
3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (2.0.15)
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Requirement already satisfied: terminado>=0.8.3 in d:\tech\ai\envs\aa-501-m5\lib\site-packages (from jupyter-server<
3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (0.17.1)
Requirement already satisfied: websocket-client>=1.7 in d:\tech\ai\envs\aa-501-m5\lib\site-packages (from jupyter-se
rver<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (1.8.0)
Requirement already satisfied: async-lru>=1.0.0 in d:\tech\ai\envs\aa-501-m5\lib\site-packages (from jupyterlab<4.5,
>=4.4.4->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (2.0.4)
Requirement already satisfied: httpx>=0.25.0 in d:\tech\ai\envs\aa-501-m5\lib\site-packages (from jupyterlab<4.5,>=
4.4.4->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (0.28.1)
Requirement already satisfied: ipykernel>=6.5.0 in d:\tech\ai\envs\aa-501-m5\lib\site-packages (from jupyterlab<4.5,
>=4.4.4->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (6.29.5)
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5,>=4.4.4->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (2.2.5)
Requirement already satisfied: setuptools>=41.1.0 in d:\tech\ai\envs\aa-501-m5\lib\site-packages (from jupyterlab<4.
5,>=4.4.4->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (78.1.1)
Requirement already satisfied: tomli>=1.2.2 in d:\tech\ai\envs\aa-501-m5\lib\site-packages (from jupyterlab<4.5,>=4.
4.4->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (2.0.1)
Requirement already satisfied: babel>=2.10 in d:\tech\ai\envs\aa-501-m5\lib\site-packages (from jupyterlab-server<3,
>=2.27.1->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (2.16.0)
Requirement already satisfied: json5>=0.9.0 in d:\tech\ai\envs\aa-501-m5\lib\site-packages (from jupyterlab-server<
3,>=2.27.1->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (0.9.25)
Requirement already satisfied: jsonschema>=4.18.0 in d:\tech\ai\envs\aa-501-m5\lib\site-packages (from jupyterlab-se
rver<3,>=2.27.1->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (4.25.0)
Requirement already satisfied: requests>=2.31 in d:\tech\ai\envs\aa-501-m5\lib\site-packages (from jupyterlab-server
<3,>=2.27.1->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (2.32.4)
Requirement already satisfied: sniffio>=1.1 in d:\tech\ai\envs\aa-501-m5\lib\site-packages (from anyio>=3.1.0->jupyt

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er-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~3.6.10->ipywidgets<8.0,>=7.5->dataprep) (1.3.0)

Requirement already satisfied: argon2-cffi-bindings in d:\tech\ai\envs\aaai-501-m5\lib\site-packages (from argon2-cffi>=21.1->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~3.6.10->ipywidgets<8.0,>=7.5->dataprep) (21.2.0)

Requirement already satisfied: certifi in d:\tech\ai\envs\aaai-501-m5\lib\site-packages (from httpx>=0.25.0->jupyterlab<4.5,>=4.4.4->notebook>=4.4.1->widgetsnbextension~3.6.10->ipywidgets<8.0,>=7.5->dataprep) (2025.7.14)

Requirement already satisfied: httpcore==1.* in d:\tech\ai\envs\aaai-501-m5\lib\site-packages (from httpx>=0.25.0->jupyterlab<4.5,>=4.4.4->notebook>=4.4.1->widgetsnbextension~3.6.10->ipywidgets<8.0,>=7.5->dataprep) (1.0.9)

Requirement already satisfied: h11>=0.16 in d:\tech\ai\envs\aaai-501-m5\lib\site-packages (from httpcore==1.*->httpx>=0.25.0->jupyterlab<4.5,>=4.4.4->notebook>=4.4.1->widgetsnbextension~3.6.10->ipywidgets<8.0,>=7.5->dataprep) (0.16.0)

Requirement already satisfied: debugpy>=1.6.5 in d:\tech\ai\envs\aaai-501-m5\lib\site-packages (from ipykernel>=6.5.0->jupyterlab<4.5,>=4.4.4->notebook>=4.4.1->widgetsnbextension~3.6.10->ipywidgets<8.0,>=7.5->dataprep) (1.8.11)

Requirement already satisfied: nest-asyncio in d:\tech\ai\envs\aaai-501-m5\lib\site-packages (from ipykernel>=6.5.0->jupyterlab<4.5,>=4.4.4->notebook>=4.4.1->widgetsnbextension~3.6.10->ipywidgets<8.0,>=7.5->dataprep) (1.6.0)

Requirement already satisfied: psutil in d:\tech\ai\envs\aaai-501-m5\lib\site-packages (from ipykernel>=6.5.0->jupyterlab<4.5,>=4.4.4->notebook>=4.4.1->widgetsnbextension~3.6.10->ipywidgets<8.0,>=7.5->dataprep) (5.9.0)

Requirement already satisfied: jsonschema-specifications>=2023.03.6 in d:\tech\ai\envs\aaai-501-m5\lib\site-packages (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.27.1->notebook>=4.4.1->widgetsnbextension~3.6.10->ipywidgets<8.0,>=7.5->dataprep) (2023.7.1)

Requirement already satisfied: referencing>=0.28.4 in d:\tech\ai\envs\aaai-501-m5\lib\site-packages (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.27.1->notebook>=4.4.1->widgetsnbextension~3.6.10->ipywidgets<8.0,>=7.5->dataprep) (0.30.2)

Requirement already satisfied: rpds-py>=0.7.1 in d:\tech\ai\envs\aaai-501-m5\lib\site-packages (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.27.1->notebook>=4.4.1->widgetsnbextension~3.6.10->ipywidgets<8.0,>=7.5->dataprep) (0.22.3)

Requirement already satisfied: platformdirs>=2.5 in d:\tech\ai\envs\aaai-501-m5\lib\site-packages (from jupyter-core!=5.0.*,>=4.12->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~3.6.10->ipywidgets<8.0,>=7.5->dataprep) (4.3.7)

Requirement already satisfied: pywin32>=300 in d:\tech\ai\envs\aaai-501-m5\lib\site-packages (from jupyter-core!=5.0.*,>=4.12->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~3.6.10->ipywidgets<8.0,>=7.5->dataprep) (308)

Requirement already satisfied: python-json-logger>=2.0.4 in d:\tech\ai\envs\aaai-501-m5\lib\site-packages (from jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~3.6.10->ipywidgets<8.0,>=7.5->dataprep) (3.2.1)

Requirement already satisfied: rfc3339-validator in d:\tech\ai\envs\aaai-501-m5\lib\site-packages (from jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~3.6.10->ipywidgets<8.0,>=7.5->dataprep) (0.1.4)

Requirement already satisfied: rfc3986-validator>=0.1.1 in d:\tech\ai\envs\aaai-501-m5\lib\site-packages (from jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~3.6.10->ipywidgets<8.0,>=7.5->dataprep) (0.1.1)

Requirement already satisfied: fqdn in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~3.6.1

0->ipywidgets<8.0,>=7.5->dataprep) (1.5.1)
Requirement already satisfied: isoduration in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (20.11.0)
Requirement already satisfied: jsonpointer>1.13 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (3.0.0)
Requirement already satisfied: rfc3987-syntax>=1.1.0 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (1.1.0)
Requirement already satisfied: uri-template in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (1.3.0)
Requirement already satisfied: webcolors>=24.6.0 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (24.11.1)
Requirement already satisfied: beautifulsoup4 in d:\tech\ai\envs\ai-501-m5\lib\site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (4.13.4)
Requirement already satisfied: bleach!=5.0.0 in d:\tech\ai\envs\ai-501-m5\lib\site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (6.2.0)
Requirement already satisfied: defusedxml in d:\tech\ai\envs\ai-501-m5\lib\site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (0.7.1)
Requirement already satisfied: jupyterlab-pygments in d:\tech\ai\envs\ai-501-m5\lib\site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (0.3.0)
Requirement already satisfied: mistune<4,>=2.0.3 in d:\tech\ai\envs\ai-501-m5\lib\site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (3.1.2)
Requirement already satisfied: nbclient>=0.5.0 in d:\tech\ai\envs\ai-501-m5\lib\site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (0.10.2)
Requirement already satisfied: pandocfilters>=1.4.1 in d:\tech\ai\envs\ai-501-m5\lib\site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (1.5.0)
Requirement already satisfied: webencodings in d:\tech\ai\envs\ai-501-m5\lib\site-packages (from bleach!=5.0.0->nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (0.5.1)
Requirement already satisfied: tinycss2<1.5,>=1.1.0 in d:\tech\ai\envs\ai-501-m5\lib\site-packages (from bleach!=5.0.0->nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (1.4.0)
Requirement already satisfied: fastjsonschema>=2.15 in d:\tech\ai\envs\ai-501-m5\lib\site-packages (from nbformat>=5.3.0->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (2.20.0)
Requirement already satisfied: locket in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from partd>=1.2.0->dask>=2022.3.0->dask[array,dataframe,delayed]>=2022.3.0->dataprep) (1.0.0)

Requirement already satisfied: charset_normalizer<4,>=2 in d:\tech\ai\envs\aaai-501-m5\lib\site-packages (from requests>=2.31->jupyterlab-server<3,>=2.27.1->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (3.3.2)

Requirement already satisfied: urllib3<3,>=1.21.1 in d:\tech\ai\envs\aaai-501-m5\lib\site-packages (from requests>=2.31->jupyterlab-server<3,>=2.27.1->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (2.5.0)

Requirement already satisfied: lark>=1.2.2 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from rfc3987-syntax>=1.1.0->jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (1.3.1)

Requirement already satisfied: cffi>=1.0.1 in d:\tech\ai\envs\aaai-501-m5\lib\site-packages (from argon2-cffi-bindings->argon2-cffi>=21.1->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (1.17.1)

Requirement already satisfied: pycparser in d:\tech\ai\envs\aaai-501-m5\lib\site-packages (from cffi>=1.0.1->argon2-cffi-bindings->argon2-cffi>=21.1->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (2.21)

Requirement already satisfied: soupsieve>1.2 in d:\tech\ai\envs\aaai-501-m5\lib\site-packages (from beautifulsoup4->nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (2.5)

Requirement already satisfied: arrow>=0.15.0 in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from isoduration->jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (1.4.0)

Requirement already satisfied: tzdata in c:\users\jagdi\appdata\roaming\python\python39\site-packages (from arrow>=0.15.0->isoduration->jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.10->ipywidgets<8.0,>=7.5->dataprep) (2026.2)

Requirement already satisfied: contourpy>=1.0.1 in d:\tech\ai\envs\aaai-501-m5\lib\site-packages (from matplotlib->wordcloud<2.0,>=1.8->dataprep) (1.2.0)

Requirement already satisfied: cyclor>=0.10 in d:\tech\ai\envs\aaai-501-m5\lib\site-packages (from matplotlib->wordcloud<2.0,>=1.8->dataprep) (0.11.0)

Requirement already satisfied: fonttools>=4.22.0 in d:\tech\ai\envs\aaai-501-m5\lib\site-packages (from matplotlib->wordcloud<2.0,>=1.8->dataprep) (4.55.3)

Requirement already satisfied: kiwisolver>=1.0.1 in d:\tech\ai\envs\aaai-501-m5\lib\site-packages (from matplotlib->wordcloud<2.0,>=1.8->dataprep) (1.4.4)

Requirement already satisfied: importlib-resources>=3.2.0 in d:\tech\ai\envs\aaai-501-m5\lib\site-packages (from matplotlib->wordcloud<2.0,>=1.8->dataprep) (6.4.0)

Note: you may need to restart the kernel to use updated packages.

```
In [101... import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import HistGradientBoostingClassifier
from sklearn.metrics import roc_auc_score, classification_report, f1_score, average_precision_score, accuracy_score,
from sklearn.preprocessing import LabelEncoder, OrdinalEncoder
from sklearn.impute import SimpleImputer
from sklearn.pipeline import Pipeline
```

```

from sklearn.compose import ColumnTransformer
from dataprep.eda import plot, plot_correlation, plot_missing, create_report
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

```

1.2 - Load dataset

```

In [3]: df = pd.read_csv('train_data.csv')

# Visualize the data on TARGET variable
plot(df, "TARGET").show()

```

0%| | 0/75 [00:00<?, ?it/s]

Stats Bar Chart Pie Chart Word Cloud Word Frequency Word Length Value Table

Overview			Sample		
Approximate Distinct Count	2		1st row	0	
Approximate Unique (%)	0.0%		2nd row	0	
Missing	0		3rd row	0	
Missing (%)	0.0%		4th row	0	
Memory Size	10147830		5th row	0	
Length			Letter		
Mean	1		Count	0	
Standard Deviation	0		Lowercase Letter	0	
Median	1		Space Separator	0	
Minimum	1		Uppercase Letter	0	
Maximum	1		Dash Punctuation	0	
			Decimal Number	153755	

```

C:\Users\Jagdi\AppData\Roaming\Python\Python39\site-packages\dask\core.py:127: RuntimeWarning: invalid value encountered in divide
    return func(*(_execute_task(a, cache) for a in args))
C:\Users\Jagdi\AppData\Roaming\Python\Python39\site-packages\dask\core.py:127: RuntimeWarning: invalid value encountered in divide
    return func(*(_execute_task(a, cache) for a in args))
C:\Users\Jagdi\AppData\Roaming\Python\Python39\site-packages\dataprep\eda\utils.py:392: RuntimeWarning: ks_2samp: Exact calculation unsuccessful. Switching to method=asympt.
    return cast(Tuple[float, float], ks_2samp_(data1, data2))
C:\Users\Jagdi\AppData\Roaming\Python\Python39\site-packages\dataprep\eda\utils.py:392: RuntimeWarning: ks_2samp: Exact calculation unsuccessful. Switching to method=asympt.
    return cast(Tuple[float, float], ks_2samp_(data1, data2))
C:\Users\Jagdi\AppData\Roaming\Python\Python39\site-packages\dataprep\eda\utils.py:392: RuntimeWarning: ks_2samp: Exact calculation unsuccessful. Switching to method=asympt.
    return cast(Tuple[float, float], ks_2samp_(data1, data2))
C:\Users\Jagdi\AppData\Roaming\Python\Python39\site-packages\dataprep\eda\utils.py:392: RuntimeWarning: ks_2samp: Exact calculation unsuccessful. Switching to method=asympt.
    return cast(Tuple[float, float], ks_2samp_(data1, data2))
C:\Users\Jagdi\AppData\Roaming\Python\Python39\site-packages\dask\core.py:127: RuntimeWarning: invalid value encountered in divide
    return func(*(_execute_task(a, cache) for a in args))
d:\TECH\AI\envs\aa1-501-m5\lib\site-packages\scipy\stats\_stats_py.py:5278: RuntimeWarning: overflow encountered in scalar multiply
    (2 * xtie * ytie) / m + x0 * y0 / (9 * m * (size - 2)))
C:\Users\Jagdi\AppData\Roaming\Python\Python39\site-packages\numpy\lib\function_base.py:2897: RuntimeWarning: invalid value encountered in divide
    c /= stddev[:, None]
C:\Users\Jagdi\AppData\Roaming\Python\Python39\site-packages\numpy\lib\function_base.py:2898: RuntimeWarning: invalid value encountered in divide
    c /= stddev[None, :]
C:\Users\Jagdi\AppData\Roaming\Python\Python39\site-packages\dask\core.py:127: RuntimeWarning: invalid value encountered in divide
    return func(*(_execute_task(a, cache) for a in args))
d:\TECH\AI\envs\aa1-501-m5\lib\site-packages\scipy\stats\_stats_py.py:5279: RuntimeWarning: invalid value encountered in sqrt
    z = con_minus_dis / np.sqrt(var)
C:\Users\Jagdi\AppData\Roaming\Python\Python39\site-packages\numpy\lib\function_base.py:2897: RuntimeWarning: invalid value encountered in divide
    c /= stddev[:, None]
C:\Users\Jagdi\AppData\Roaming\Python\Python39\site-packages\numpy\lib\function_base.py:2898: RuntimeWarning: invalid value encountered in divide
    c /= stddev[None, :]

```

```

d:\TECH\AI\envs\aa1-501-m5\lib\site-packages\scipy\stats\_stats_py.py:5278: RuntimeWarning: overflow encountered in s
calar multiply
    (2 * xtie * ytie) / m + x0 * y0 / (9 * m * (size - 2)))
d:\TECH\AI\envs\aa1-501-m5\lib\site-packages\scipy\stats\_stats_py.py:5279: RuntimeWarning: invalid value encountered
in sqrt
    z = con_minus_dis / np.sqrt(var)
C:\Users\Jagdi\AppData\Roaming\Python\Python39\site-packages\numpy\lib\function_base.py:2897: RuntimeWarning: invalid
value encountered in divide
    c /= stddev[:, None]
C:\Users\Jagdi\AppData\Roaming\Python\Python39\site-packages\numpy\lib\function_base.py:2898: RuntimeWarning: invalid
value encountered in divide
    c /= stddev[None, :]
C:\Users\Jagdi\AppData\Roaming\Python\Python39\site-packages\dask\core.py:127: RuntimeWarning: invalid value encounte
red in divide
    return func(*(_execute_task(a, cache) for a in args))
d:\TECH\AI\envs\aa1-501-m5\lib\site-packages\scipy\stats\_stats_py.py:5278: RuntimeWarning: overflow encountered in s
calar multiply
    (2 * xtie * ytie) / m + x0 * y0 / (9 * m * (size - 2)))
d:\TECH\AI\envs\aa1-501-m5\lib\site-packages\scipy\stats\_stats_py.py:5279: RuntimeWarning: invalid value encountered
in sqrt
    z = con_minus_dis / np.sqrt(var)
C:\Users\Jagdi\AppData\Roaming\Python\Python39\site-packages\numpy\lib\function_base.py:2897: RuntimeWarning: invalid
value encountered in divide
    c /= stddev[:, None]
C:\Users\Jagdi\AppData\Roaming\Python\Python39\site-packages\numpy\lib\function_base.py:2898: RuntimeWarning: invalid
value encountered in divide
    c /= stddev[None, :]
d:\TECH\AI\envs\aa1-501-m5\lib\site-packages\scipy\stats\_stats_py.py:5278: RuntimeWarning: overflow encountered in s
calar multiply
    (2 * xtie * ytie) / m + x0 * y0 / (9 * m * (size - 2)))
d:\TECH\AI\envs\aa1-501-m5\lib\site-packages\scipy\stats\_stats_py.py:5279: RuntimeWarning: invalid value encountered
in sqrt
    z = con_minus_dis / np.sqrt(var)

```

1.3 - Minimal feature engineering

```

In [4]: ## Remove customer ID and Target variable from the dataset
x = df.drop(columns=['SK_ID_CURR', 'TARGET'])

## Target variable
y = df['TARGET']

```


In [5]: `x.head()`

Out[5]:

	NAME_CONTRACT_TYPE	CODE_GENDER	FLAG_OWN_CAR	FLAG_OWN_REALTY	CNT_CHILDREN	AMT_INCOME_TOTAL	AMT_C
0	Cash loans	F	N	Y	1	157500.0	90
1	Cash loans	F	N	Y	1	90000.0	73
2	Cash loans	F	Y	Y	0	189000.0	179
3	Cash loans	M	Y	N	0	175500.0	49
4	Cash loans	F	Y	Y	0	270000.0	172

5 rows × 120 columns



In [6]: `y.head()`

Out[6]:

0	0
1	0
2	0
3	0
4	0

Name: TARGET, dtype: int64

Lets convert all the object type data to numbers. This is because Machine learning algorithms works better with numbers. Machine learning algorithms cannot directly process text values.

In [7]:

```

### Convert all object type columns to numbers and then apply label encoding
for col in x.select_dtypes(include=['object']).columns:
    le = LabelEncoder()      ## Create a new LabelEncoder object for each column
    x[col] = le.fit_transform(x[col].astype(str))
x.head()

```

Out[7]:

	NAME_CONTRACT_TYPE	CODE_GENDER	FLAG_OWN_CAR	FLAG_OWN_REALTY	CNT_CHILDREN	AMT_INCOME_TOTAL	AMT_C	
0		0	0	0	1	1	157500.0	90
1		0	0	0	1	1	90000.0	73
2		0	0	1	1	0	189000.0	179
3		0	1	1	0	0	175500.0	49
4		0	0	1	1	0	270000.0	172

5 rows × 120 columns



1.4 - Split the data set for training and Testing

Split the dataset - 80% data for training and 20% for testing

```
In [8]: x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2, random_state=42)
```

1.5 - Train Baseline Model

Lets select HistGradientBoosting Model for training, as the dataset contains lot of missing values. This model is often used when dataset contains missing values.

```
In [9]: ### max_iter: is the maximum number of iterations for the boosting process. # Setting it to 100 means that the algo
### This parameter helps control overfitting and can be adjusted based on the size and complexity of the dataset.
### random_state: 42 ensures that the results are reproducible.

model = HistGradientBoostingClassifier(random_state=42, max_iter=100)
model.fit(x_train, y_train)
```

```
d:\TECH\AI\envs\aa-501-m5\lib\site-packages\joblib\externals\loky\backend\context.py:136: UserWarning: Could not find the number of physical cores for the following reason:
[WinError 2] The system cannot find the file specified
Returning the number of logical cores instead. You can silence this warning by setting LOKY_MAX_CPU_COUNT to the number of cores you want to use.
  warnings.warn(
  File "d:\TECH\AI\envs\aa-501-m5\lib\site-packages\joblib\externals\loky\backend\context.py", line 257, in _count_physical_cores
    cpu_info = subprocess.run(
  File "d:\TECH\AI\envs\aa-501-m5\lib\subprocess.py", line 505, in run
    with Popen(*popenargs, **kwargs) as process:
  File "d:\TECH\AI\envs\aa-501-m5\lib\subprocess.py", line 951, in __init__
    self._execute_child(args, executable, preexec_fn, close_fds,
  File "d:\TECH\AI\envs\aa-501-m5\lib\subprocess.py", line 1436, in _execute_child
    hp, ht, pid, tid = _winapi.CreateProcess(executable, args,
```

Out[9]:

```
▼ HistGradientBoostingClassifier ⓘ ?
HistGradientBoostingClassifier(random_state=42)
```

1.5 - Evaluate the model

```
In [10]: y_pred = model.predict(x_test)
y_pred_proba = model.predict_proba(x_test)[: , 1]

roc_auc = roc_auc_score(y_test, y_pred_proba)
class_report = classification_report(y_test, y_pred)
f1 = f1_score(y_test, y_pred)

print("Model Evaluation Metrics:")
print("-" * 30)
print(f"ROC AUC Score: {roc_auc}")
print(f"F1 Score: {f1}")
print(f"Classification Report:\n{class_report}")
```

Model Evaluation Metrics:

ROC AUC Score: 0.7452743633559354

F1 Score: 0.03304484657749803

Classification Report:

	precision	recall	f1-score	support
0	0.92	1.00	0.96	28294
1	0.49	0.02	0.03	2457
accuracy			0.92	30751
macro avg	0.71	0.51	0.50	30751
weighted avg	0.89	0.92	0.88	30751

Evaluation Report:

The metrics shows that the model is struggling with class imbalance. This is because

1. The classification report shows the class 0 has 28294 samples, where as class 1 has only 2457 samples ie only 8% of total samples.
2. As class 0 is dominant , the overall accuracy 92% (0.92) is completely misleading. so any dummy model can predict the outcome as 0 without learning anything.
3. **Recall Definit** = 0.02 = The model catches only 2% of actual class.
4. **Precision** for class 1 is only 0.49. ie for class 1 prediction is wrong for half of the time. precision = TP / (TP + FP)
5. **F1-score**: for class 1 is only 0.03, ie the models predictive capability is almost 0.
6. **ROC AUC Score**: 0.745 - Models decent ability to seperate the classes if we change the decision threshold.

2.2 - Apply Feature engineering

Lets apply three distinct feature engineering techniques - Anamolies, Encoding, handling missing values and resampling.

2.1 - Load dataset

```
In [142... df2 = pd.read_csv('./train_data.csv')
df2.head()
```

Out[142...

	SK_ID_CURR	TARGET	NAME_CONTRACT_TYPE	CODE_GENDER	FLAG_OWN_CAR	FLAG_OWN_REALTY	CNT_CHILDREN	AMT_I
0	410704	0	Cash loans	F	N	Y	1	
1	381230	0	Cash loans	F	N	Y	1	
2	450177	0	Cash loans	F	Y	Y	0	
3	332445	0	Cash loans	M	Y	N	0	
4	357429	0	Cash loans	F	Y	Y	0	

5 rows × 122 columns



In [143...

```

### Dataset shape
print(f'Dataset Shape : {df2.shape}')
### Dataset TARGET variable value counts
print(f'Target Value Counts:\n{df2["TARGET"].value_counts()}')

### Column Datatype
num_cols = df2.select_dtypes(include=['number']).columns.tolist()
cat_cols = df2.select_dtypes(include=['object']).columns.tolist()
print(f'Numeric Columns count : len{num_cols}')
print(f'Categorical Columns count : len{cat_cols}')

### Categoris count in each categorical column
for c in cat_cols:
    print(f'Unique values in {c} : {df2[c].nunique()}')

```

Dataset Shape : (153755, 122)

Target Value Counts:

0 141343

1 12412

Name: TARGET, dtype: int64

Numeric Columns count : len['SK_ID_CURR', 'TARGET', 'CNT_CHILDREN', 'AMT_INCOME_TOTAL', 'AMT_CREDIT', 'AMT_ANNUITY', 'AMT_GOODS_PRICE', 'REGION_POPULATION_RELATIVE', 'DAYS_BIRTH', 'DAYS_EMPLOYED', 'DAYS_REGISTRATION', 'DAYS_ID_PUBLISH', 'OWN_CAR_AGE', 'FLAG_MOBIL', 'FLAG_EMP_PHONE', 'FLAG_WORK_PHONE', 'FLAG_CONT_MOBILE', 'FLAG_PHONE', 'FLAG_EMAIL', 'CNT_FAM_MEMBERS', 'REGION_RATING_CLIENT', 'REGION_RATING_CLIENT_W_CITY', 'HOUR_APPR_PROCESS_START', 'REG_REGION_NOT_LIVE_REGION', 'REG_REGION_NOT_WORK_REGION', 'LIVE_REGION_NOT_WORK_REGION', 'REG_CITY_NOT_LIVE_CITY', 'REG_CITY_NOT_WORK_CITY', 'LIVE_CITY_NOT_WORK_CITY', 'EXT_SOURCE_1', 'EXT_SOURCE_2', 'EXT_SOURCE_3', 'APARTMENTS_AVG', 'BASEMENTAREA_AVG', 'YEARS_BEGINEXPLUATATION_AVG', 'YEARS_BUILD_AVG', 'COMMONAREA_AVG', 'ELEVATORS_AVG', 'ENTRANCES_AVG', 'FLOORSMAX_AVG', 'FLOORSMIN_AVG', 'LANDAREA_AVG', 'LIVINGAPARTMENTS_AVG', 'LIVINGAREA_AVG', 'NONLIVINGAPARTMENTS_AVG', 'NONLIVINGAREA_AVG', 'APARTMENTS_MODE', 'BASEMENTAREA_MODE', 'YEARS_BEGINEXPLUATATION_MODE', 'YEARS_BUILD_MODE', 'COMMONAREA_MODE', 'ELEVATORS_MODE', 'ENTRANCES_MODE', 'FLOORSMAX_MODE', 'FLOORSMIN_MODE', 'LANDAREA_MODE', 'LIVINGAPARTMENTS_MODE', 'LIVINGAREA_MODE', 'NONLIVINGAPARTMENTS_MODE', 'NONLIVINGAREA_MODE', 'APARTMENTS_MEDI', 'BASEMENTAREA_MEDI', 'YEARS_BEGINEXPLUATATION_MEDI', 'YEARS_BUILD_MEDI', 'COMMONAREA_MEDI', 'ELEVATORS_MEDI', 'ENTRANCES_MEDI', 'FLOORSMAX_MEDI', 'FLOORSMIN_MEDI', 'LANDAREA_MEDI', 'LIVINGAPARTMENTS_MEDI', 'LIVINGAREA_MEDI', 'NONLIVINGAPARTMENTS_MEDI', 'NONLIVINGAREA_MEDI', 'TOTALAREA_MODE', 'OBS_30_CNT_SOCIAL_CIRCLE', 'DEF_30_CNT_SOCIAL_CIRCLE', 'OBS_60_CNT_SOCIAL_CIRCLE', 'DEF_60_CNT_SOCIAL_CIRCLE', 'DAYS_LAST_PHONE_CHANGE', 'FLAG_DOCUMENT_2', 'FLAG_DOCUMENT_3', 'FLAG_DOCUMENT_4', 'FLAG_DOCUMENT_5', 'FLAG_DOCUMENT_6', 'FLAG_DOCUMENT_7', 'FLAG_DOCUMENT_8', 'FLAG_DOCUMENT_9', 'FLAG_DOCUMENT_10', 'FLAG_DOCUMENT_11', 'FLAG_DOCUMENT_12', 'FLAG_DOCUMENT_13', 'FLAG_DOCUMENT_14', 'FLAG_DOCUMENT_15', 'FLAG_DOCUMENT_16', 'FLAG_DOCUMENT_17', 'FLAG_DOCUMENT_18', 'FLAG_DOCUMENT_19', 'FLAG_DOCUMENT_20', 'FLAG_DOCUMENT_21', 'AMT_REQ_CREDIT_BUREAU_HOUR', 'AMT_REQ_CREDIT_BUREAU_DAY', 'AMT_REQ_CREDIT_BUREAU_WEEK', 'AMT_REQ_CREDIT_BUREAU_MON', 'AMT_REQ_CREDIT_BUREAU_QRT', 'AMT_REQ_CREDIT_BUREAU_YEAR']

Categorical Columns count : len['NAME_CONTRACT_TYPE', 'CODE_GENDER', 'FLAG_OWN_CAR', 'FLAG_OWN_REALTY', 'NAME_TYPE_SUITE', 'NAME_INCOME_TYPE', 'NAME_EDUCATION_TYPE', 'NAME_FAMILY_STATUS', 'NAME_HOUSING_TYPE', 'OCCUPATION_TYPE', 'WEEKDAY_APPR_PROCESS_START', 'ORGANIZATION_TYPE', 'FONDKAPREMONT_MODE', 'HOUSETYPE_MODE', 'WALLSMATERIAL_MODE', 'EMERGENCY_STATE_MODE']

Unique values in NAME_CONTRACT_TYPE : 2

Unique values in CODE_GENDER : 2

Unique values in FLAG_OWN_CAR : 2

Unique values in FLAG_OWN_REALTY : 2

Unique values in NAME_TYPE_SUITE : 7

Unique values in NAME_INCOME_TYPE : 8

Unique values in NAME_EDUCATION_TYPE : 5

Unique values in NAME_FAMILY_STATUS : 6

Unique values in NAME_HOUSING_TYPE : 6

Unique values in OCCUPATION_TYPE : 18

Unique values in WEEKDAY_APPR_PROCESS_START : 7

Unique values in ORGANIZATION_TYPE : 58

Unique values in FONDKAPREMONT_MODE : 4

Unique values in HOUSETYPE_MODE : 3
Unique values in WALLSMATERIAL_MODE : 7
Unique values in EMERGENCYSTATE_MODE : 2

2.2 - Data Visualization using Dataprep

Generate the report using dataprep for data analysis

```
In [ ]: ### Create diretory to store the plots in html format using DataPrep  
from pathlib import Path  
EDA_DIR = Path('./eda_plots')  
EDA_DIR.mkdir(exist_ok=True)  
dataprep_report = create_report(df2, title='Home credit dataset EDA report')  
dataprep_report.save(EDA_DIR / 'eda_report.html')  
dataprep_report.show()
```

```
100%|#####| 3/3 [00:00<?, ?it/s]  
0%|          | 0/34009 [00:00<?, ?it/s]
```

```
C:\Users\Jagdi\AppData\Roaming\Python\Python39\site-packages\dataprep\eda\distribution\render.py:274: FutureWarning:
The frame.append method is deprecated and will be removed from pandas in a future version. Use pandas.concat instead.
    df = df.append(pd.DataFrame({col: [nrows - npresent]}, index=["Others"])))
C:\Users\Jagdi\AppData\Roaming\Python\Python39\site-packages\dataprep\eda\distribution\render.py:274: FutureWarning:
The frame.append method is deprecated and will be removed from pandas in a future version. Use pandas.concat instead.
    df = df.append(pd.DataFrame({col: [nrows - npresent]}, index=["Others"])))
C:\Users\Jagdi\AppData\Roaming\Python\Python39\site-packages\dataprep\eda\distribution\render.py:274: FutureWarning:
The frame.append method is deprecated and will be removed from pandas in a future version. Use pandas.concat instead.
    df = df.append(pd.DataFrame({col: [nrows - npresent]}, index=["Others"])))
C:\Users\Jagdi\AppData\Roaming\Python\Python39\site-packages\dataprep\eda\distribution\render.py:274: FutureWarning:
The frame.append method is deprecated and will be removed from pandas in a future version. Use pandas.concat instead.
    df = df.append(pd.DataFrame({col: [nrows - npresent]}, index=["Others"])))
C:\Users\Jagdi\AppData\Roaming\Python\Python39\site-packages\dataprep\eda\distribution\render.py:274: FutureWarning:
The frame.append method is deprecated and will be removed from pandas in a future version. Use pandas.concat instead.
    df = df.append(pd.DataFrame({col: [nrows - npresent]}, index=["Others"])))
C:\Users\Jagdi\AppData\Roaming\Python\Python39\site-packages\dataprep\eda\distribution\render.py:274: FutureWarning:
The frame.append method is deprecated and will be removed from pandas in a future version. Use pandas.concat instead.
    df = df.append(pd.DataFrame({col: [nrows - npresent]}, index=["Others"])))
C:\Users\Jagdi\AppData\Roaming\Python\Python39\site-packages\dataprep\eda\distribution\render.py:274: FutureWarning:
The frame.append method is deprecated and will be removed from pandas in a future version. Use pandas.concat instead.
    df = df.append(pd.DataFrame({col: [nrows - npresent]}, index=["Others"])))
C:\Users\Jagdi\AppData\Roaming\Python\Python39\site-packages\dataprep\eda\distribution\render.py:274: FutureWarning:
The frame.append method is deprecated and will be removed from pandas in a future version. Use pandas.concat instead.
    df = df.append(pd.DataFrame({col: [nrows - npresent]}, index=["Others"])))
E-1019 (DUPLICATE_FACTORS): FactorRange must specify a unique list of categorical factors for an axis: duplicate fact
ors found: 'REG_REGION_NO...ION', 'REG_CITY_NOT...ITY'
Report has been saved to eda_plots\eda_report.html!
```


Overview

Dataset Statistics

Number of Variables	122
Number of Rows	153755
Missing Cells	4.5782×10 ⁰⁶
Missing Cells (%)	24.4%
Duplicate Rows	0
Duplicate Rows (%)	0.0%
Total Size in Memory	268.3 MB
Average Row Size in Memory	1.8 KB
Variable Types	Numerical: 68 Categorical: 54

Dataset Insights

`APARTMENTS_AVG` and `APARTMENTS_MODE` have similar distributions

Similar Distribution

`APARTMENTS_AVG` and `APARTMENTS_MEDI` have similar distributions

Similar Distribution

`BASEMENTAREA_AVG` and `BASEMENTAREA_MODE` have similar distributions

Similar Distribution

`BASEMENTAREA_AVG` and `BASEMENTAREA_MEDI` have similar distributions

Similar Distribution

`YEARS_BEGINEXPLUATATION_AVG` and `YEARS_BEGINEXPLUATATION_MODE` have similar distributions

Similar Distribution

`YEARS_BEGINEXPLUATATION_AVG` and `YEARS_BEGINEXPLUATATION_MEDI` have similar distributions

Similar Distribution

`YEARS_BUILD_AVG` and `YEARS_BUILD_MODE` have similar distributions

Similar Distribution

`YEARS_BUILD_AVG` and `YEARS_BUILD_MEDI` have similar distributions

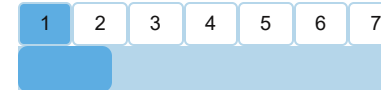
Similar Distribution

`COMMONAREA_AVG` and `COMMONAREA_MODE` have similar distributions

Similar Distribution

`COMMONAREA_AVG` and `COMMONAREA_MEDI` have similar distributions

Similar Distribution

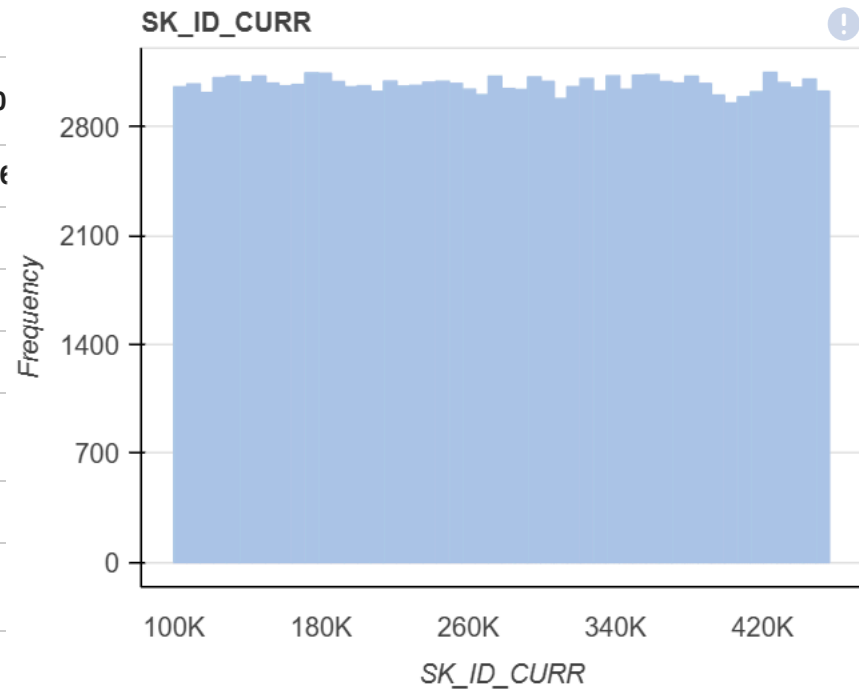


Variables

Sort by Feature order ☐ Reverse order

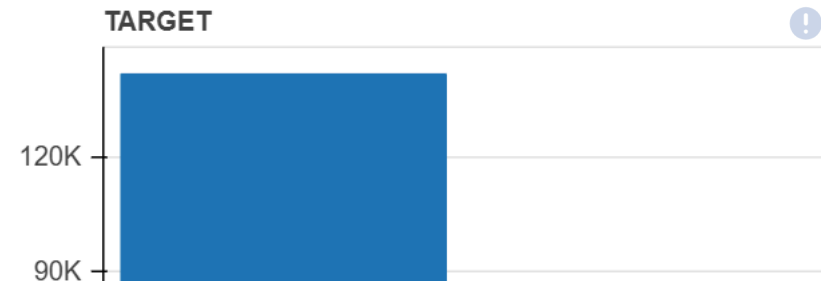
SK_ID_CURR numerical	Approximate Distinct Count	153755	Memory Size	2460080
	Approximate Unique (%)	100.0%	Mean	277867.6
	Missing	0	Minimum	100004
	Missing (%)	0.0%	Maximum	456255
	Infinite	0	Zeros	0
	Infinite (%)	0.0%	Zeros (%)	0.0%
			Negatives	0
			Negatives (%)	0.0%

Show Details



Approximate
Distinct Count

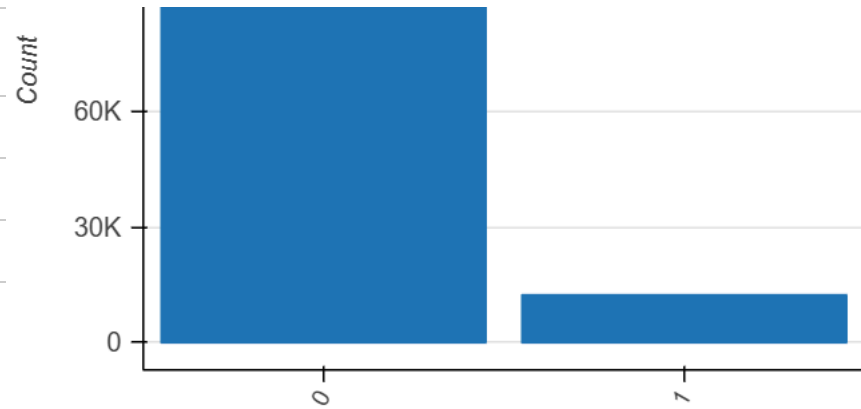
2



TARGET
categorical

Show Details

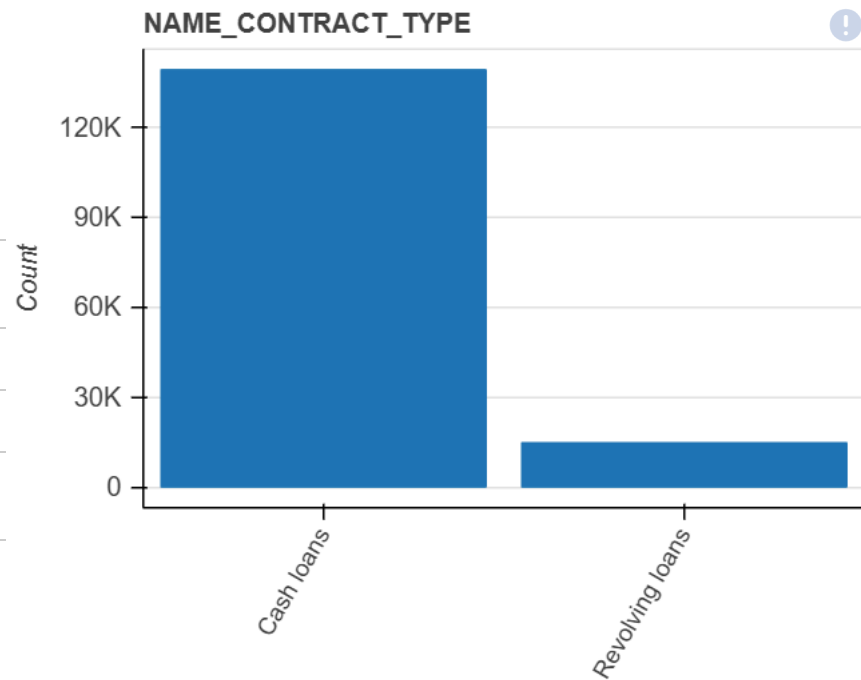
Approximate Unique (%)	0.0%
Missing	0
Missing (%)	0.0%
Memory Size	10147830



NAME_CONTRACT_...
categorical

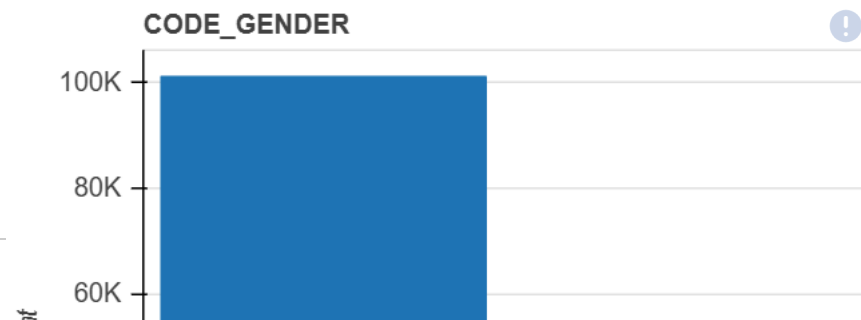
Show Details

Approximate Distinct Count	2
Approximate Unique (%)	0.0%
Missing	0
Missing (%)	0.0%
Memory Size	11604840



CODE_GENDER

Approximate Distinct Count	2
----------------------------	---

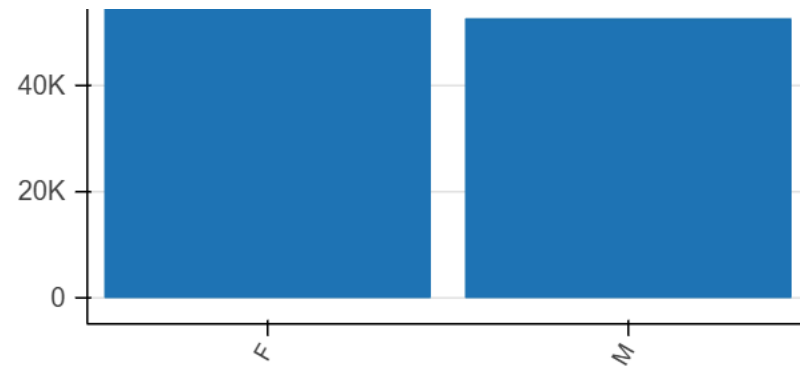


categorical

[Show Details](#)

Approximate Unique (%)	0.0%
Missing	0
Missing (%)	0.0%
Memory Size	10147830

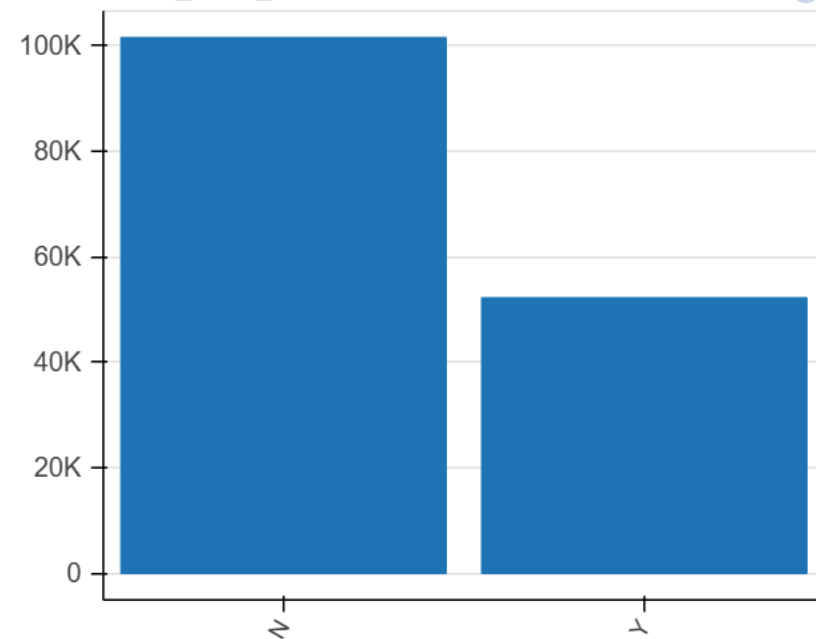
Count

FLAG_OWN_CAR
categorical[Show Details](#)

Approximate Distinct Count	2
Approximate Unique (%)	0.0%
Missing	0
Missing (%)	0.0%
Memory Size	10147830

Count

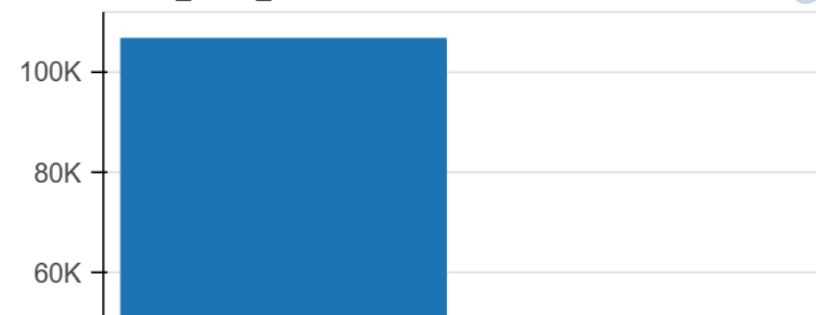
FLAG_OWN_CAR

FLAG_OWN_REALTY
categorical

Approximate Distinct Count	2
Approximate Unique (%)	0.0%

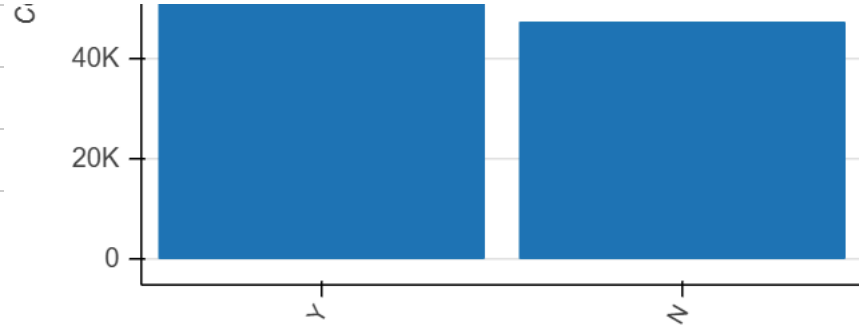
Count

FLAG_OWN_REALTY

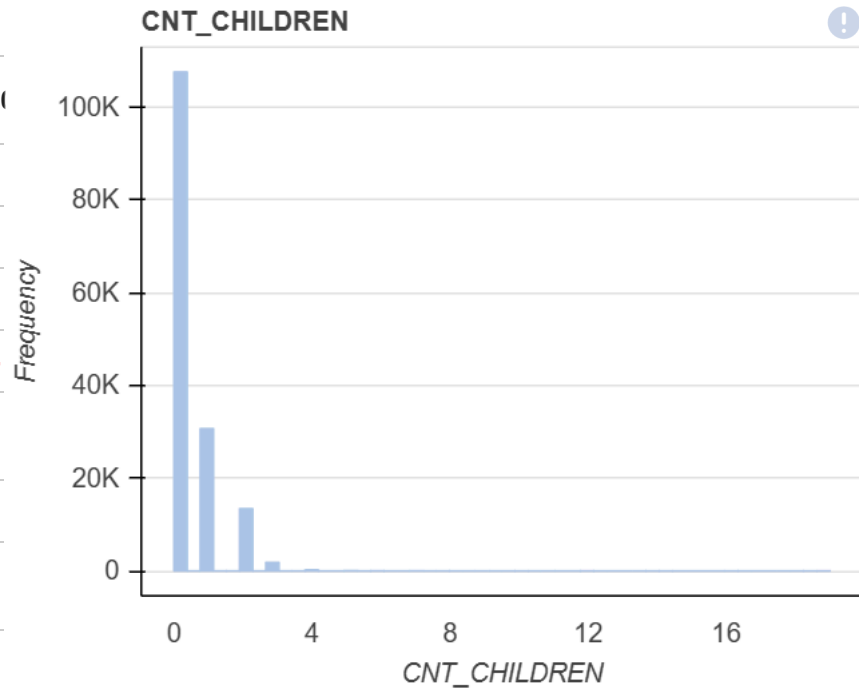


[Show Details](#)

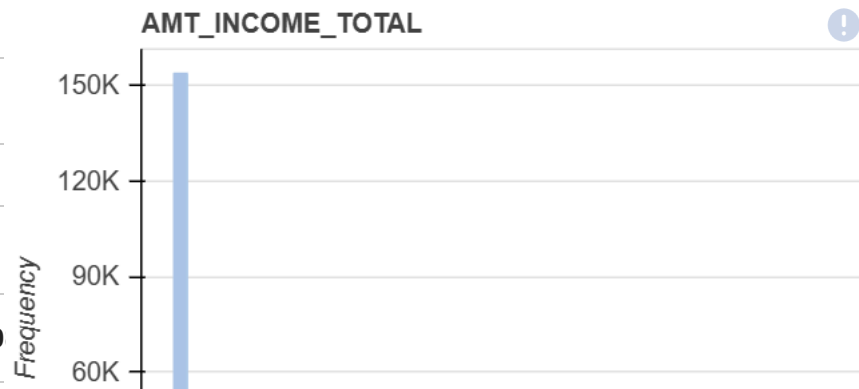
Missing	0
Missing (%)	0.0%
Memory Size	10147830

CNT_CHILDREN
numerical[Show Details](#)

Approximate Distinct Count	13	Memory Size	2460081
Approximate Unique (%)	0.0%	Mean	0.4174
Missing	0	Minimum	0
Missing (%)	0.0%	Maximum	19
Infinite	0	Zeros	107584
Infinite (%)	0.0%	Zeros (%)	70.0%
		Negatives	0
		Negatives (%)	0.0%

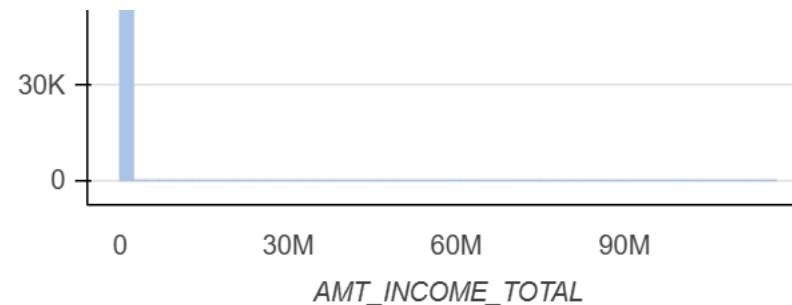
AMT_INCOME_TOTAL
numerical

Approximate Distinct Count	1596	Missing (%)	0.0%
Approximate Unique (%)	1.0%	Infinite	0
Missing	0	Infinite (%)	0.0%
		Memory Size	24600



[Show Details](#)

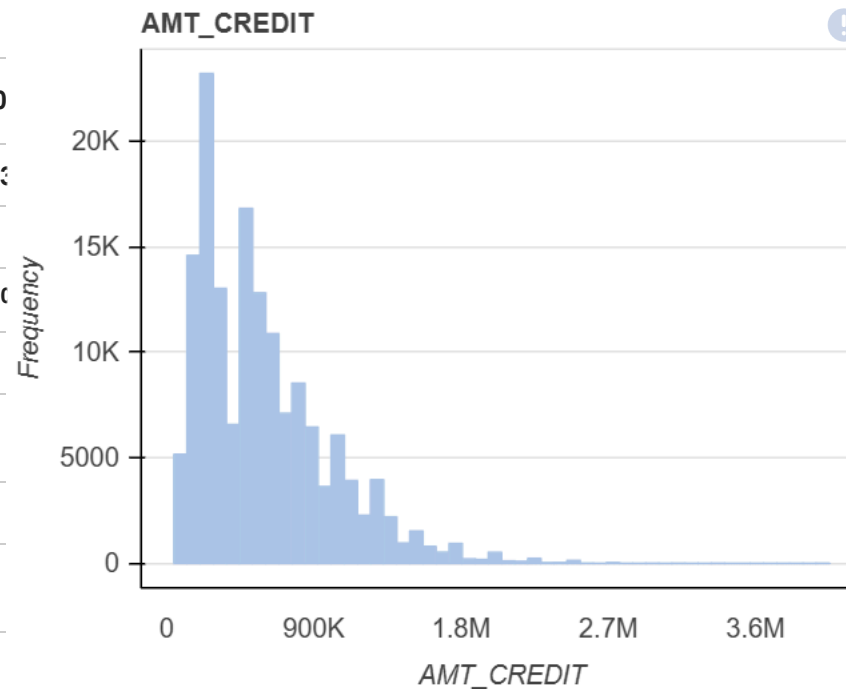
Mean	169261.08	Zeros	0.0%
Minimum	25650	Negatives	0
Maximum	1.7×10^8	Negatives	0.0%
Zeros	0	Negatives	0.0%



AMT_CREDIT
numerical

[Show Details](#)

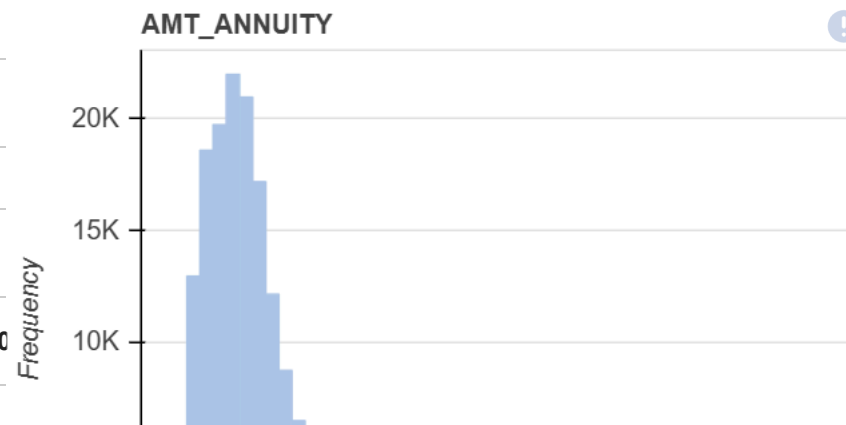
Approximate Distinct Count	4667	Memory Size	2460080
Approximate Unique (%)	3.0%	Mean	598882.3
Missing	0	Minimum	45000
Missing (%)	0.0%	Maximum	4.05×10^6
Infinite	0	Zeros	0
Infinite (%)	0.0%	Zeros (%)	0.0%
		Negatives	0
		Negatives (%)	0.0%



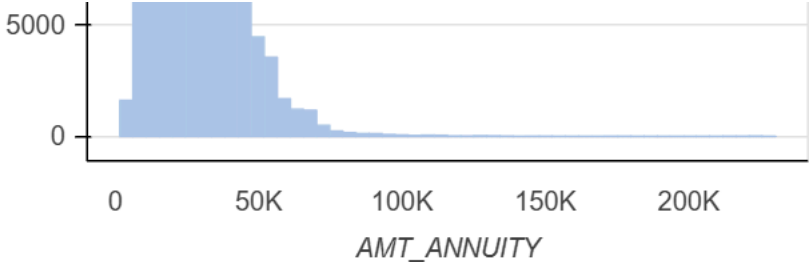
AMT_ANNUITY
numerical

[Show Details](#)

Approximate Distinct Count	11957	Missing (%)	0.0%
Approximate Unique (%)	7.8%	Infinite	0
Missing	5	Infinite (%)	0.0%
Missing (%)	0.0%	Memory Size	2460000



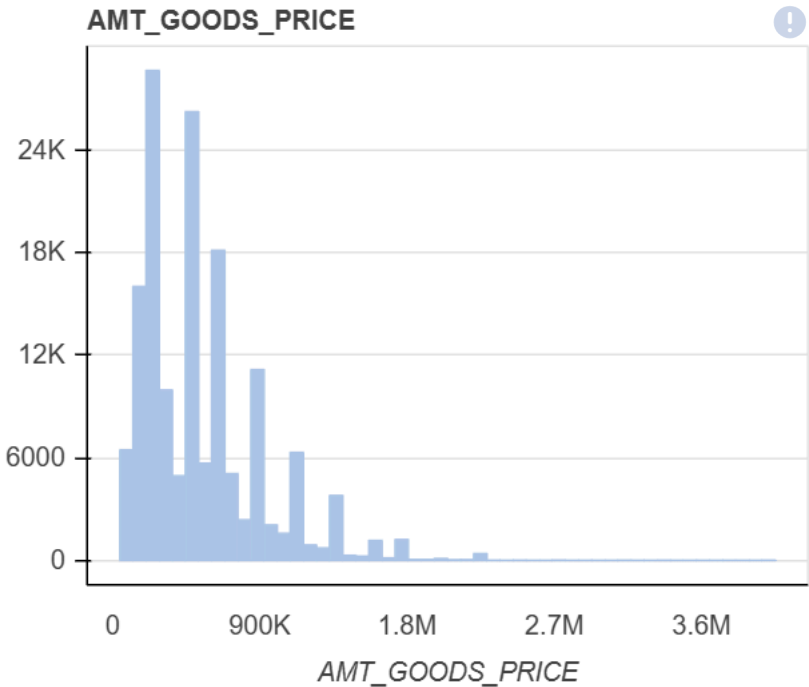
Mean	27083.127	Zeros (%)	0.0%
Minimum	1615.5	Negatives	0
Maximum	230161.5	Negatives (%)	0.0%
Zeros	0		



AMT_GOODS_PRICE
numerical

Show Details

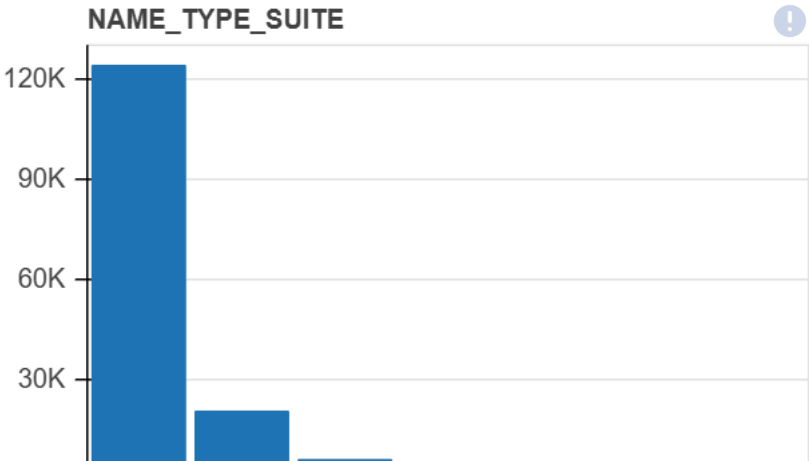
Approximate Distinct Count	739	Memory Size	24576
Approximate Unique (%)	0.5%	Mean	53830
Missing	149	Minimum	45000
Missing (%)	0.1%	Maximum	4051
Infinite	0	Zeros	0
Infinite (%)	0.0%	Zeros (%)	0.0%
		Negatives	0
		Negatives (%)	0.0%



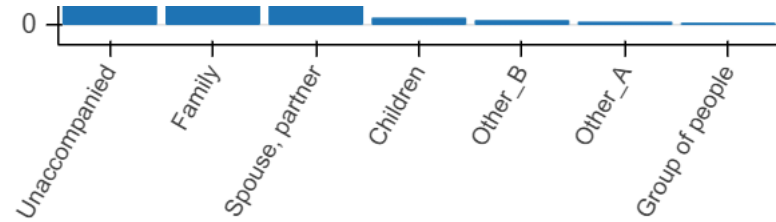
NAME_TYPE_SUITE
categorical

Show Details

Approximate Distinct Count	7
Approximate Unique (%)	0.0%
Missing	654
Missing (%)	0.4%



Memory Size	11795361
--------------------	-----------------

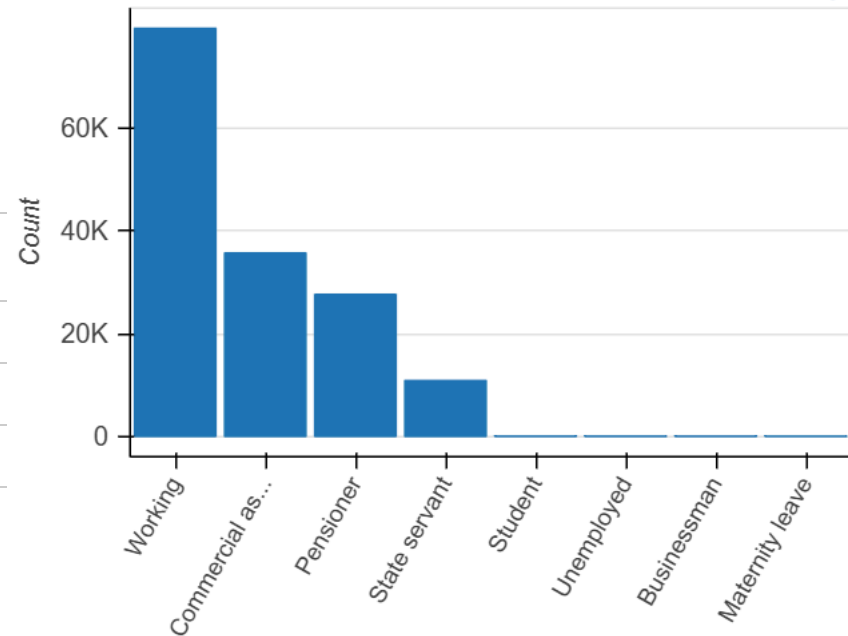


NAME_INCOME_TYPE
categorical

Show Details

Approximate Distinct Count	8
Approximate Unique (%)	0.0%
Missing	0
Missing (%)	0.0%
Memory Size	11655263

NAME_INCOME_TYPE

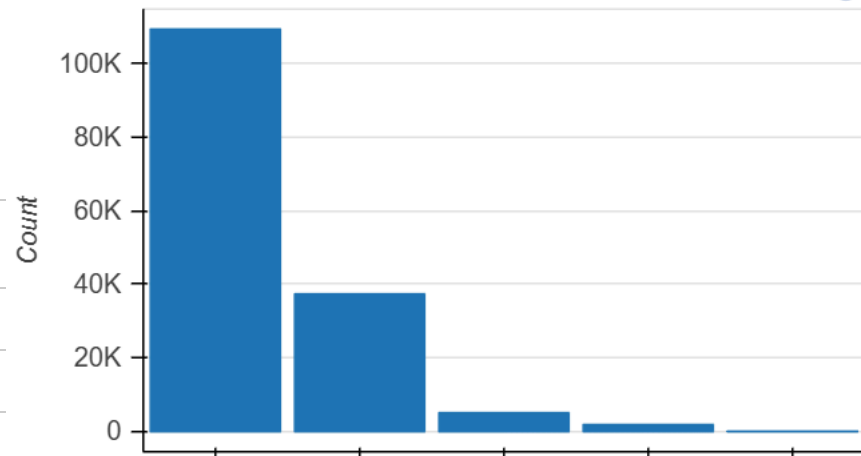


NAME_EDUCATION...
categorical

Show Details

Approximate Distinct Count	5
Approximate Unique (%)	0.0%
Missing	0
Missing (%)	0.0%

NAME_EDUCATION_TYPE



Memory Size	13877984
-------------	----------

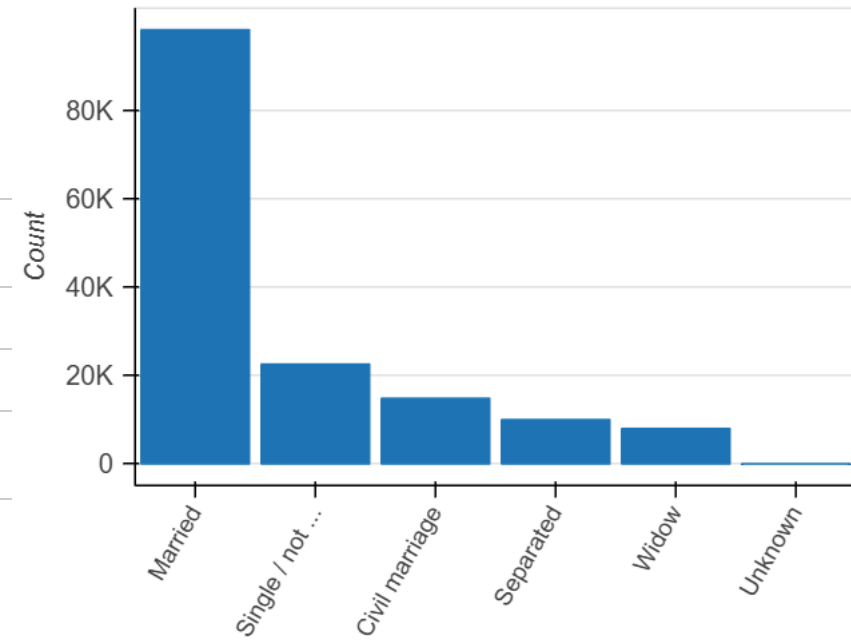
Secondary / s...
Higher educat...
Incomplete hi...
Lower secondary
Academic degree

NAME_FAMILY_STA...
categorical

Show Details

Approximate Distinct Count	6
Approximate Unique (%)	0.0%
Missing	0
Missing (%)	0.0%
Memory Size	11471730

NAME_FAMILY_STATUS

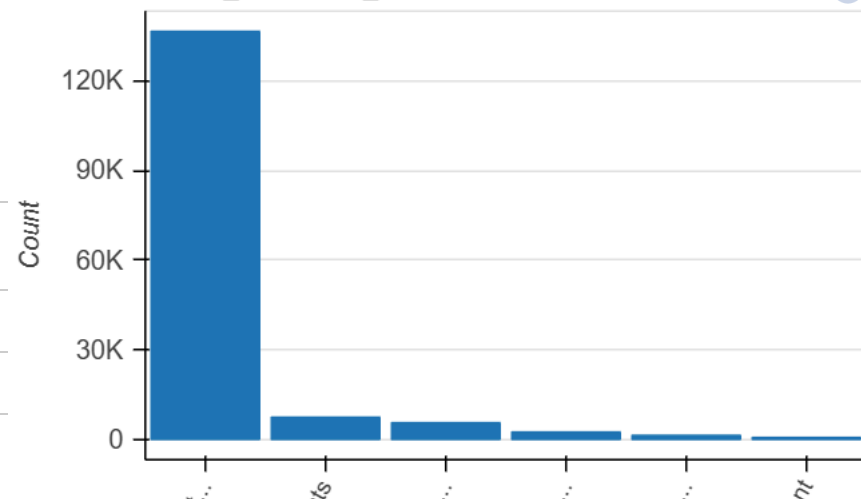


NAME_HOUSING_T...
categorical

Show Details

Approximate Distinct Count	6
Approximate Unique (%)	0.0%
Missing	0
Missing (%)	0.0%

NAME_HOUSING_TYPE



Memory Size	12577246
-------------	----------

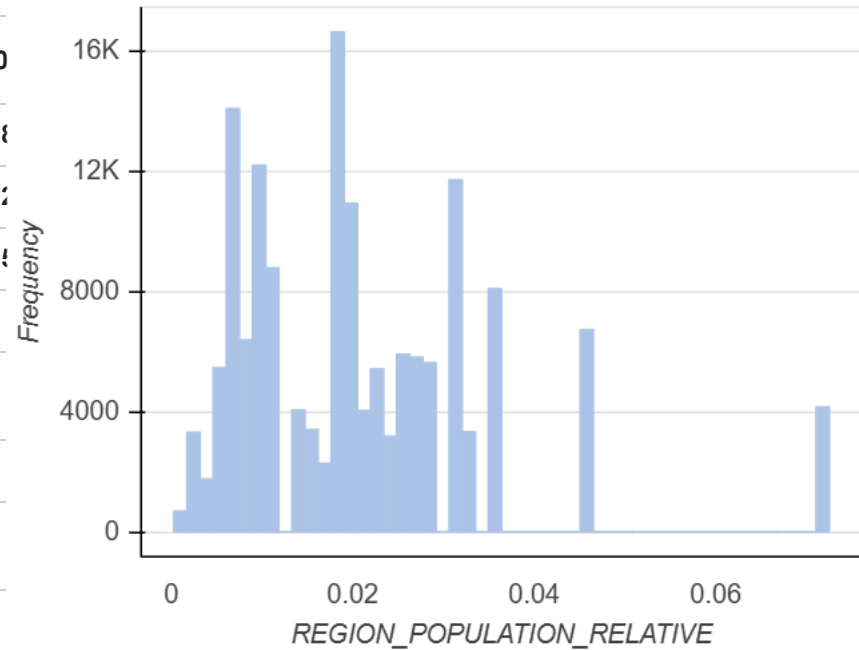
House / apart.
With paren.
Municipal apa.
Rented apartm.
Office apartm.
Co-op apartme.

REGION_POPULATI...
numerical

Show Details

Approximate Distinct Count	81	Memory Size	24600
Approximate Unique (%)	0.1%	Mean	0.0208
Missing	0	Minimum	0.000%
Missing (%)	0.0%	Maximum	0.072%
Infinite	0	Zeros	0
Infinite (%)	0.0%	Zeros (%)	0.0%
		Negatives	0
		Negatives (%)	0.0%

REGION_POPULATION_RELATIVE

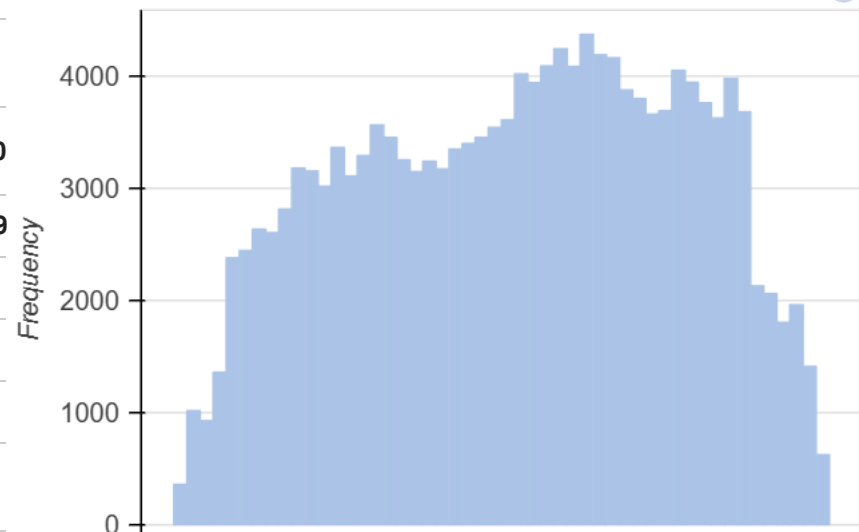


DAYS_BIRTH
numerical

Show Details

Approximate Distinct Count	17263	Infinite (%)	0.0%
Approximate Unique (%)	11.2%	Memory Size	2460080
Missing	0	Mean	-16025.9
Missing (%)	0.0%	Minimum	-25201
Infinite	0	Maximum	-7678
		Zeros	0
		Zeros (%)	0.0%

DAYS_BIRTH



Negatives	453755
Negatives (%)	100.0%

-25K -21K -17K -13K -9000

DAYS_BIRTH

DAYS_EMPLOYED
numerical

[Show Details](#)

Approximate
Distinct
Count

Approximate
Unique
(%)

Missing 0

Missing (%) 0.0%

Infinite 0

Infinite (%) 0.0%

Memory
Size 246008

Mean 63742.6

Minimum 17583

Maximum 65243

Zeros 2

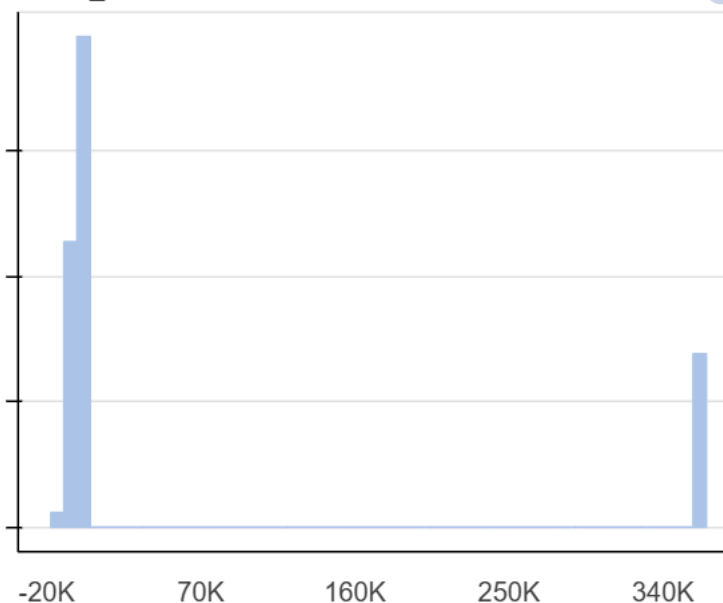
Zeros (%) 0.0%

Negatives 26100

Negatives (%) 82.0%

Frequency

DAYS_EMPLOYED



DAYS_EMPLOYED

DAYS_REGISTRATION
numerical

[Show Details](#)

Approximate
Distinct
Count

Approximate
Unique
(%)

Missing 0

Missing (%) 0.0%

Infinite 0

Infinite (%) 0.0%

Memory
Size 24600

Mean -4985.

Minimum 24671

Maximum 0

Zeros 43

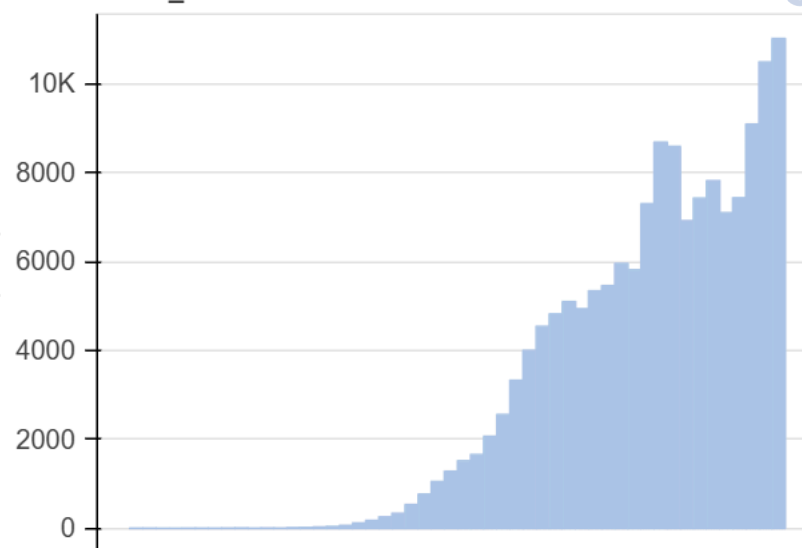
Zeros (%) 0.0%

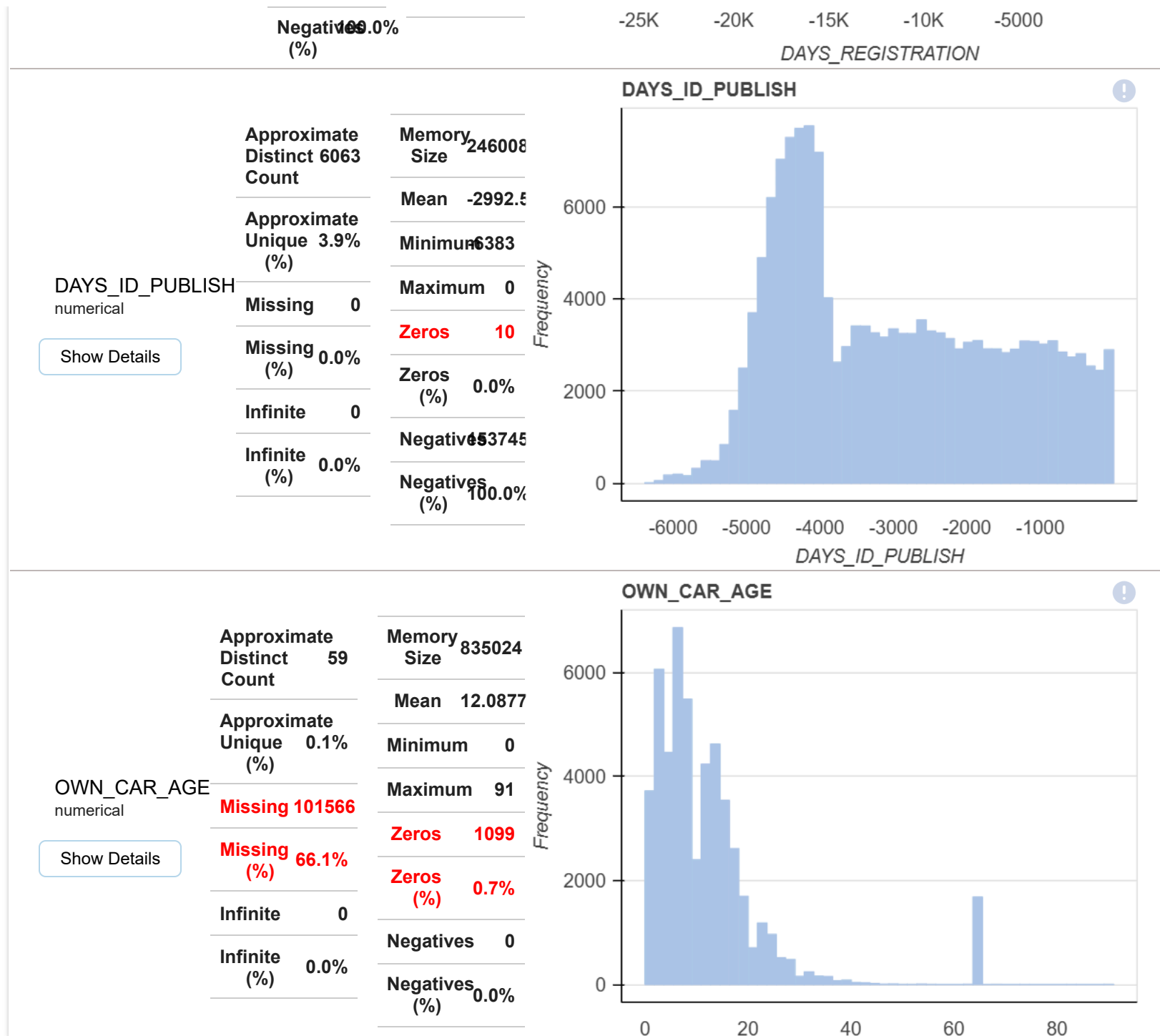
Negatives 6371

Negatives (%) 0.0%

Frequency

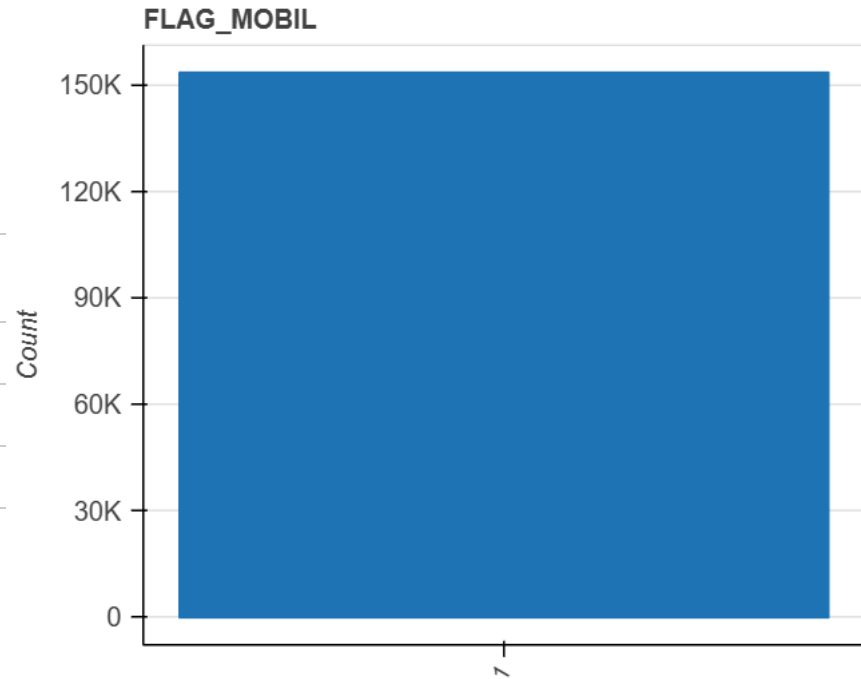
DAYS_REGISTRATION



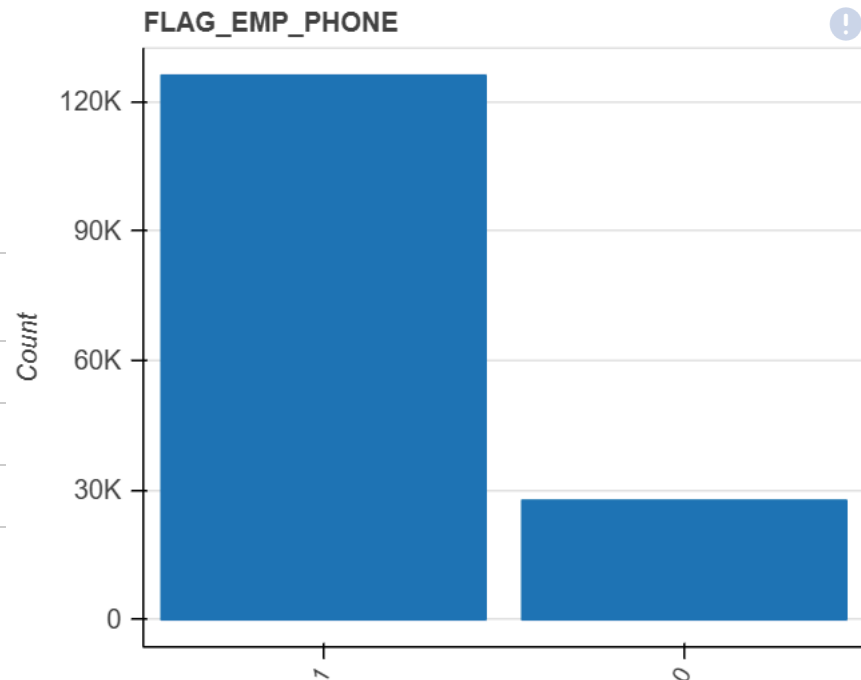


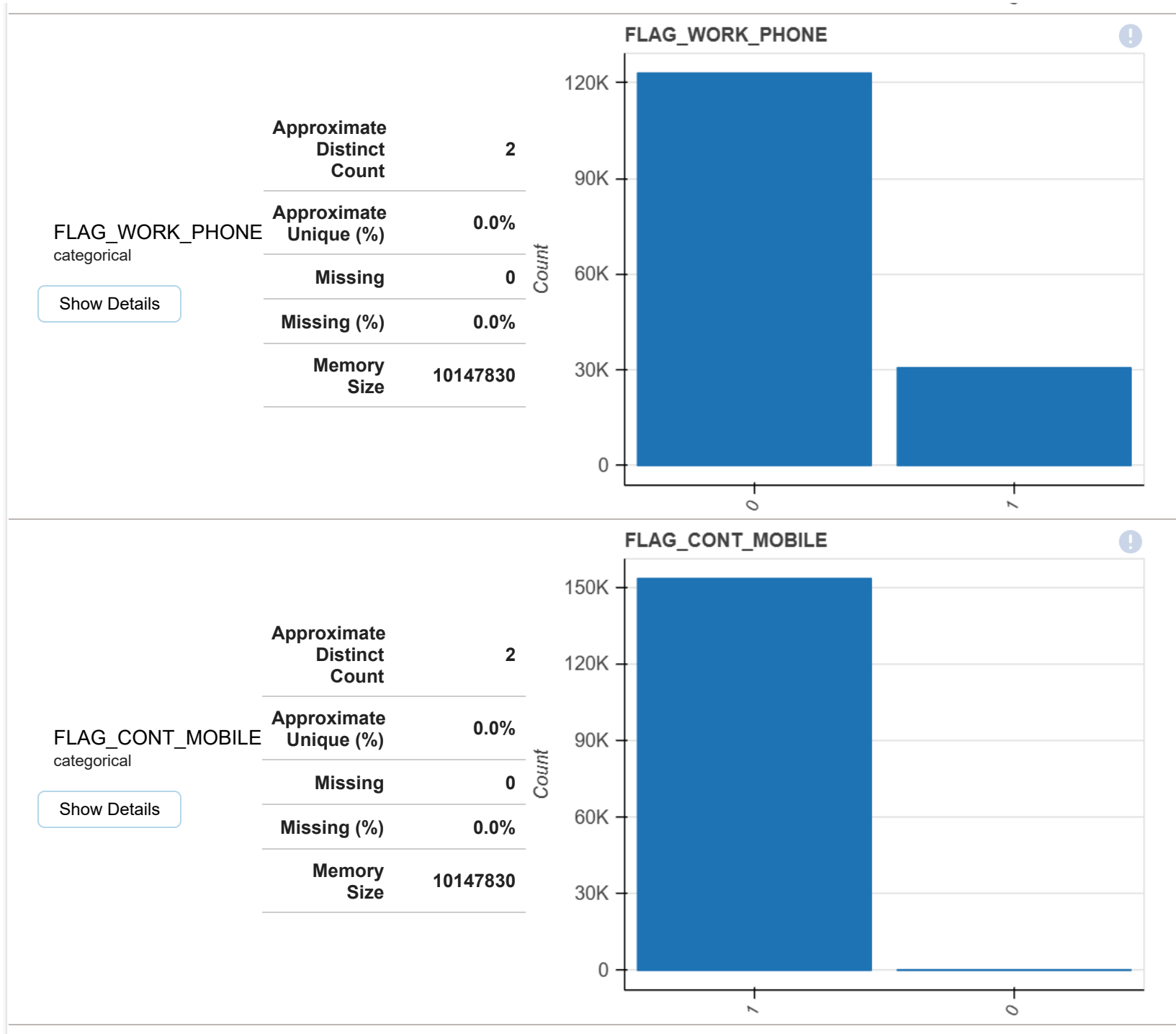
OWN_CAR_AGE

FLAG_MOBIL categorical	Approximate Distinct Count	1
	Approximate Unique (%)	0.0%
	Missing	0
	Missing (%)	0.0%
	Memory Size	10147830

[Show Details](#)

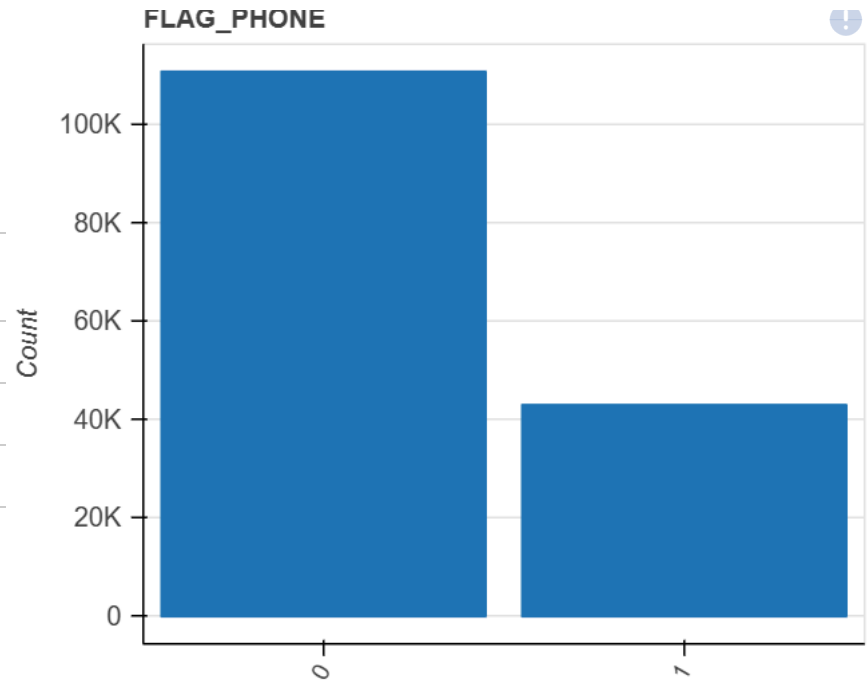
FLAG_EMP_PHONE categorical	Approximate Distinct Count	2
	Approximate Unique (%)	0.0%
	Missing	0
	Missing (%)	0.0%
	Memory Size	10147830

[Show Details](#)



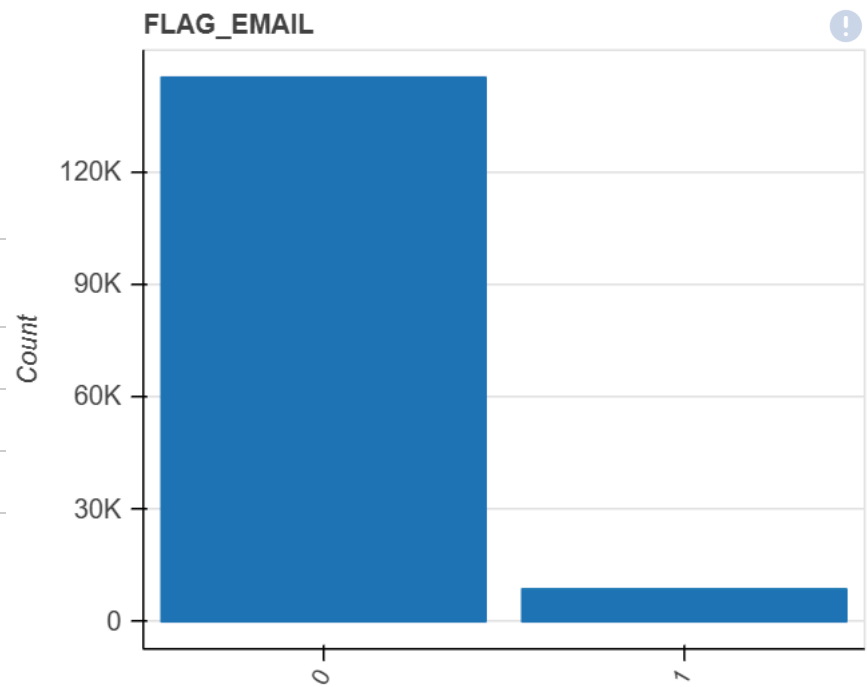
FLAG_PHONE categorical	Approximate Distinct Count	2
	Approximate Unique (%)	0.0%
	Missing	0
	Missing (%)	0.0%
	Memory Size	10147830

Show Details



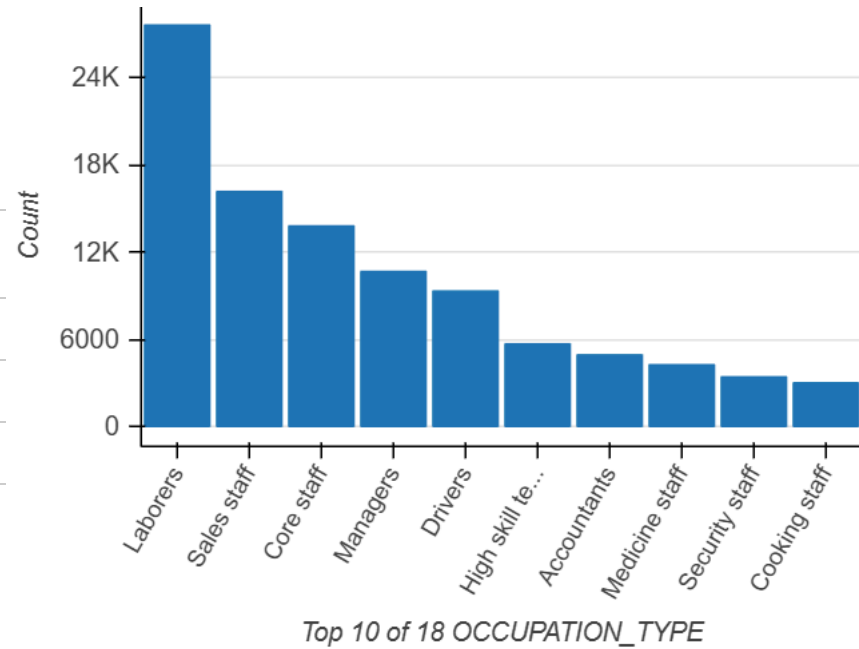
FLAG_EMAIL categorical	Approximate Distinct Count	2
	Approximate Unique (%)	0.0%
	Missing	0
	Missing (%)	0.0%
	Memory Size	10147830

Show Details

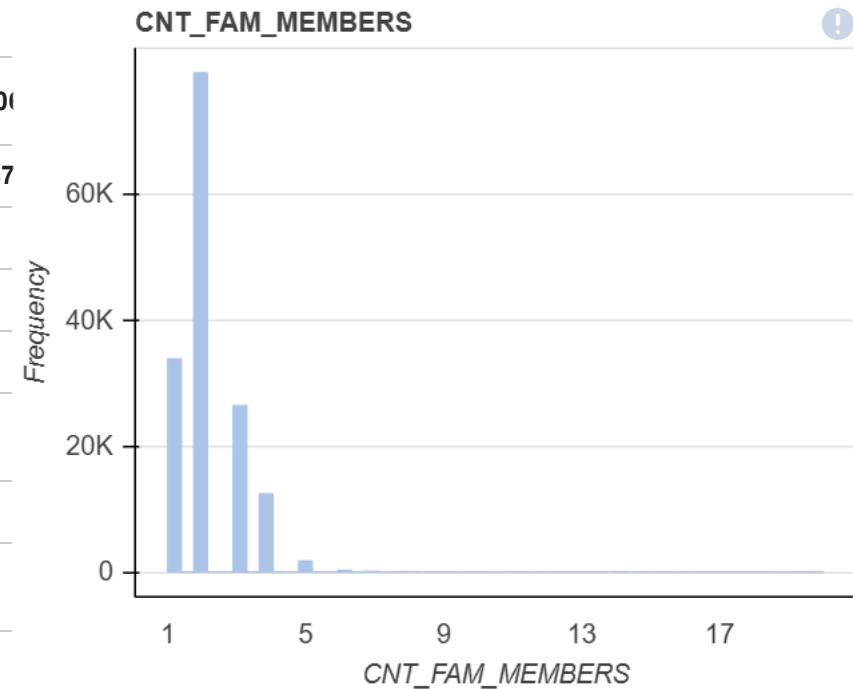


OCCUPATION_TYPE

OCCUPATION_TYPE categorical	Approximate Distinct Count	18
	Approximate Unique (%)	0.0%
	Missing	48090
	Missing (%)	31.3%
	Memory Size	7983082

[Show Details](#)


CNT_FAM_MEMBERS numerical	Approximate Distinct Count	14
	Approximate Unique (%)	0.0%
	Missing	1
	Missing (%)	0.0%
	Infinite	0
	Infinite (%)	0.0%
	Memory Size	24600
	Mean	2.1537
	Minimum	1
	Maximum	20

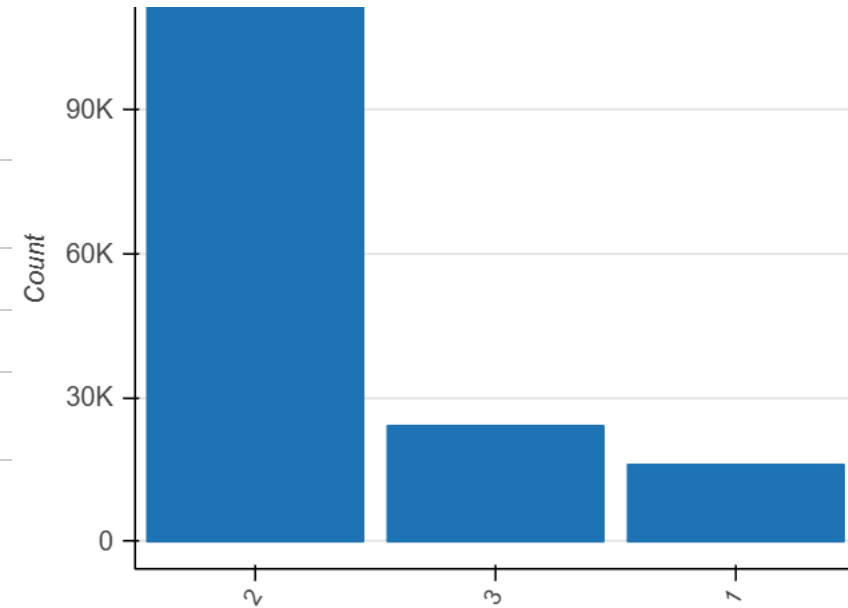
[Show Details](#)


REGION_RATING_CLIENT

REGION_RATING_C...
categorical

Show Details

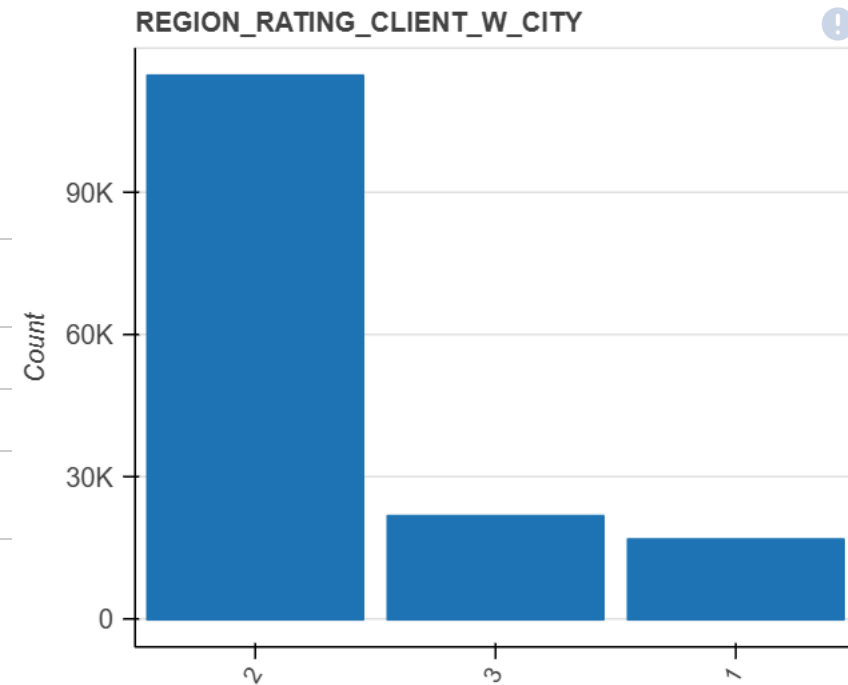
Approximate Distinct Count	3
Approximate Unique (%)	0.0%
Missing	0
Missing (%)	0.0%
Memory Size	10147830



REGION_RATING_C...
categorical

Show Details

Approximate Distinct Count	3
Approximate Unique (%)	0.0%
Missing	0
Missing (%)	0.0%
Memory Size	10147830



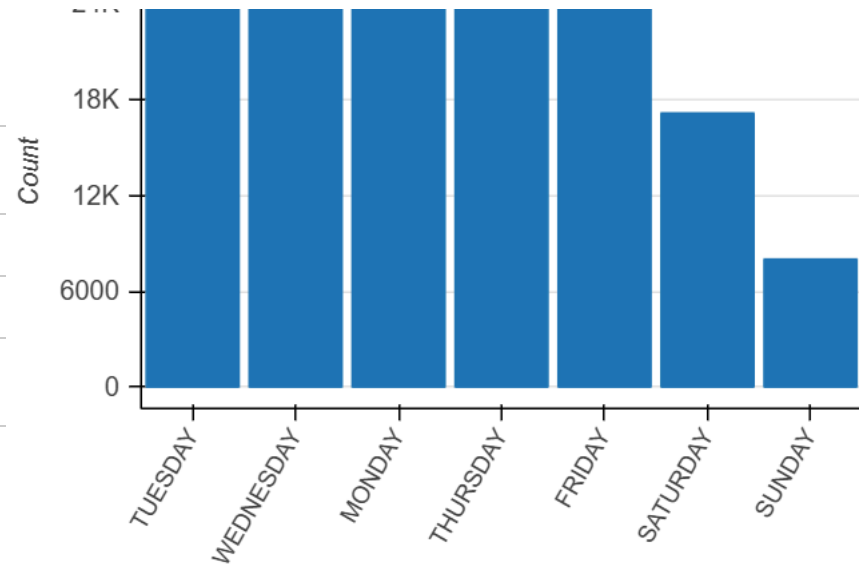
WEEKDAY_APPR_PROCESS_START



WEEKDAY_APPR_P...
categorical

Show Details

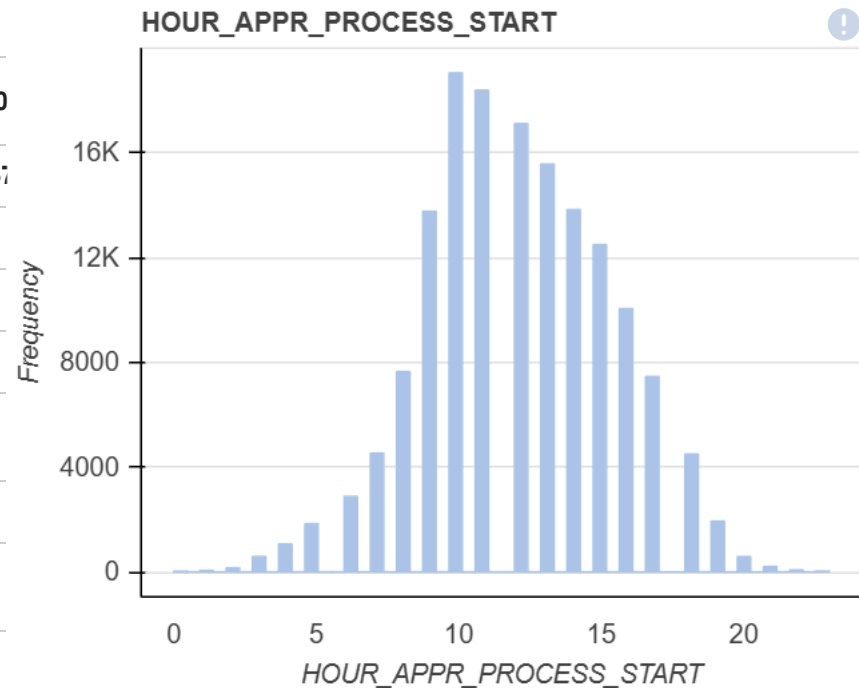
Approximate Distinct Count	7
Approximate Unique (%)	0.0%
Missing	0
Missing (%)	0.0%
Memory Size	11106062



HOUR_APPR_PROG...
numerical

Show Details

Approximate Distinct Count	24	Memory Size	24600
Approximate Unique (%)	0.0%	Mean	12.05
Missing	0	Minimum	0
Missing (%)	0.0%	Maximum	23
Infinite	0	Zeros	21
Infinite (%)	0.0%	Zeros (%)	0.0%
		Negatives	0
		Negatives (%)	0.0%



REG_REGION_NOT_LIVE_REGION



REG_REGION_NOT...

categorical

Show Details

Approximate
Distinct
Count

2

Approximate
Unique (%)

0.0%

Missing

0

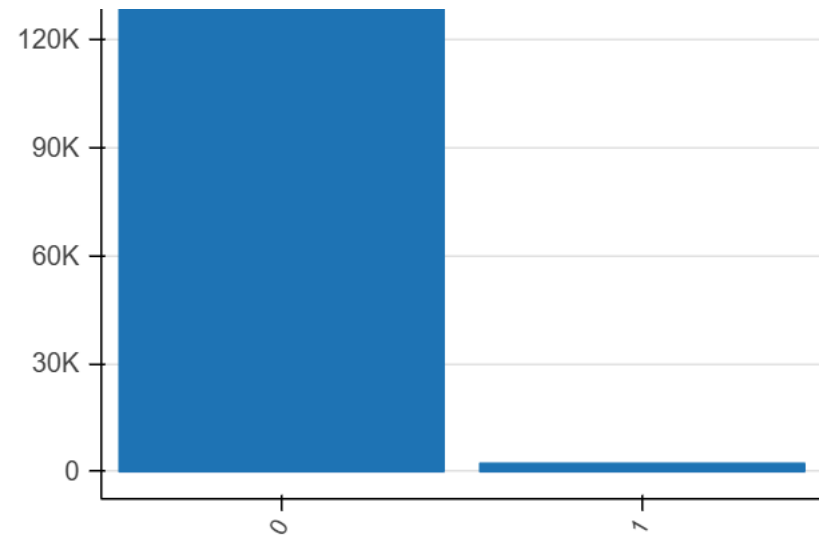
Missing (%)

0.0%

Memory
Size

10147830

Count



REG_REGION_NOT...

categorical

Show Details

Approximate
Distinct
Count

2

Approximate
Unique (%)

0.0%

Missing

0

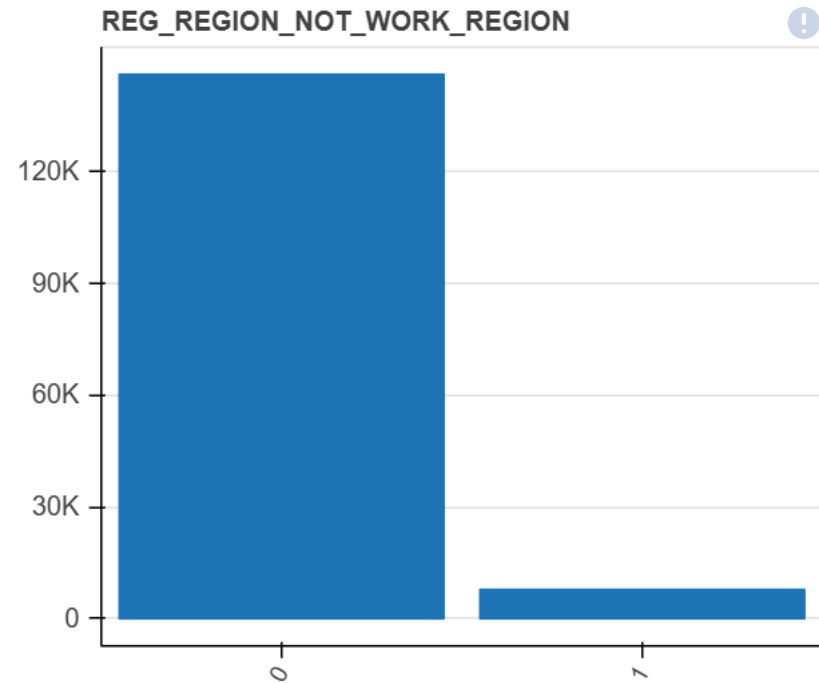
Missing (%)

0.0%

Memory
Size

10147830

Count



LIVE_REGION_NOT_WORK_REGION

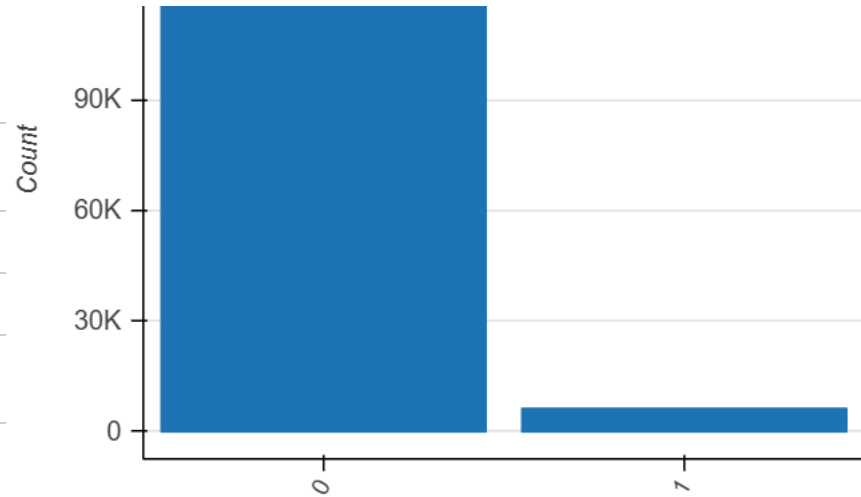


LIVE_REGION_NOT...

categorical

Show Details

Approximate Distinct Count	2
Approximate Unique (%)	0.0%
Missing	0
Missing (%)	0.0%
Memory Size	10147830

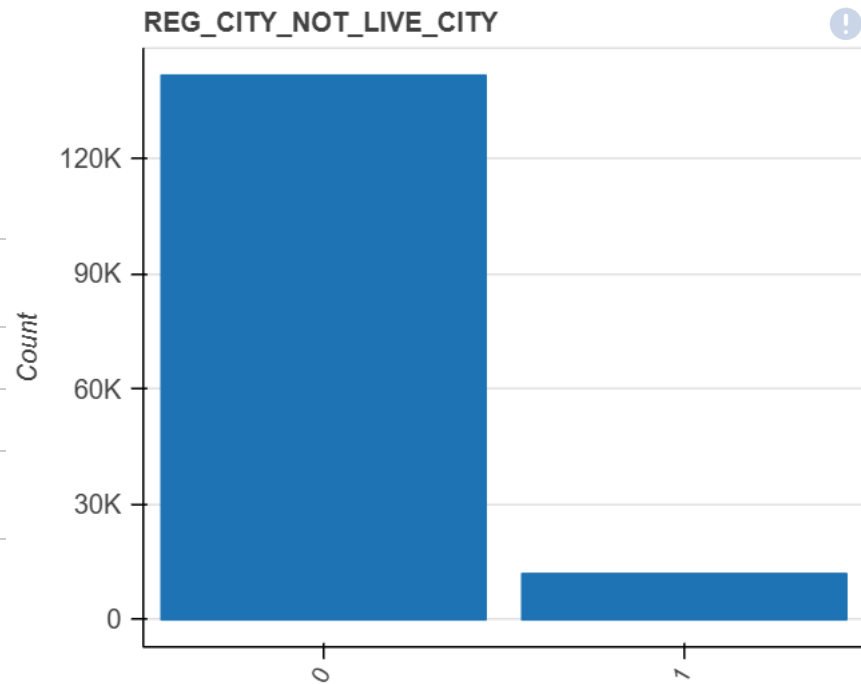


REG_CITY_NOT_LI...

categorical

Show Details

Approximate Distinct Count	2
Approximate Unique (%)	0.0%
Missing	0
Missing (%)	0.0%
Memory Size	10147830



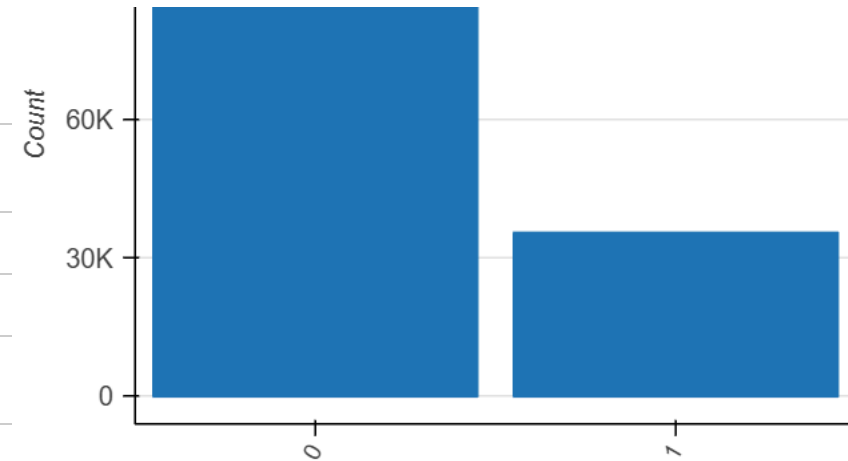
REG_CITY_NOT_WORK_CITY

90K

REG_CITY_NOT_W...
categorical

Show Details

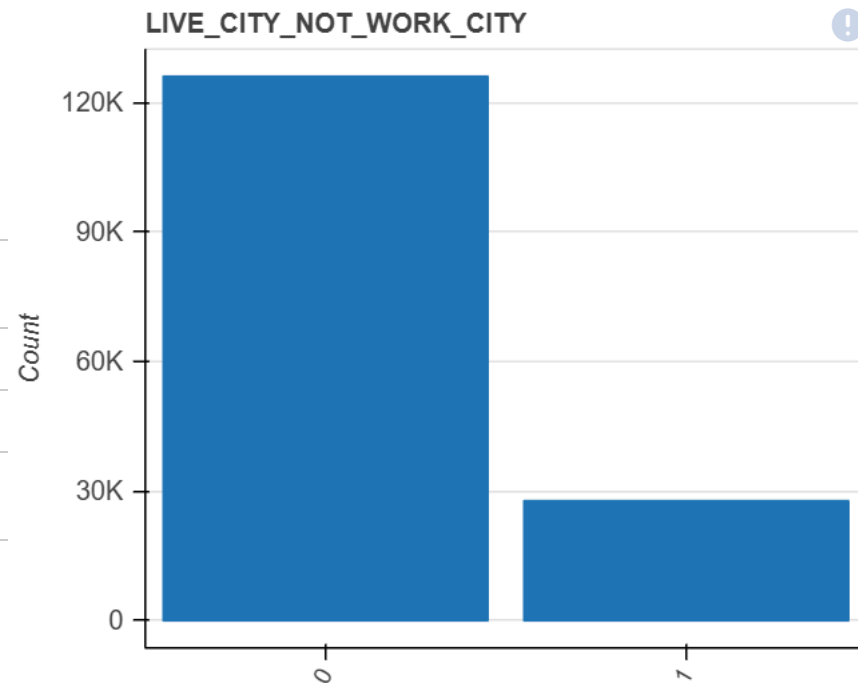
Approximate Distinct Count	2
Approximate Unique (%)	0.0%
Missing	0
Missing (%)	0.0%
Memory Size	10147830



LIVE_CITY_NOT_W...
categorical

Show Details

Approximate Distinct Count	2
Approximate Unique (%)	0.0%
Missing	0
Missing (%)	0.0%
Memory Size	10147830

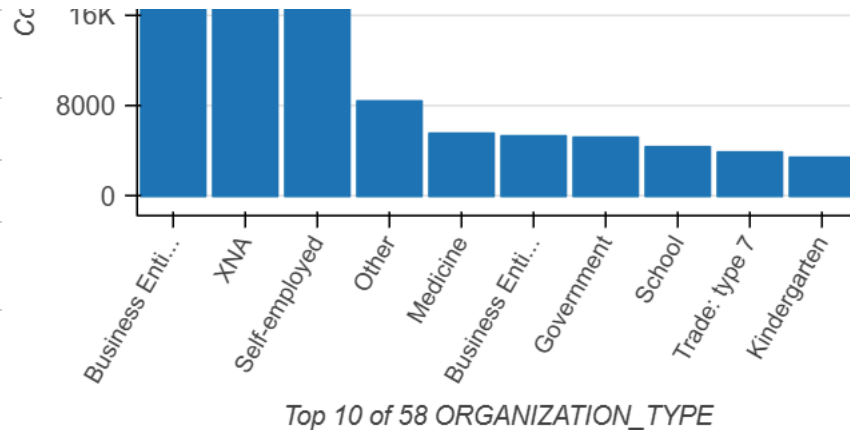


ORGANIZATION_TYPE

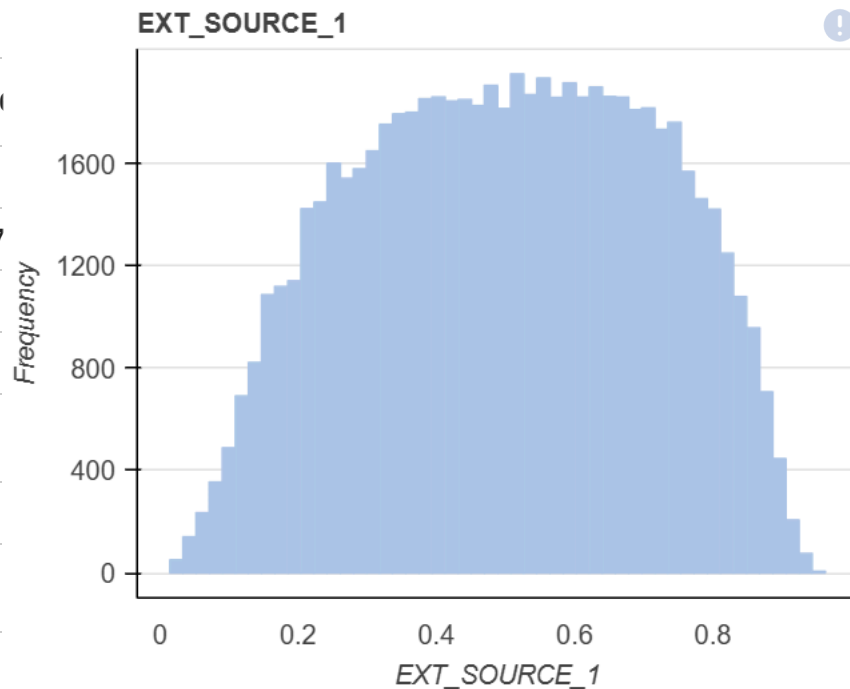
Approximate Distinct Count	58
----------------------------------	----



ORGANIZATION_TYPE categorical	Approximate Unique (%)	0.0%
	Missing	0
	Missing (%)	0.0%
	Memory Size	11920588

[Show Details](#)


EXT_SOURCE_1 numerical	Approximate Distinct Count	61894	Memory Size	1072331
	Approximate Unique (%)	92.3%	Mean	0.5012
	Missing	86734	Minimum	0.01457
	Missing (%)	56.4%	Maximum	0.9627
	Infinite	0	Zeros	0
	Infinite (%)	0.0%	Zeros (%)	0.0%
			Negatives	0
			Negatives (%)	0.0%

[Show Details](#)


Approximate Distinct Count	Missing	303
	Missing (%)	0.2%
	Infinite	0
Approximate Unique (%)		



EXT_SOURCE_2
numerical[Show Details](#)

Infinite (%) 0.0%

Memory Size 2455232

Mean 0.5143

Minimum 8.1736×10^{-08}

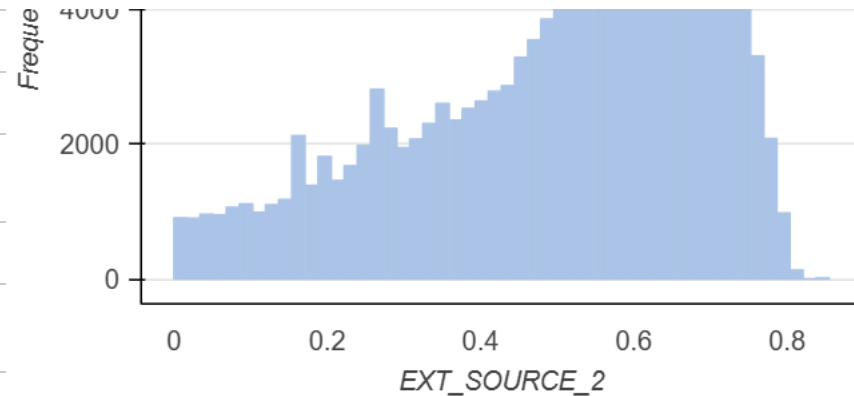
Maximum 0.855

Zeros 0

Zeros (%) 0.0%

Negatives 0

Negatives (%) 0.0%

EXT_SOURCE_3
numerical[Show Details](#)

Approximate Distinct Count 785

Approximate Unique (%) 0.6%

Missing 30529

Missing (%) 19.9%

Infinite 0

Infinite (%) 0.0%

Memory Size 1971610

Mean 0.5111

Minimum 0.00052

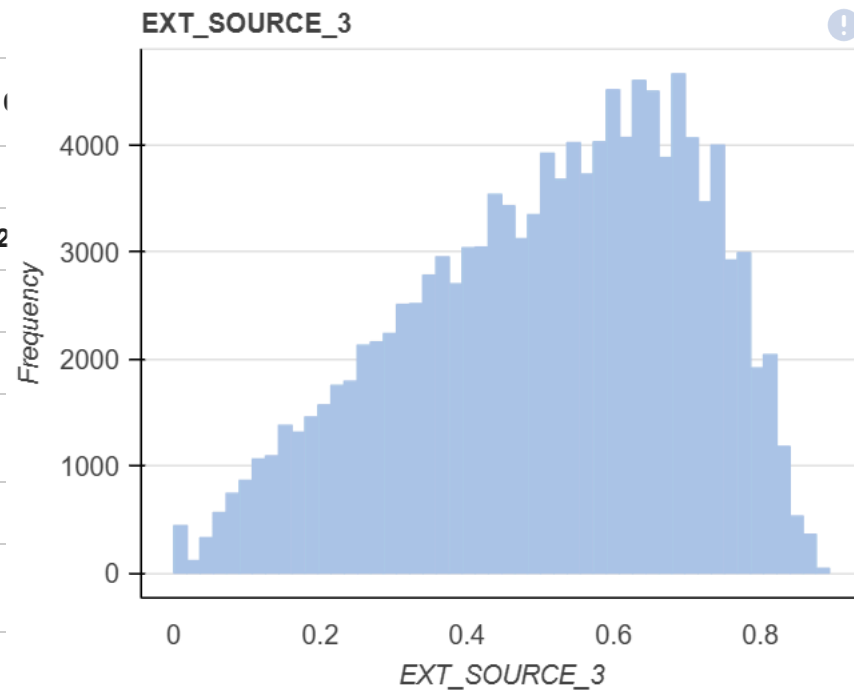
Maximum 0.894

Zeros 0

Zeros (%) 0.0%

Negatives 0

Negatives (%) 0.0%



APARTMENTS_AVG

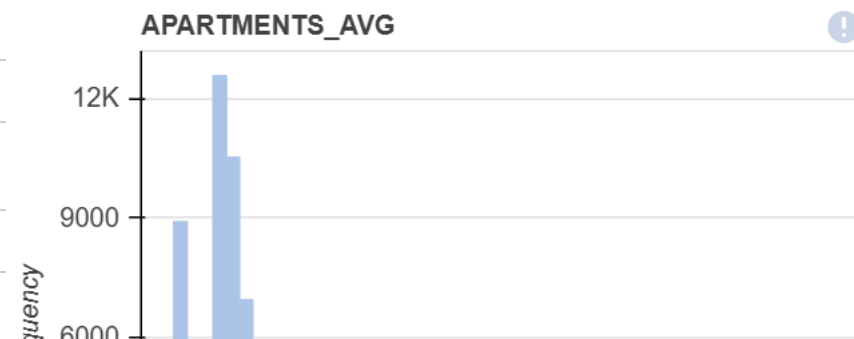
Approximate Distinct Count 2048

Approximate Unique (%) 2.7%

Missing 78090

Missing (%) 50.8%

Infinite 0



numerical

[Show Details](#)

Infinite (%) 0.0%

Zeros 362

Memory Size 1210640

Zeros (%) 0.2%

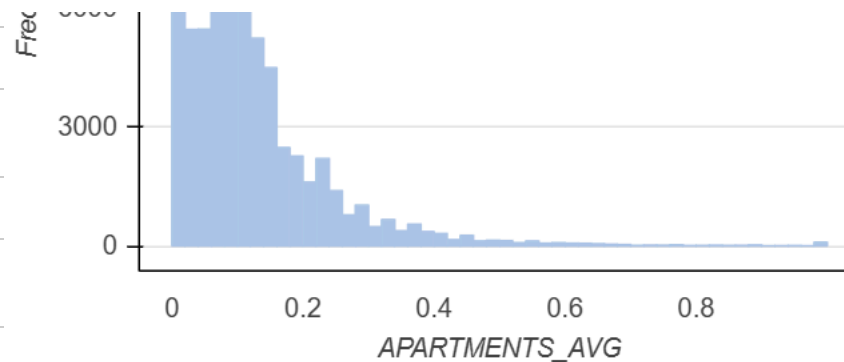
Mean 0.1173

Negatives 0

Minimum 0

Negatives (%) 0.0%

Maximum 1

BASEMENTAREA_A...
numerical[Show Details](#)

Approximate Distinct Count 3450

Memory Size 10215

Approximate Unique (%) 5.4%

Mean 0.0884

Missing 9907

Minimum 0

Missing (%) 98.5%

Maximum 1

Infinite 0

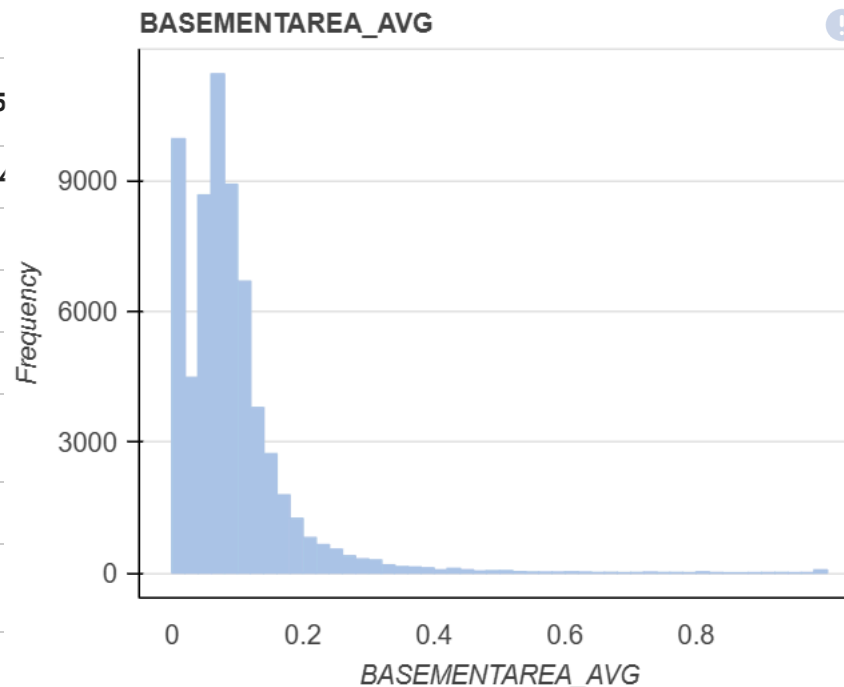
Zeros 7421

Infinite (%) 0.0%

Zeros (%) 4.8%

Negatives 0

Negatives (%) 0.0%

YEARS_BEGINEXPL...
numerical

Approximate Distinct Count 247

Missing (%) 48.8%

Approximate Unique (%) 0.3%

Infinite 0

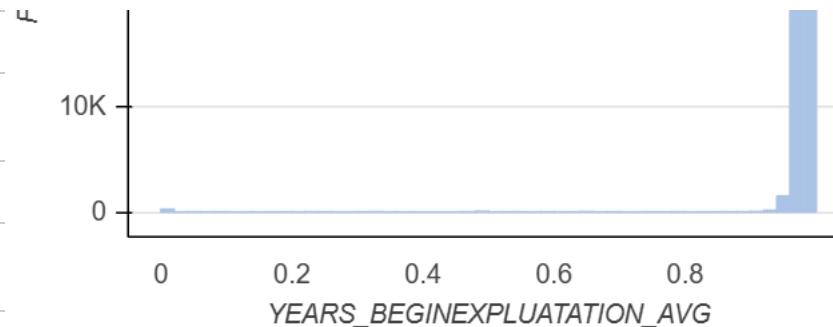
Missing 5022

Infinite (%) 0.0%

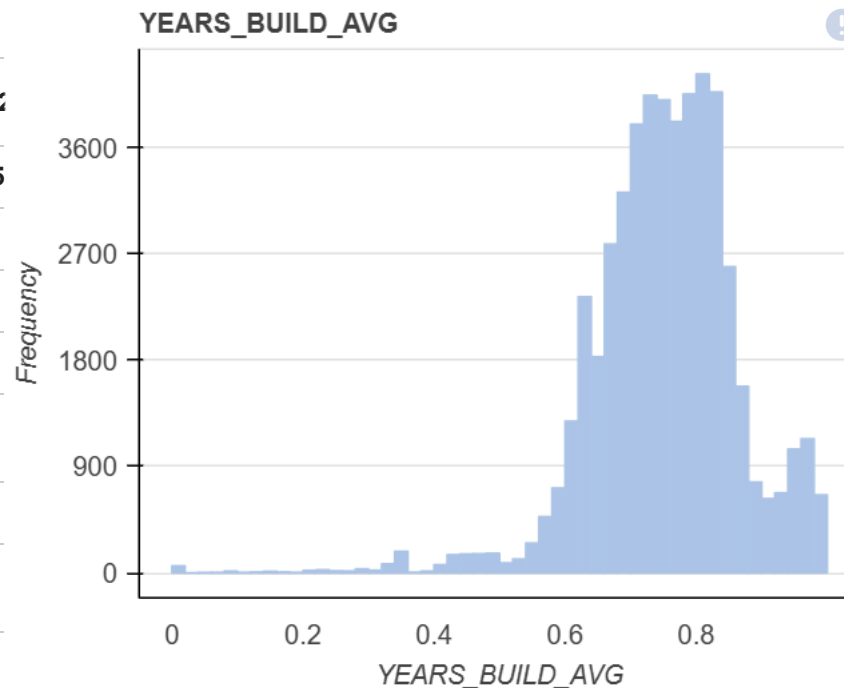


[Show Details](#)

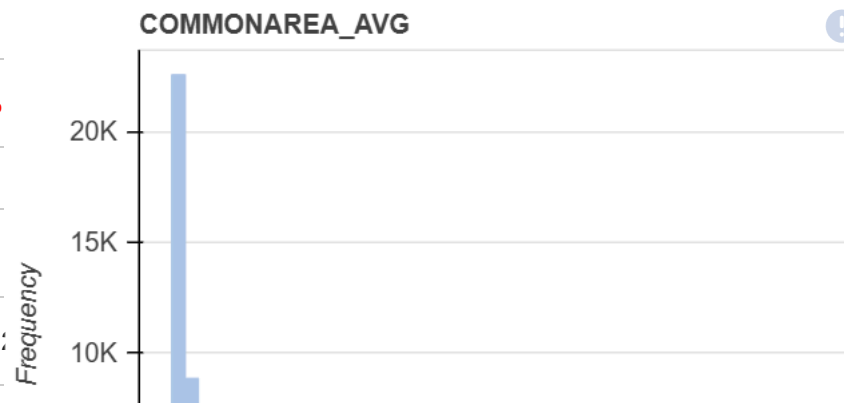
Memory Size	1259728	Zeros	245
Mean	0.9778	Zeros (%)	0.2%
Minimum	0	Negatives	0
Maximum	1	Negatives (%)	0.0%

YEARS_BUILD_AVG
numerical[Show Details](#)

Approximate Distinct Count	144	Memory Size	823552
Approximate Unique (%)	0.3%	Mean	0.7525
Missing	102283	Minimum	0
Missing (%)	66.5%	Maximum	1
Infinite	0	Zeros	55
Infinite (%)	0.0%	Zeros (%)	0.0%
		Negatives	0
		Negatives (%)	0.0%

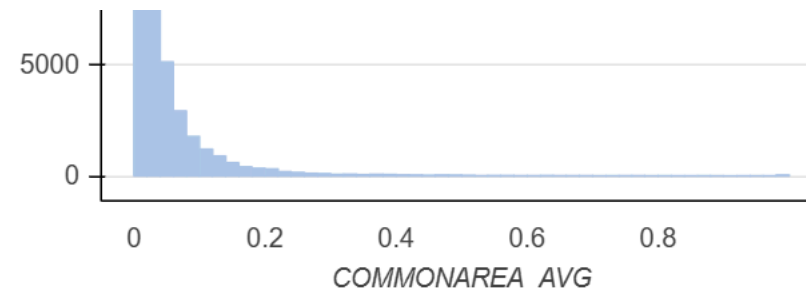
COMMONAREA_AVG
numerical

Approximate Distinct Count	2730	Missing	69.9%
Approximate Unique (%)	5.9%	Infinite	0
Missing	07523	Infinite (%)	0.0%
Missing (%)		Memory Size	739712



[Show Details](#)

Mean	0.04441	Zeros (%)	2.7%
Minimum	0	Negatives	0
Maximum	1	Negatives (%)	0.0%
Zeros	4146		

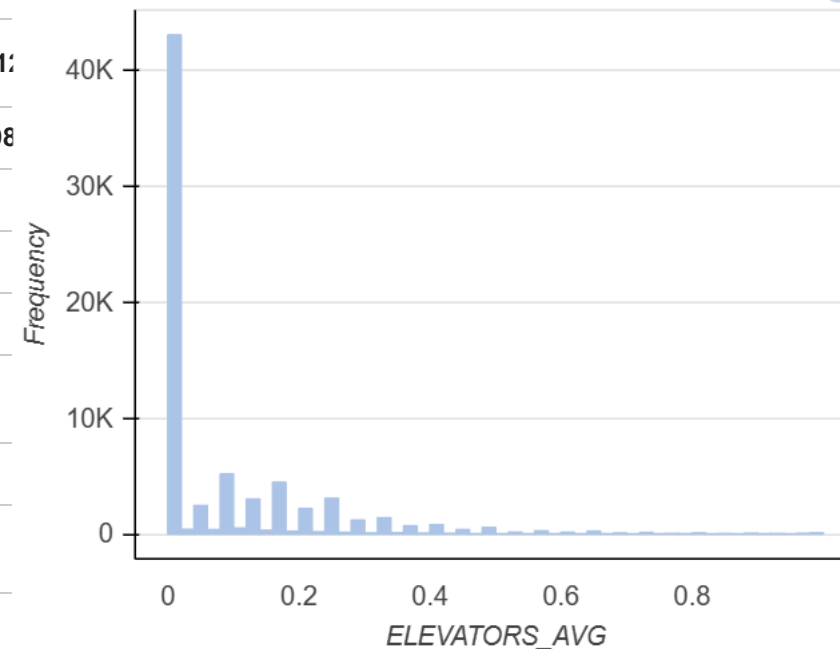


ELEVATORS_AVG
numerical

[Show Details](#)

Approximate Distinct Count	225	Memory Size	114851
Approximate Unique (%)	0.3%	Mean	0.07898
Missing	81973	Minimum	0
Missing (%)	53.3%	Maximum	1
Infinite	0	Zeros	42922
Infinite (%)	0.0%	Zeros (%)	27.9%
		Negatives	0
		Negatives (%)	0.0%

ELEVATORS_AVG

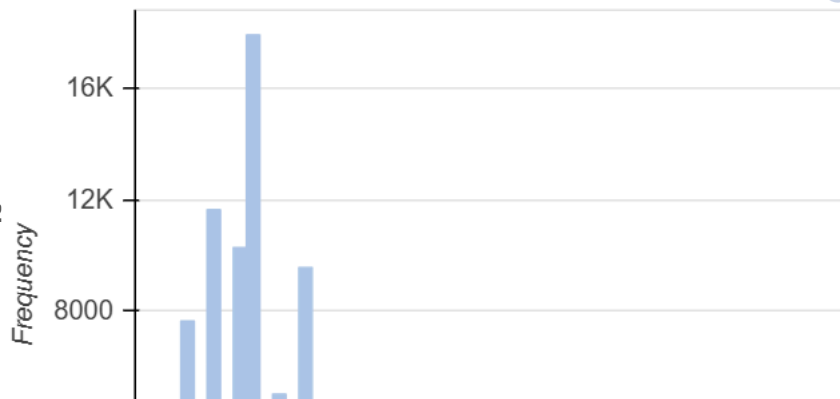


ENTRANCES_AVG
numerical

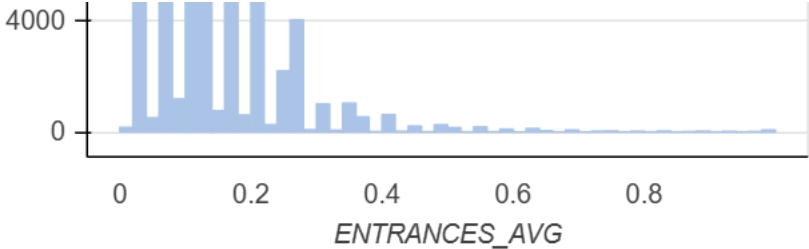
[Show Details](#)

Approximate Distinct Count	247	Infinite	0
Approximate Unique (%)	0.3%	Infinite (%)	0.0%
Missing	77410	Memory Size	122152
Missing (%)	50.3%	Mean	0.1498
		Minimum	0

ENTRANCES_AVG



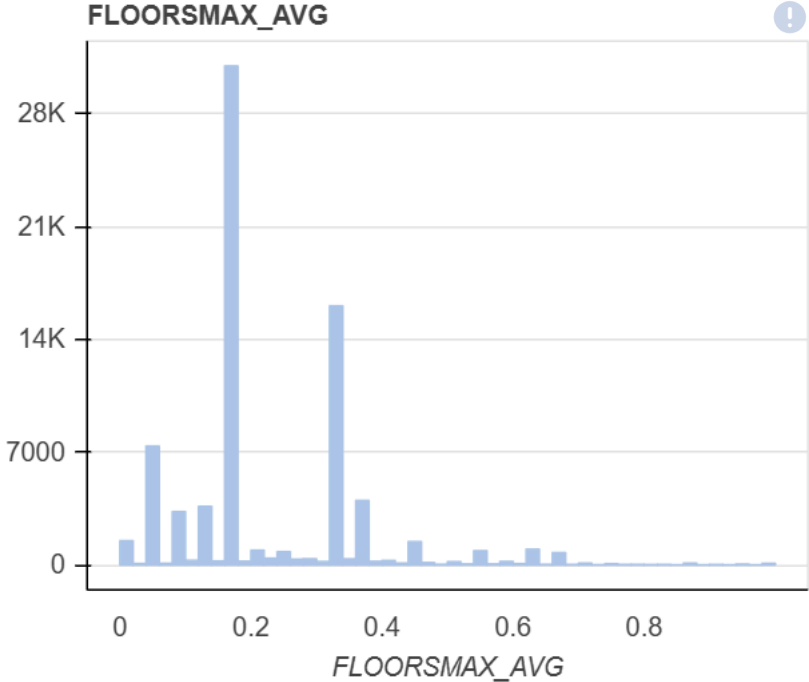
Maximum	1	Negatives	0
Zeros	153	Negatives (%)	0.0%
Zeros (%)	0.1%		



FLOORSMAX_AVG
numerical

Show Details

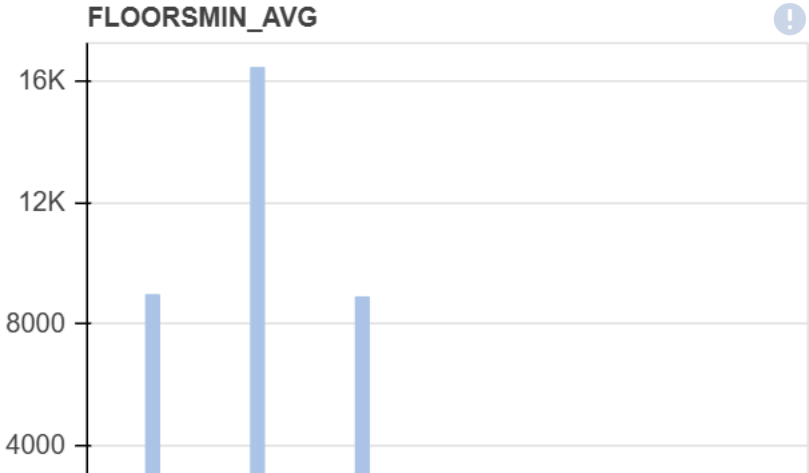
Approximate Distinct Count	335	Memory Size	123579
Approximate Unique (%)	0.4%	Mean	0.2261
Missing	76518	Minimum	0
Missing (%)	49.8%	Maximum	1
Infinite	0	Zeros	1475
Infinite (%)	0.0%	Zeros (%)	1.0%
		Negatives	0
		Negatives (%)	0.0%



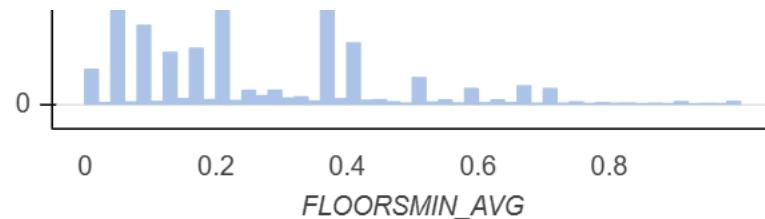
FLOORSMIN_AVG
numerical

Show Details

Approximate Distinct Count	259	Infinite	0
Approximate Unique (%)	0.5%	Infinite (%)	0.0%
Missing	104424	Memory Size	789296
Missing (%)	67.9%	Mean	0.2317
		Minimum	0
		Maximum	1



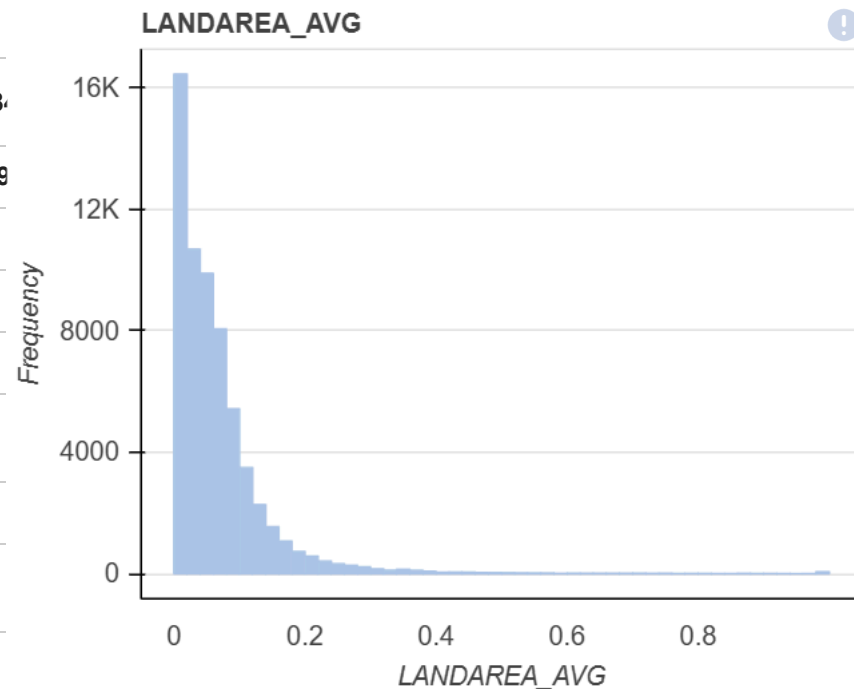
Zeros	1129	Negatives	0
Zeros (%)	0.7%	Negatives (%)	0.0%



LANDAREA_AVG
numerical

Show Details

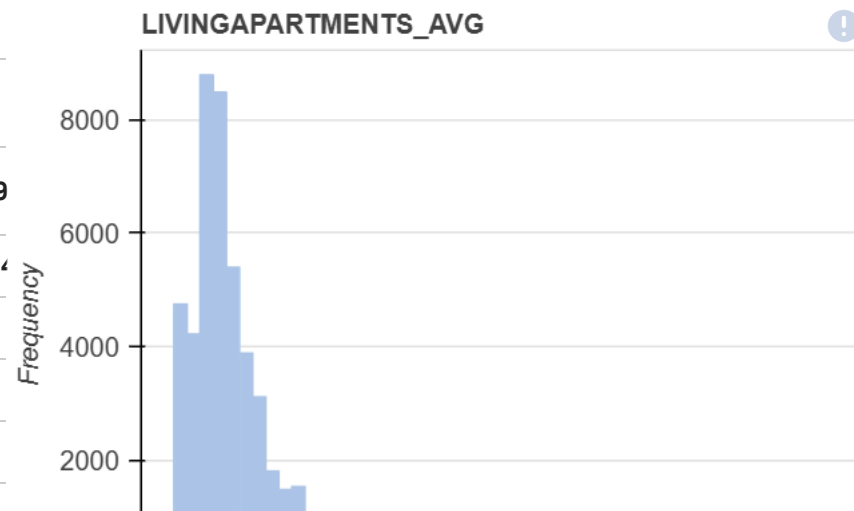
Approximate Distinct Count	3174	Memory Size	1000784
Approximate Unique (%)	5.1%	Mean	0.06629
Missing	91206	Minimum	0
Missing (%)	59.3%	Maximum	1
Infinite	0	Zeros	7754
Infinite (%)	0.0%	Zeros (%)	5.0%
		Negatives	0
		Negatives (%)	0.0%



LIVINGAPARTMENT...
numerical

Show Details

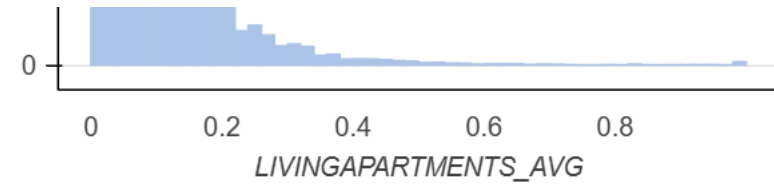
Approximate Distinct Count	623	Infinite (%)	0.0%
Approximate Unique (%)	8.3%	Memory Size	77699
Missing	405193	Mean	0.1004
Missing (%)	88.4%	Minimum	0
Infinite	0	Maximum	1
		Zeros	218



Zeros
(%) 0.1%

Negatives
0

Negatives
(%) 0.0%



LIVINGAREA_AVG
numerical

Show Details

Approximate
Distinct 4762
Count

Approximate
Unique 6.2%
(%)

Missing 77133

Missing
(%) 50.2%

Infinite 0

Infinite
(%) 0.0%

Memory
Size 122595

Mean 0.1074

Minimum 0

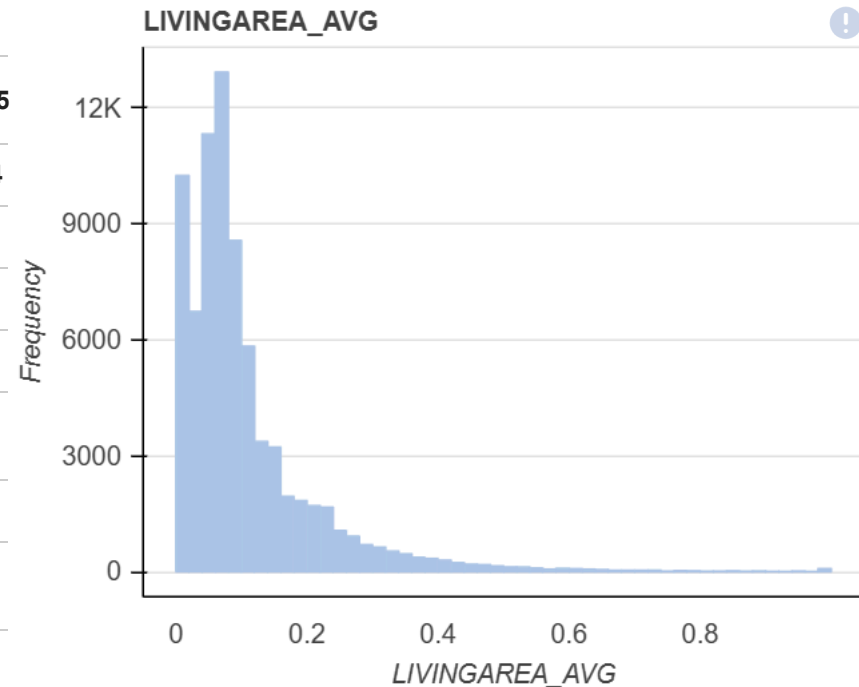
Maximum 1

Zeros 146

Zeros
(%) 0.1%

Negatives 0

Negatives
(%) 0.0%



NONLIVINGAPARTM...
numerical

Show Details

Approximate
Distinct 312
Count

Approximate
Unique 0.7%
(%)

Missing 06918

Missing
(%) 69.5%

Infinite 0

Infinite
(%) 0.0%

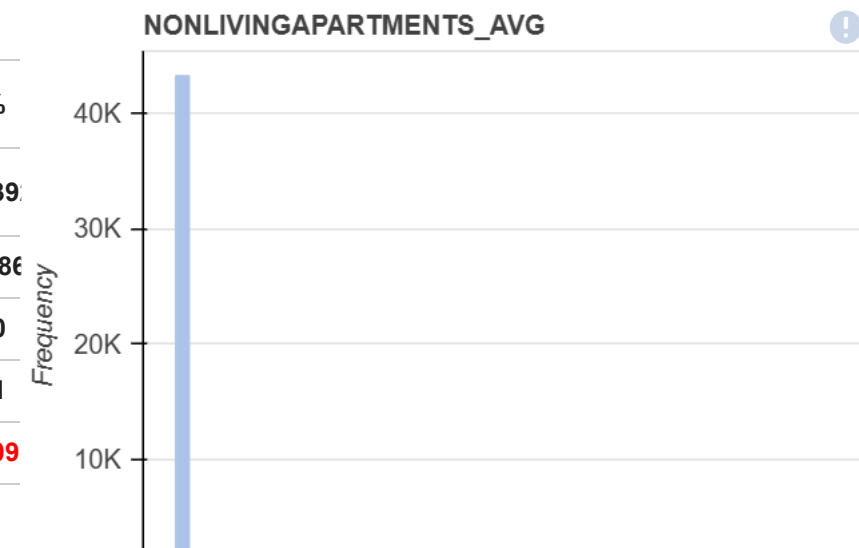
Memory
Size 74939

Mean 0.0086

Minimum 0

Maximum 1

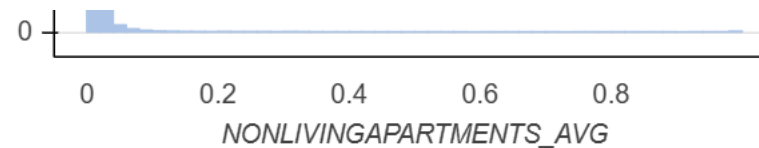
Zeros 27309



Zeros
(%) 17.8%

Negatives
(%) 0.0%

Negatives 0



NONLIVINGAREA_A...
numerical

Show Details

Approximate
Distinct 2806
Count

Memory
Size 11023:

Approximate
Unique 4.1%
(%)

Mean 0.0282

Minimum 0

Maximum 1

Missing 4860

Missing
(%) 55.2%

Zeros 29485

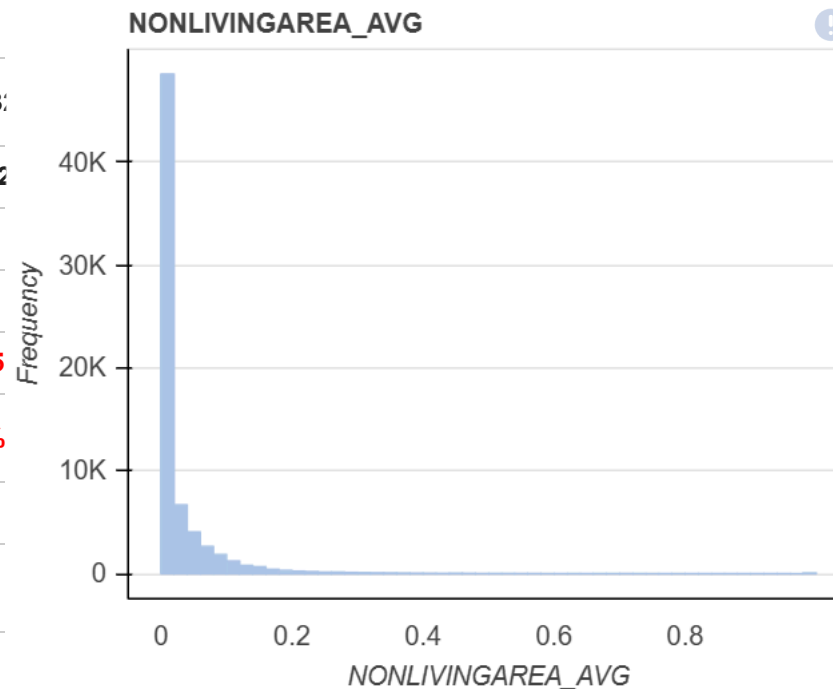
Zeros
(%) 19.2%

Infinite 0

Negatives 0

Infinite
(%) 0.0%

Negatives
(%) 0.0%



APARTMENTS_MODE
numerical

Show Details

Approximate
Distinct 717
Count

Infinite
(%) 0.0%

Approximate
Unique 0.9%
(%)

Memory
Size 12106

Mean 0.1142

Minimum 0

Maximum 1

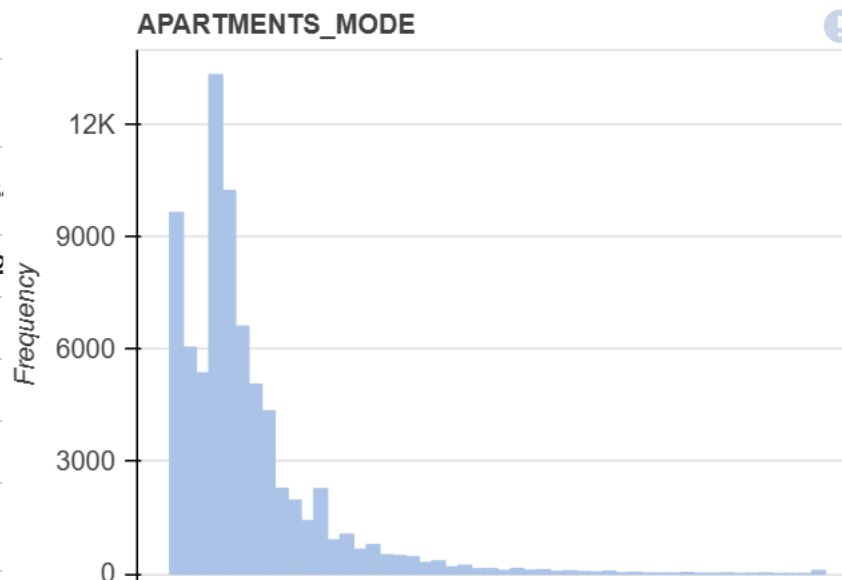
Missing 8090

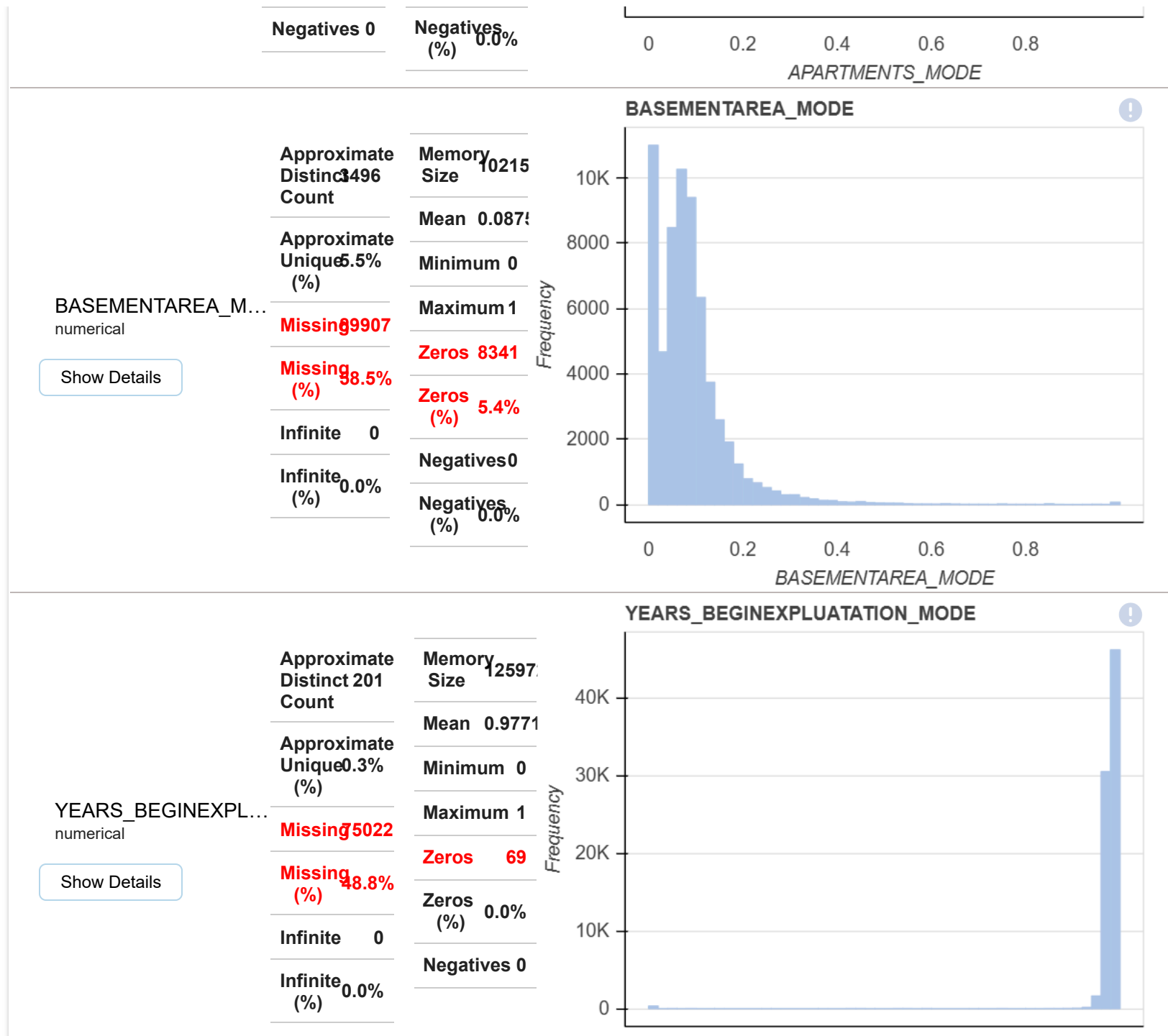
Missing
(%) 50.8%

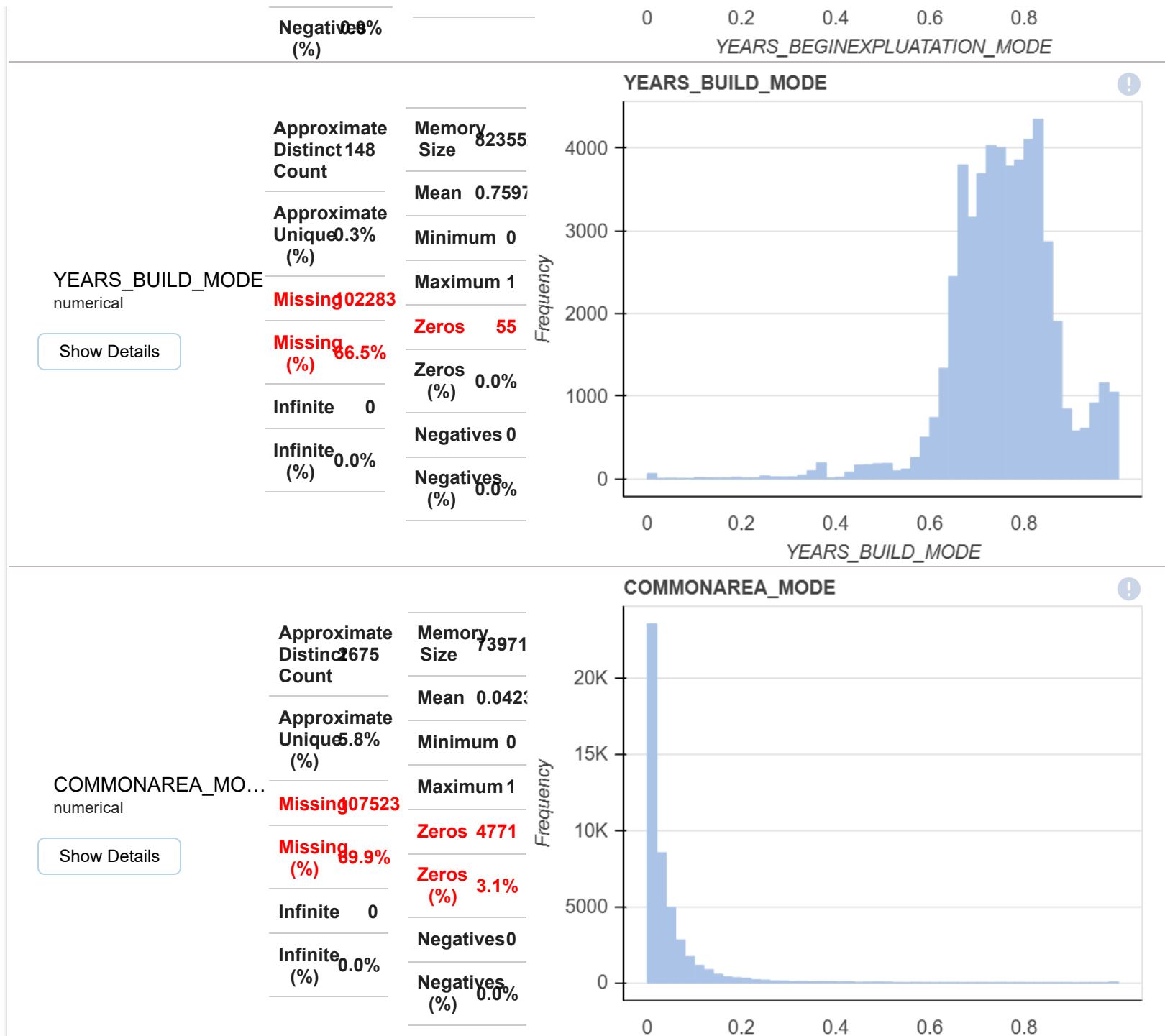
Zeros 490

Zeros
(%) 0.3%

Infinite 0







COMMONAREA_MODE

ELEVATORS_MODE
numerical[Show Details](#)Approximate
Distinct Count 26Approximate
Unique 0.0%
(%)

Missing 81973

Missing 53.3%
(%)

Infinite 0

Infinite 0.0%
(%)

Memory Size 114851

Mean 0.0745

Minimum 0

Maximum 1

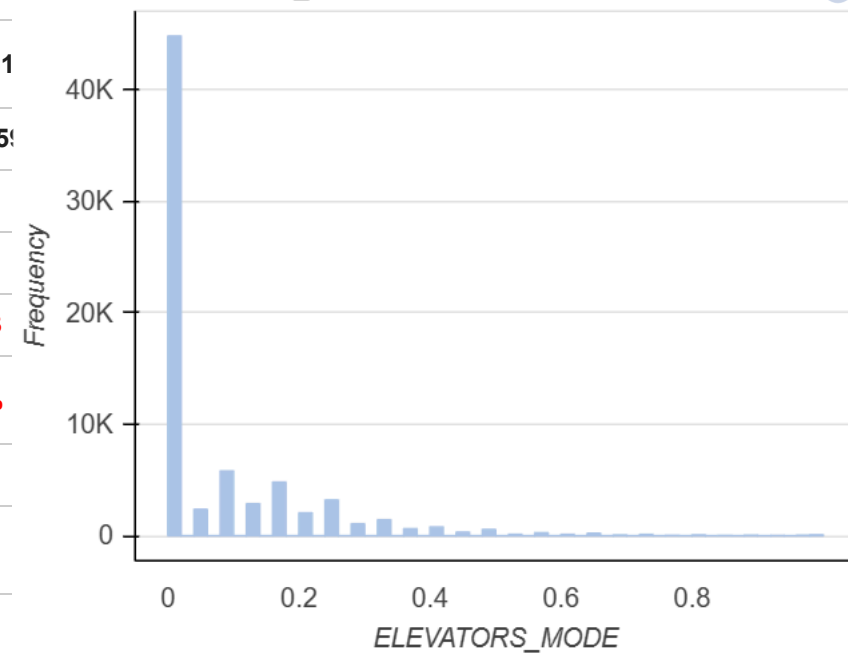
Zeros 44813

Zeros 29.1%
(%)

Negatives 0

Negatives 0.0%
(%)

ELEVATORS_MODE

ENTRANCES_MODE
numerical[Show Details](#)Approximate
Distinct Count 30Approximate
Unique 0.0%
(%)

Missing 77410

Missing 50.3%
(%)

Infinite 0

Infinite 0.0%
(%)

Memory Size 12215

Mean 0.1453

Minimum 0

Maximum 1

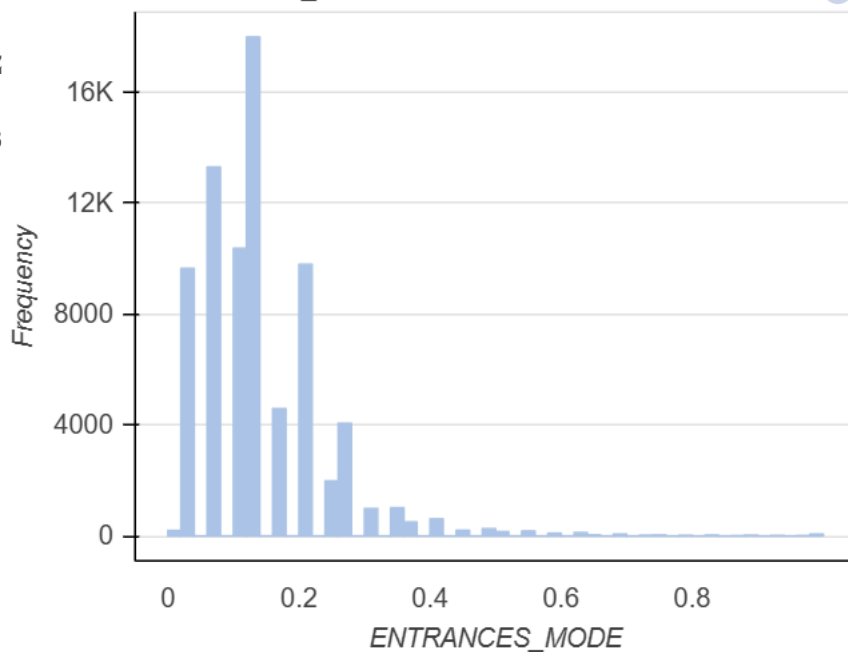
Zeros 192

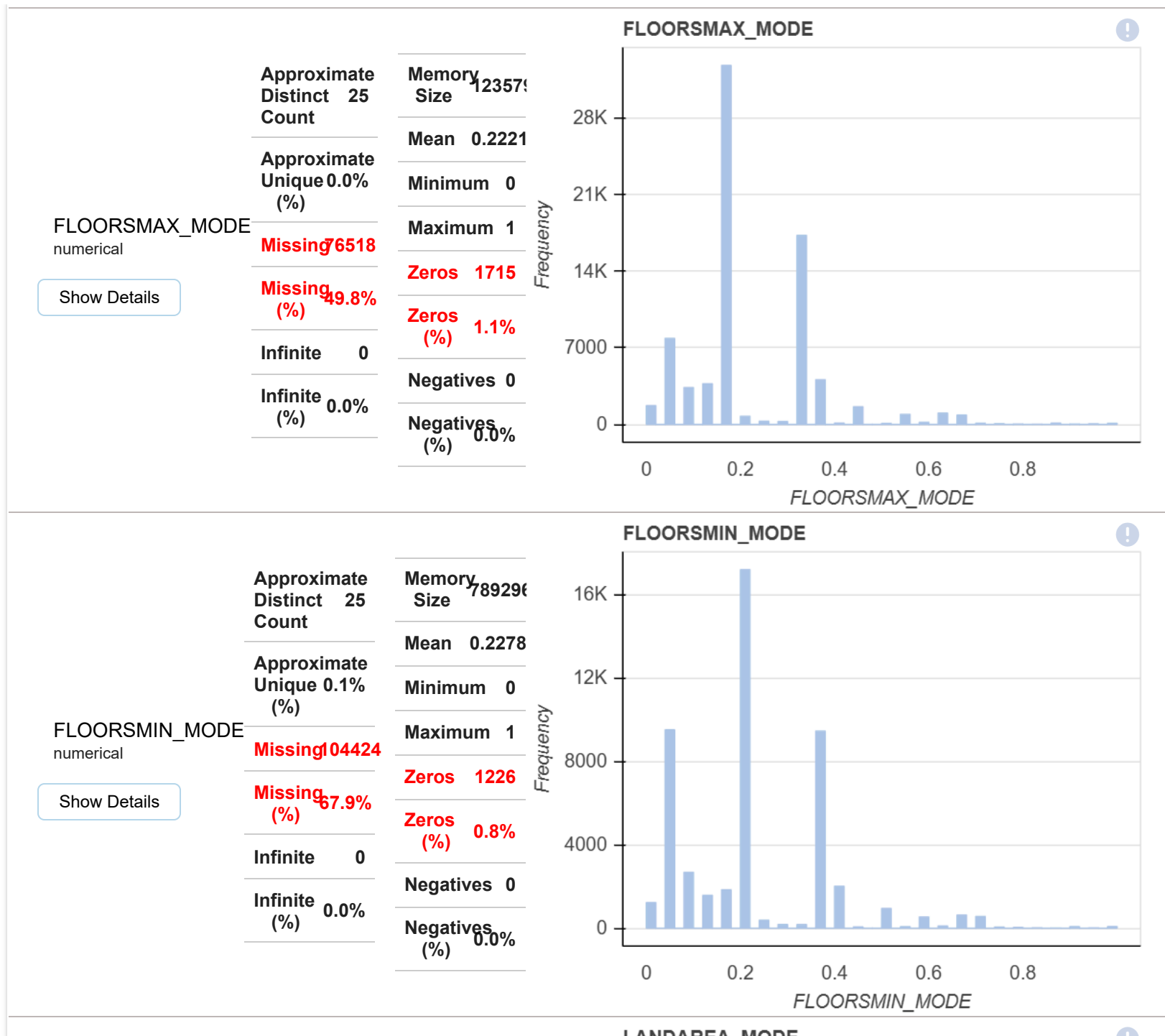
Zeros 0.1%
(%)

Negatives 0

Negatives 0.0%
(%)

ENTRANCES_MODE

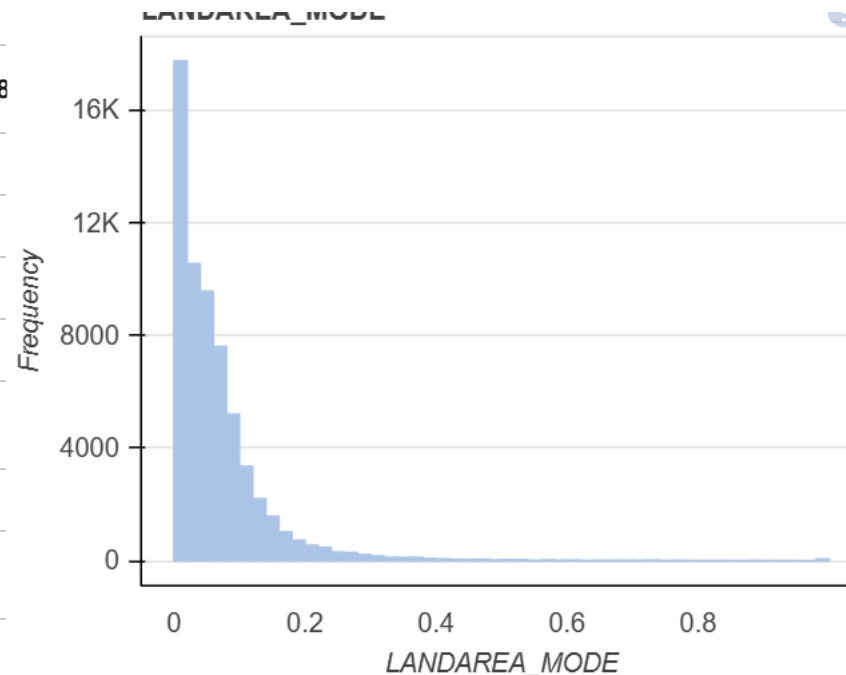




LANDAREA_MODE
numerical

Show Details

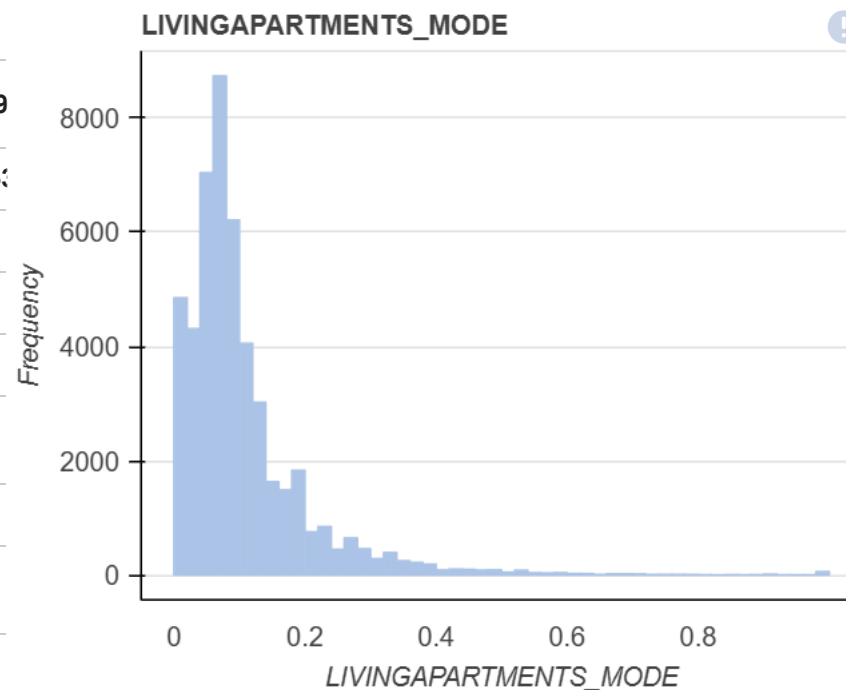
Approximate Distinct Count	3210	Memory Size	100078
Approximate Unique (%)	5.1%	Mean	0.065
Missing	91206	Minimum	0
Missing (%)	59.3%	Maximum	1
Infinite	0	Zeros	8644
Infinite (%)	0.0%	Zeros (%)	5.6%
		Negatives	0
		Negatives (%)	0.0%



LIVINGAPARTMENT...
numerical

Show Details

Approximate Distinct Count	686	Memory Size	77699
Approximate Unique (%)	1.4%	Mean	0.105%
Missing	405193	Minimum	0
Missing (%)	88.4%	Maximum	1
Infinite	0	Zeros	263
Infinite (%)	0.0%	Zeros (%)	0.2%
		Negatives	0
		Negatives (%)	0.0%



LIVINGAREA_MODE

LIVINGAREA_MODE
numerical

Show Details

Approximate
Distinct
Count

Approximate
Unique
(%)

Missing

Missing
(%)

Infinite

Infinite
(%)

Memory
Size

Mean

Minimum

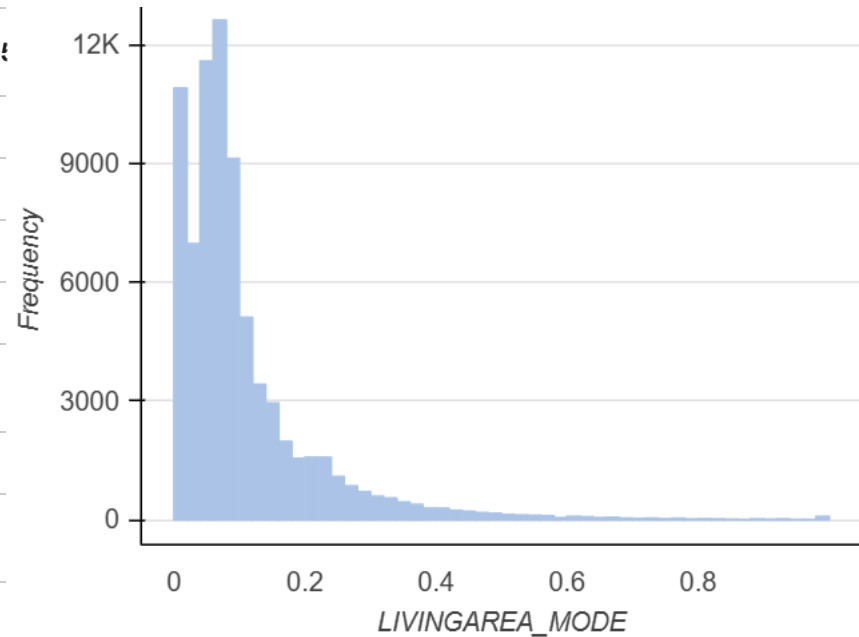
Maximum

Zeros

Zeros
(%)

Negatives

Negatives
(%)



NONLIVINGAPARTM...
numerical

Show Details

Approximate
Distinct
Count

Approximate
Unique
(%)

Missing

Missing
(%)

Infinite

Infinite
(%)

Memory
Size

Mean

Minimum

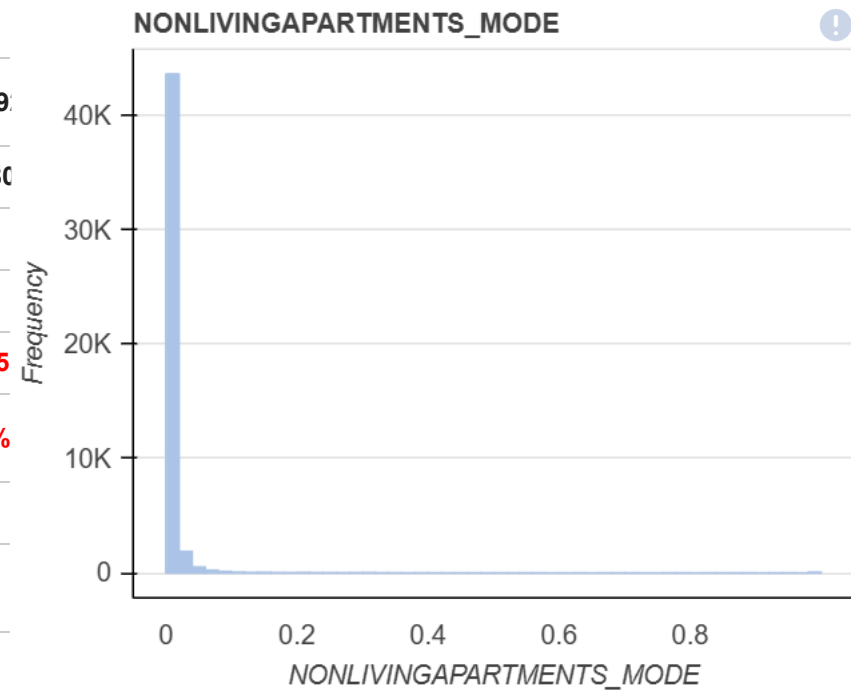
Maximum

Zeros

Zeros
(%)

Negatives

Negatives
(%)



NONLIVINGAREA_MODE

50K

NONLIVINGAREA_M...
numerical

Show Details

Approximate
Distinct
Count

Approximate
Unique
(%)

Missing
(%)

Missing
(%)

Infinite
(%)

Infinite
(%)

Memory
Size

Mean

Minimum

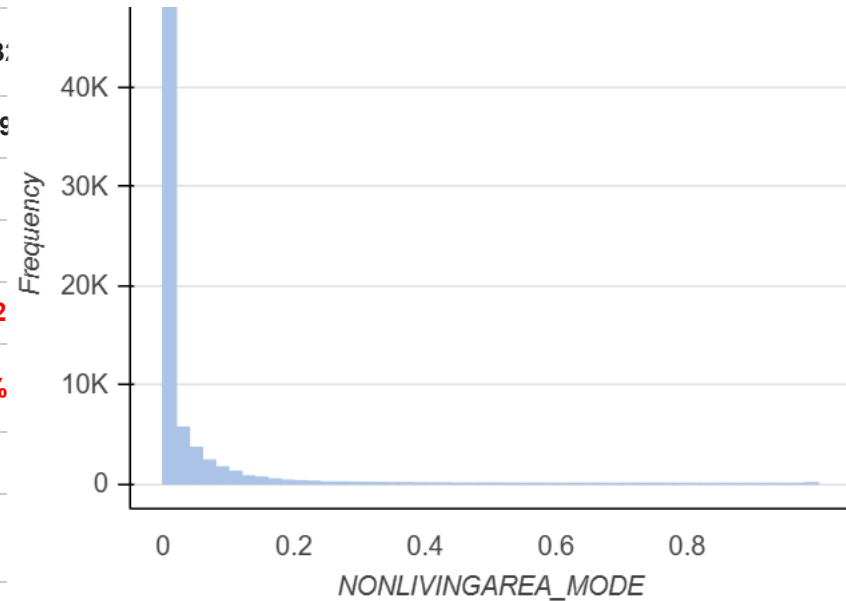
Maximum

Zeros

Zeros
(%)

Negatives

Negatives
(%)



APARTMENTS_MEDI
numerical

Show Details

Approximate
Distinct
Count

Approximate
Unique
(%)

Missing
(%)

Missing
(%)

Infinite
(%)

Infinite
(%)

Memory
Size

Mean

Minimum

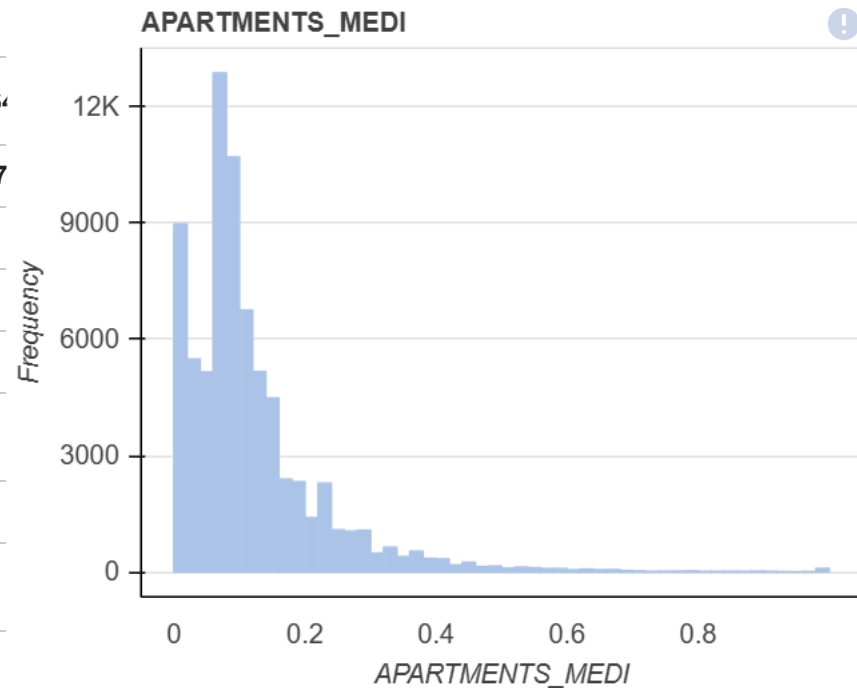
Maximum

Zeros

Zeros
(%)

Negatives

Negatives
(%)



BASEMENTAREA_MEDI

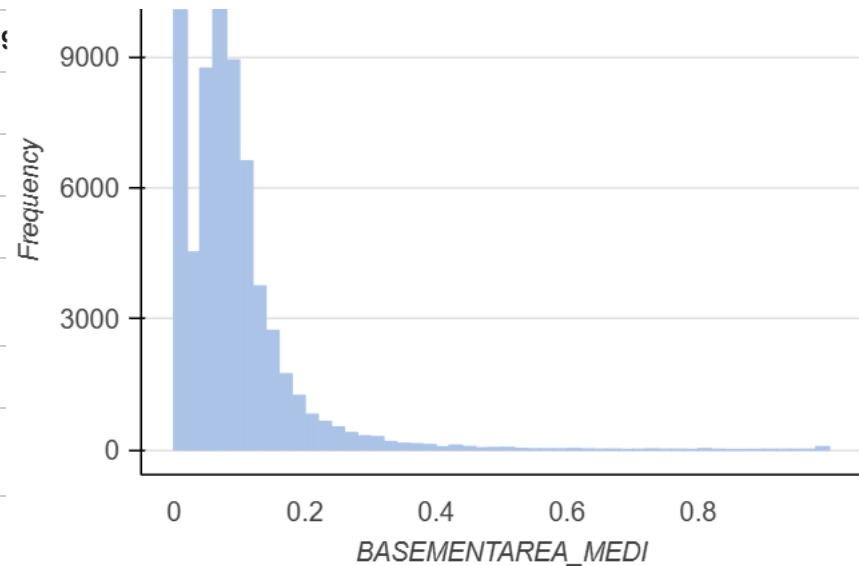
Approximate
Distinct
Count



BASEMENTAREA_MEDI
numerical

Show Details

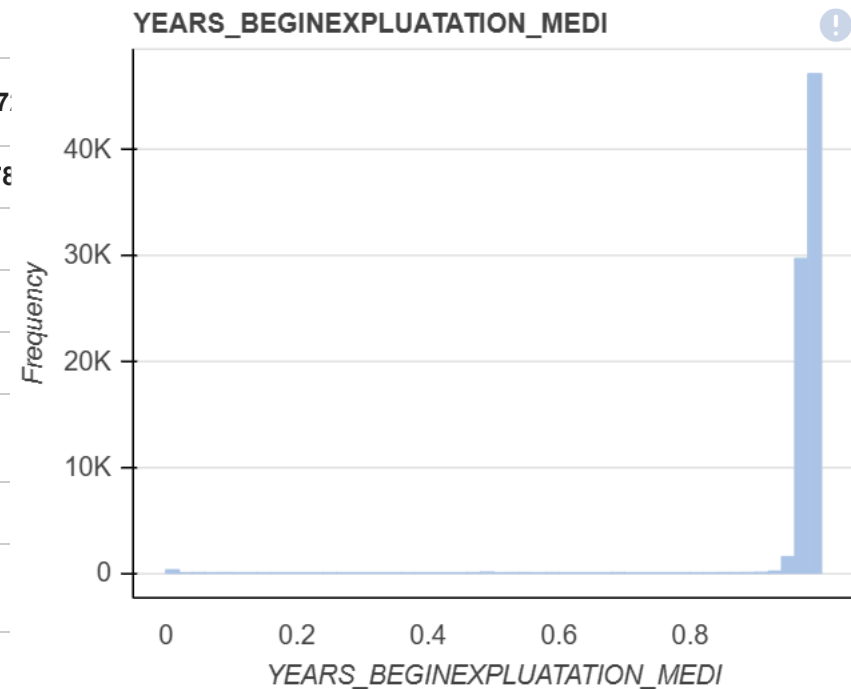
Approximate Unique Count	Mean
5.4%	0.0875
(%)	
Minimum	0
Maximum	1
Missing Count	Zeros
99907	7537
98.5%	(%)
(%)	4.9%
Infinite	Zeros
0	(%)
Infinite	Negatives
0.0%	0
(%)	
Memory Size	Negatives
1021568	(%)
	0.0%



YEARS_BEGINEXPLUATION_MEDI
numerical

Show Details

Approximate Distinct Count	Memory Size
217	12597
Approximate Unique Count	Mean
0.3%	0.9778
(%)	
Minimum	0
Maximum	1
Missing Count	Zeros
5022	264
48.8%	(%)
(%)	0.2%
Infinite	Negatives
0	0
Infinite	Negatives
0.0%	(%)
(%)	0.0%



YEARS_BUILD_MEDI



YEARS_BUILD_MEDI
numerical

Show Details

Approximate
Distinct 146
Count

Approximate
Unique 0.3%
(%)

Missing 02283

Missing 66.5%
(%)

Infinite 0

Infinite 0.0%
(%)

Memory
Size 82355

Mean 0.7557

Minimum 0

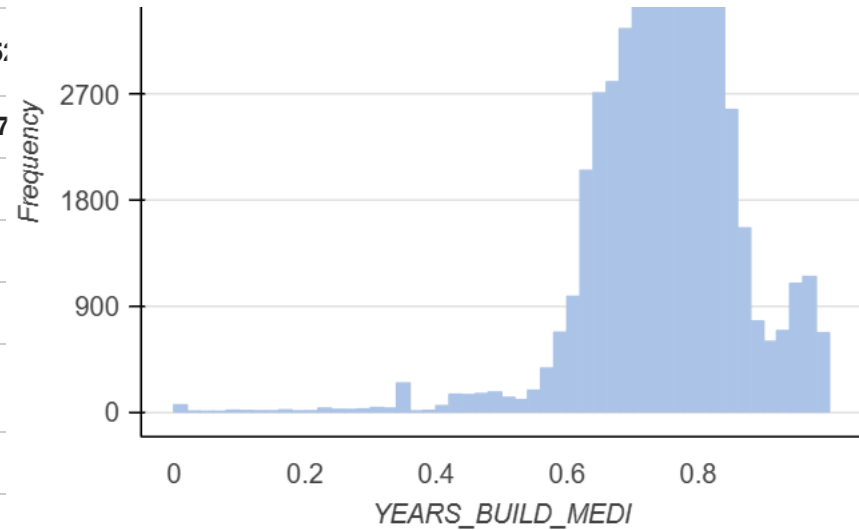
Maximum 1

Zeros 55

Zeros 0.0%
(%)

Negatives 0

Negatives 0.0%
(%)



COMMONAREA_MEDI
numerical

Show Details

Approximate
Distinct 2731
Count

Approximate
Unique 5.9%
(%)

Missing 07523

Missing 89.9%
(%)

Infinite 0

Infinite 0.0%
(%)

Memory
Size 73971

Mean 0.0443

Minimum 0

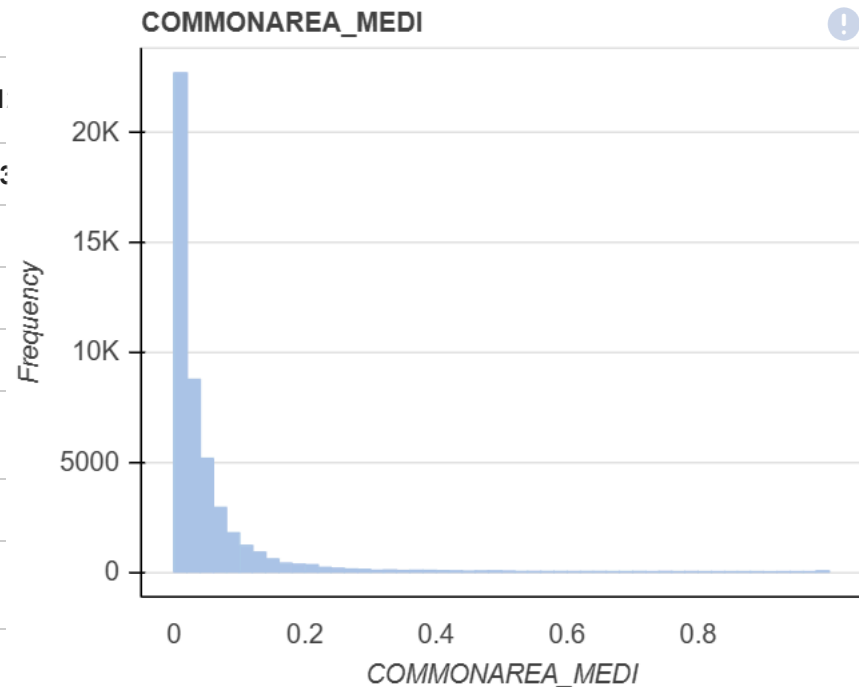
Maximum 1

Zeros 4270

Zeros 2.8%
(%)

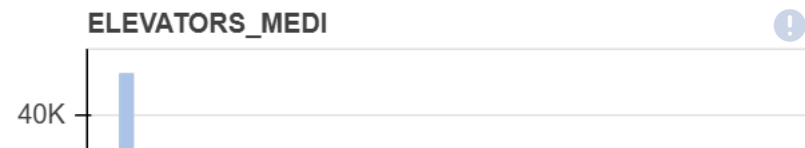
Negatives 0

Negatives 0.0%
(%)



Approximate
Distinct 46

Count



ELEVATORS_MEDI
numerical

Show Details

Approximate
Unique 0.1%
(%)

Missing 81973

Missing 53.3%
(%)

Infinite 0

Infinite 0.0%
(%)

Memory
Size 1148512

Mean 0.07805

Minimum 0

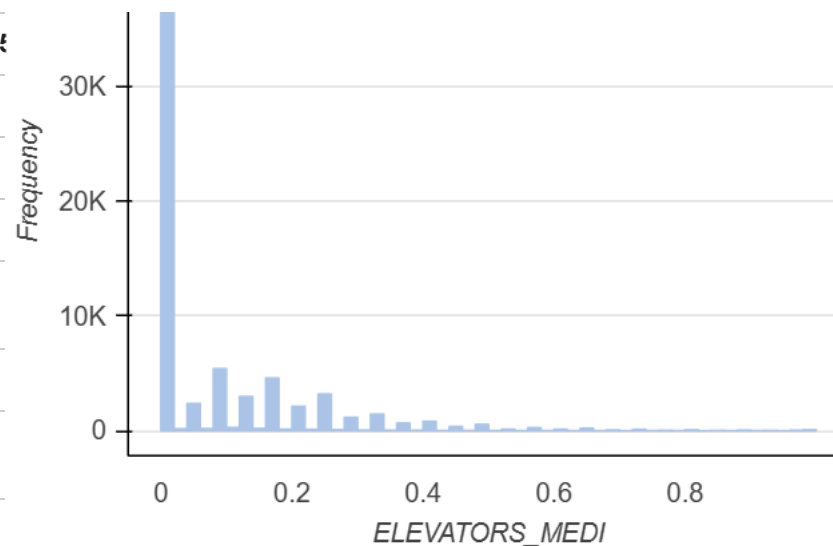
Maximum 1

Zeros 43585

Zeros 28.3%
(%)

Negatives 0

Negatives 0.0%
(%)



ENTRANCES_MEDI
numerical

Show Details

Approximate
Distinct 45
Count

Approximate
Unique 0.1%
(%)

Missing 77410

Missing 50.3%
(%)

Infinite 0

Infinite 0.0%
(%)

Memory
Size 122152

Mean 0.1492

Minimum 0

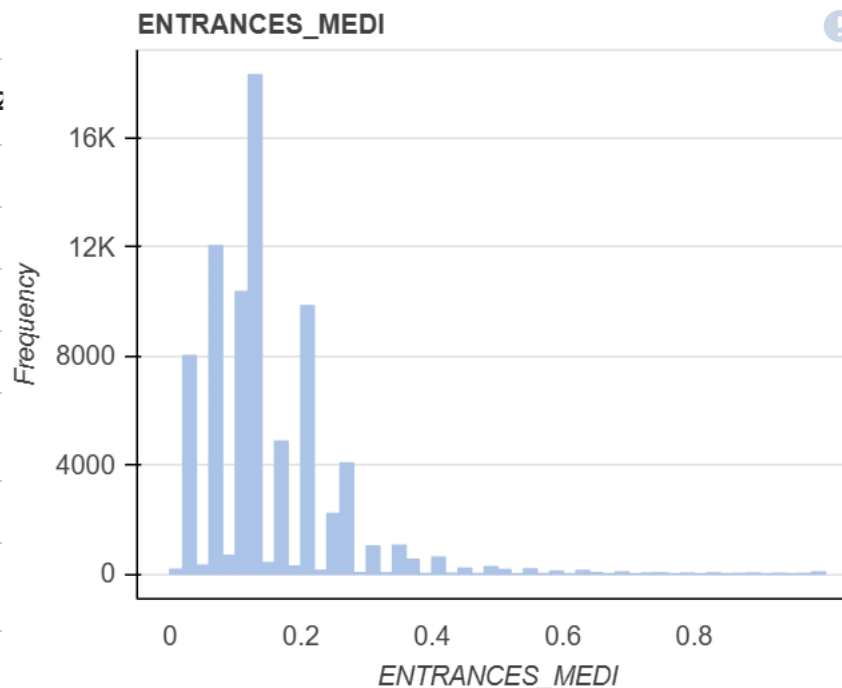
Maximum 1

Zeros 157

Zeros 0.1%
(%)

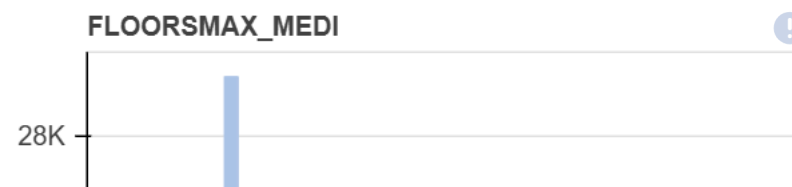
Negatives 0

Negatives 0.0%
(%)



Approximate
Distinct 49
Count

Approximate
Unique 0.1%
(%)



FLOORSMAX_MEDI
numerical

Show Details

Missing 76518

Missing (%) 49.8%

Infinite 0

Infinite (%) 0.0%

Memory Size 1235792

Mean 0.2256

Minimum 0

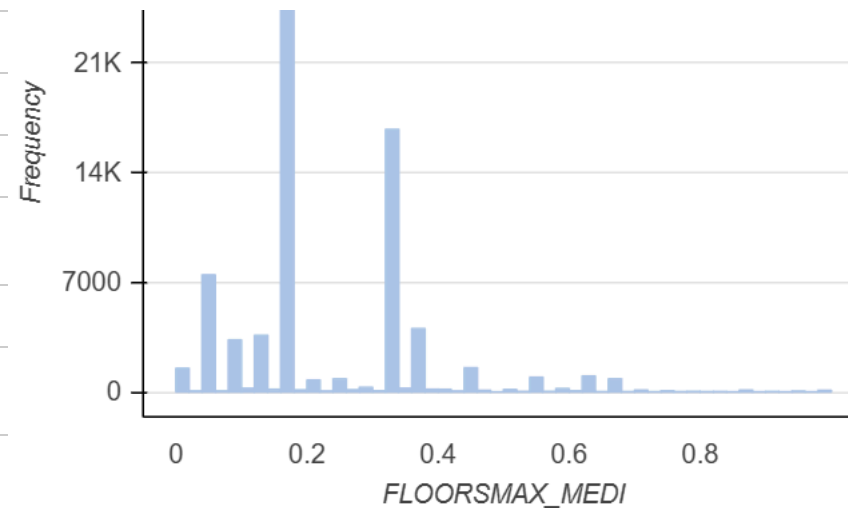
Maximum 1

Zeros 1506

Zeros (%) 1.0%

Negatives 0

Negatives (%) 0.0%



FLOORSMIN_MEDI
numerical

Show Details

Approximate
Distinct Count 47

Approximate
Unique (%) 0.1%

Missing 104424

Missing (%) 67.9%

Infinite 0

Infinite (%) 0.0%

Memory Size 789296

Mean 0.2314

Minimum 0

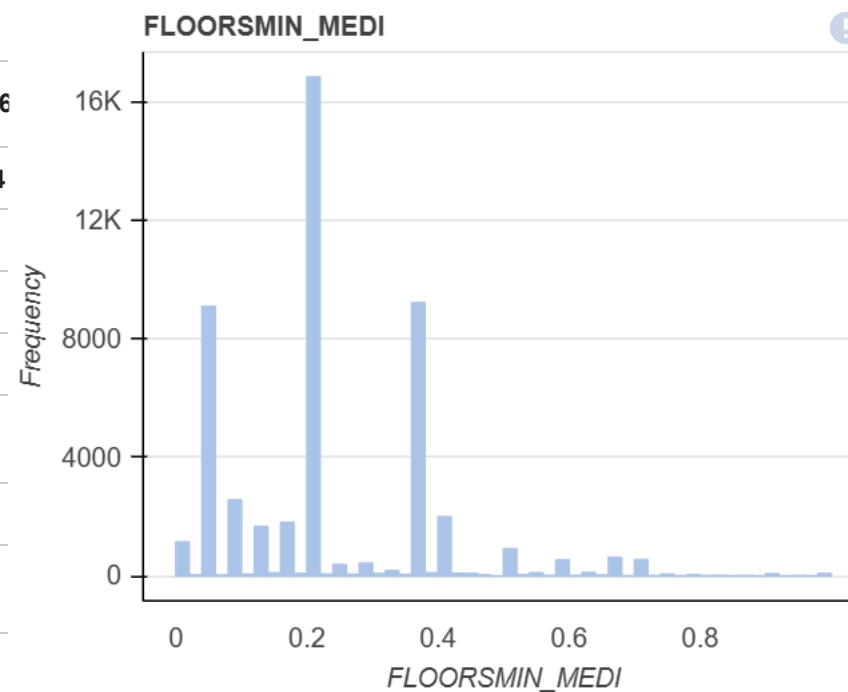
Maximum 1

Zeros 1143

Zeros (%) 0.7%

Negatives 0

Negatives (%) 0.0%



Approximate
Distinct Count 3209

Approximate
Unique (%) 5.1%

LANDAREA_MEDI

16K

12K

LANDAREA_MEDI
numerical

Show Details

Missing 91206

Missing (%) 59.3%

Infinite 0

Infinite (%) 0.0%

Memory Size 1000784

Mean 0.06713

Minimum 0

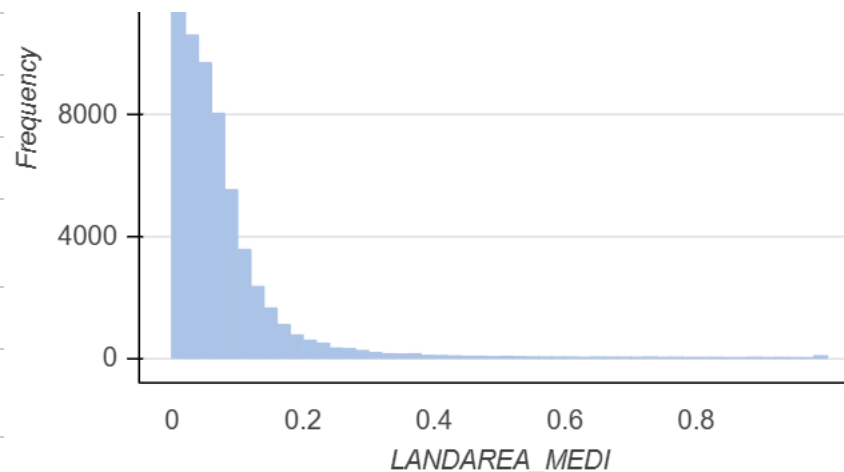
Maximum 1

Zeros 7907

Zeros (%) 5.1%

Negatives 0

Negatives (%) 0.0%



LIVINGAPARTMENT...
numerical

Show Details

Approximate
Distinct Count 1010

Approximate
Unique (%) 2.1%

Missing 405193

Missing (%) 88.4%

Infinite 0

Infinite (%) 0.0%

Memory Size 77699

Mean 0.1014

Minimum 0

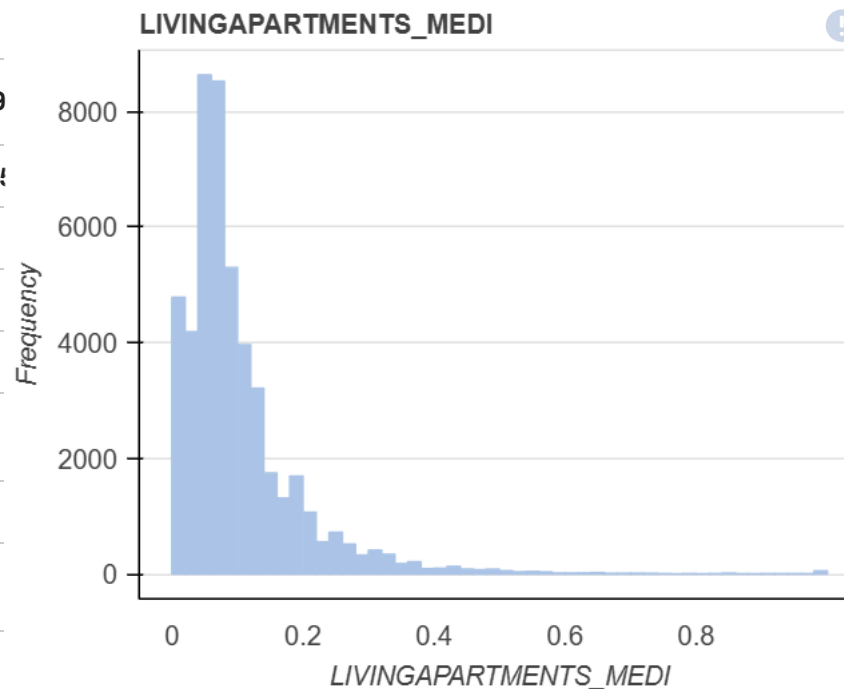
Maximum 1

Zeros 225

Zeros (%) 0.1%

Negatives 0

Negatives (%) 0.0%



Approximate
Distinct Count 4831

Approximate
Unique (%) 0.2%

(%)
Missing 77133

Missing (%) 50.2%

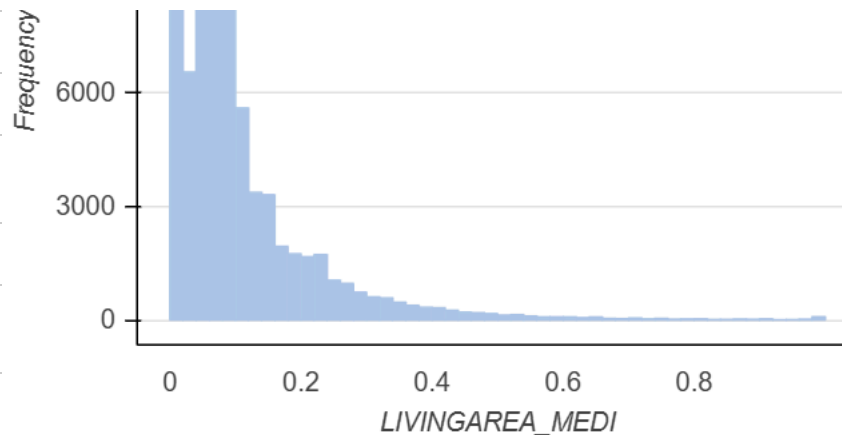


LIVINGAREA_MEDI

numerical

[Show Details](#)

Infinite	0	Maximum	1
Infinite (%)	0.0%	Zeros	153
Memory Size	1225952	Zeros (%)	0.1%
Mean	0.1086	Negatives	0
Minimum	0	Negatives (%)	0.0%

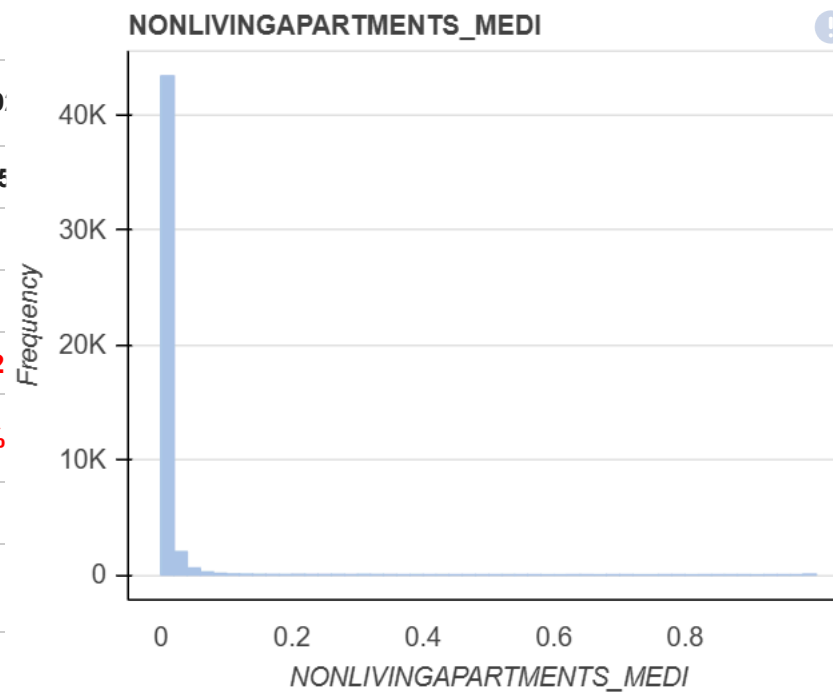


NONLIVINGAPARTM...

numerical

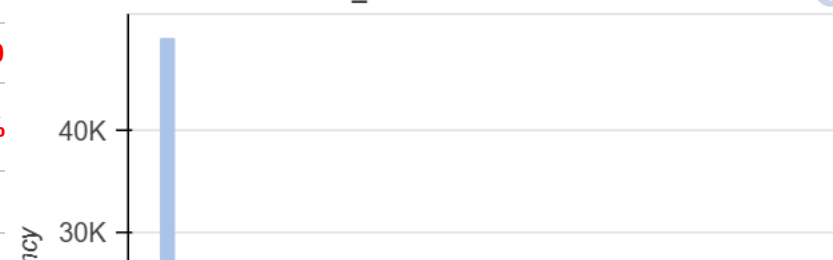
[Show Details](#)

Approximate Distinct Count	168	Memory Size	74939
Approximate Unique (%)	0.4%	Mean	0.0085
Missing	06918	Minimum	0
Missing (%)	69.5%	Maximum	1
Infinite	0	Zeros	28082
Infinite (%)	0.0%	Zeros (%)	18.3%
		Negatives	0
		Negatives (%)	0.0%



NONLIVINGAREA_MEDI

Approximate Distinct Count	2841	Missing	4860
Approximate Unique (%)	4.1%	Missing (%)	55.2%
		Infinite	0



NONLIVINGAREA_M...
numerical

Show Details

Infinite
(%) 0.0%

Memory
Size 1102320

Mean 0.02812

Minimum 0

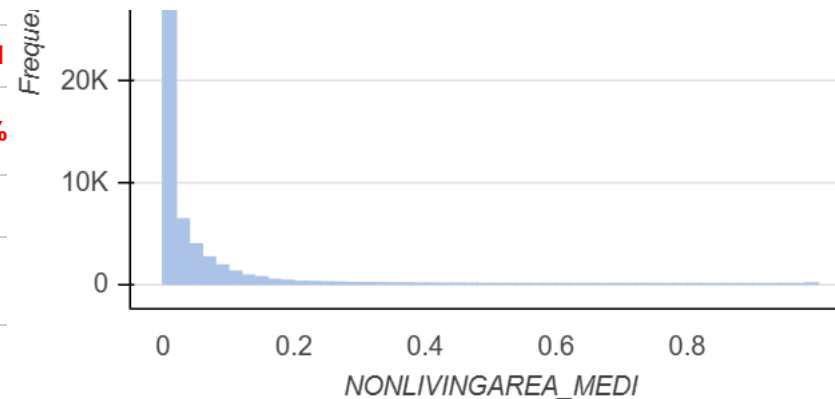
Maximum 1

Zeros 30581

Zeros (%) 19.9%

Negatives 0

Negatives
(%) 0.0%



FONDKAPREMONT...
categorical

Show Details

Approximate
Distinct
Count 4

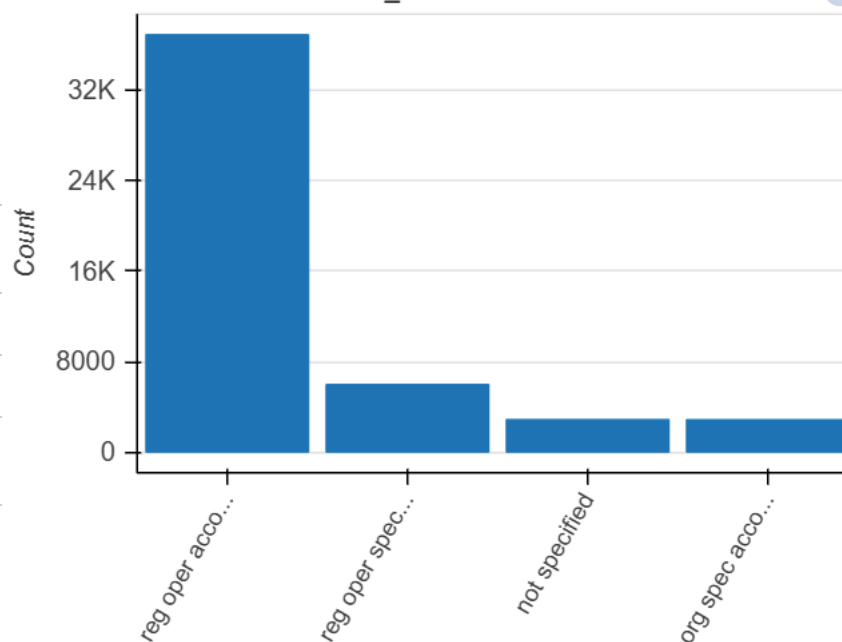
Approximate
Unique (%) 0.0%

Missing 105177

Missing (%) 68.4%

Memory
Size 3956054

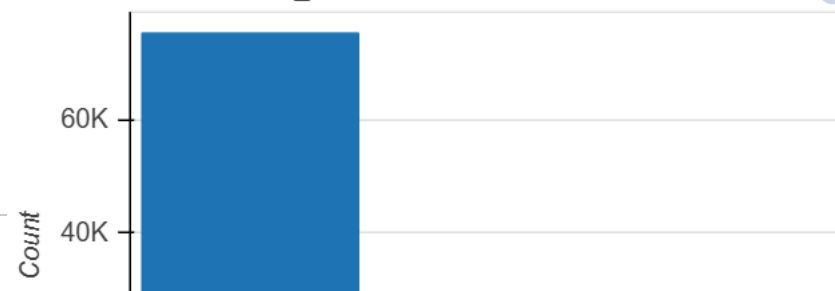
FONDKAPREMONT_MODE



HOUSETYPE_MODE

Approximate
Distinct
Count 3

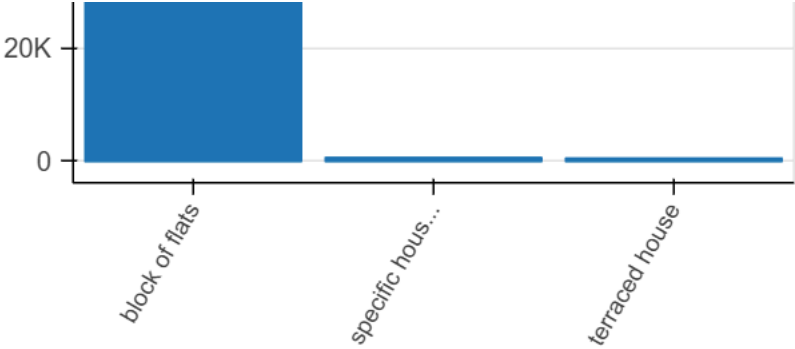
HOUSETYPE_MODE



categorical

Show Details

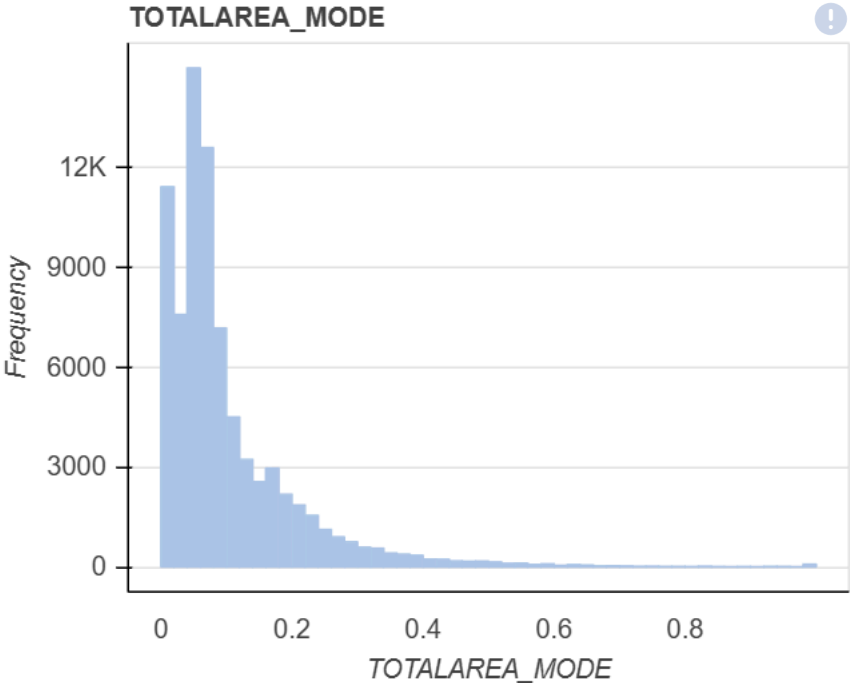
Approximate Unique (%)	0.0%
Missing	77137
Missing (%)	50.2%
Memory Size	6054252



TOTALAREA_MODE
numerical

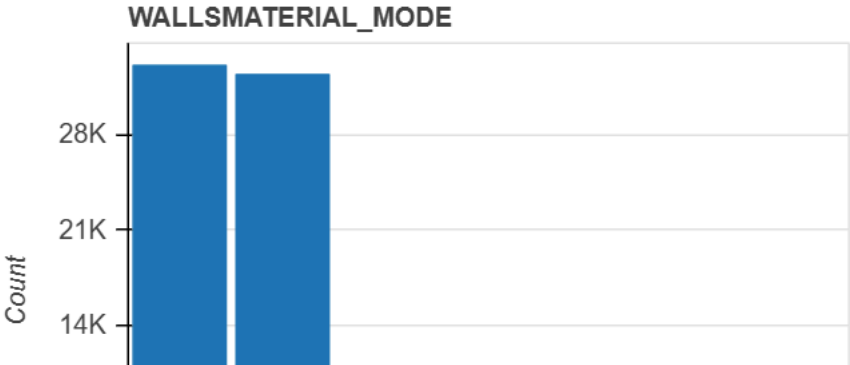
Show Details

Approximate Distinct Count	4705	Memory Size	127220
Approximate Unique (%)	5.9%	Mean	0.1026
Missing	74242	Minimum	0
Missing (%)	48.3%	Maximum	1
Infinite	0	Zeros	292
Infinite (%)	0.0%	Zeros (%)	0.2%
		Negatives	0
		Negatives (%)	0.0%



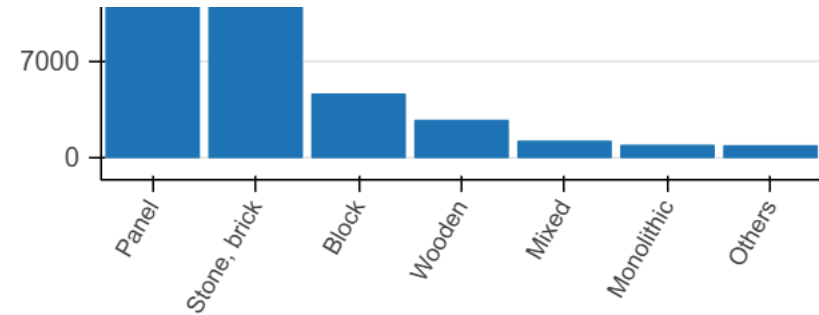
WALLSMATERIAL_M...
categorical

Approximate Distinct Count	7
Approximate Unique (%)	0.0%



[Show Details](#)

Missing	78142
Missing (%)	50.8%
Memory Size	5527425

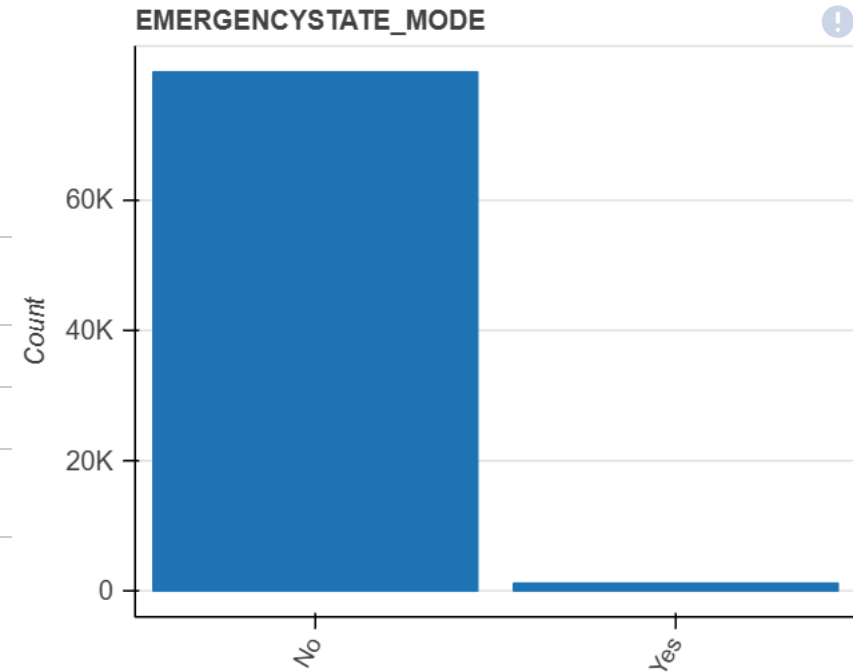


EMERGENCYSTATE...

categorical

[Show Details](#)

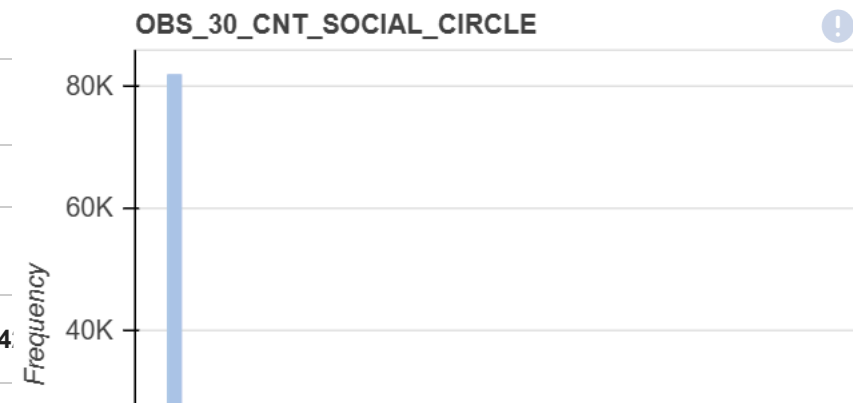
Approximate Distinct Count	2
Approximate Unique (%)	0.0%
Missing	72886
Missing (%)	47.4%
Memory Size	5419394



OBS_30_CNT_SOCIAL...

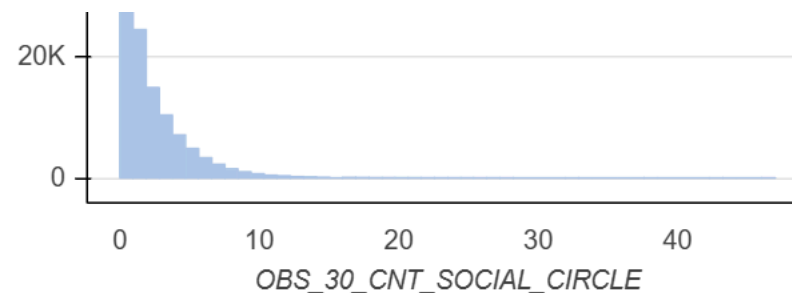
numerical

Approximate Distinct 30 Count	Missing 8.4%
Infinite	0
Approximate Unique (%)	0.0%
Infinite (%)	0.0%
Missing	541
Memory Size	24514

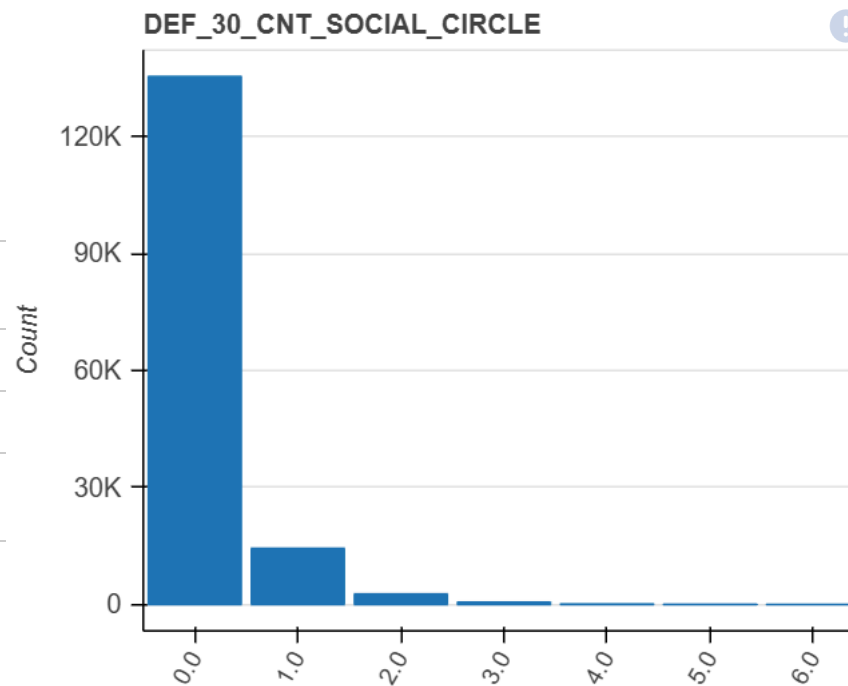


[Show Details](#)

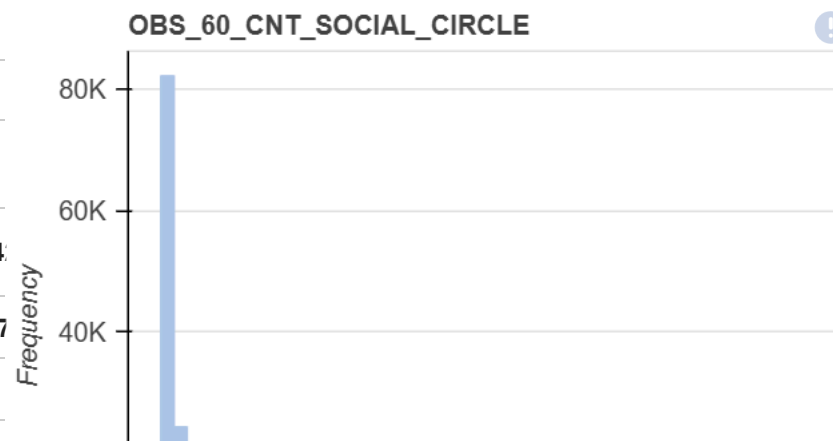
Mean	1.4274	Zeros (%)	53.2%
Minimum	0	Negatives	0
Maximum	47	Negatives (%)	0.0%
Zeros	81854		

DEF_30_CNT_SOCIAL_CIRCLE
categorical[Show Details](#)

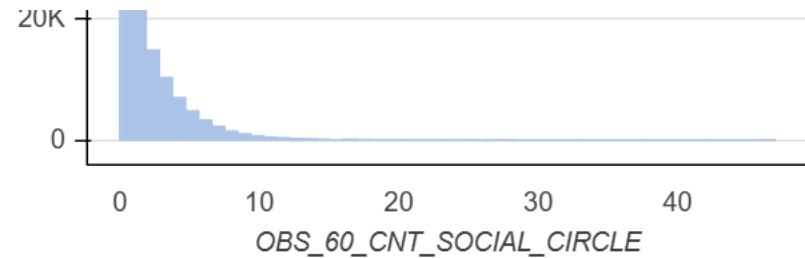
Approximate Distinct Count	7
Approximate Unique (%)	0.0%
Missing	541
Missing (%)	0.4%
Memory Size	10418552

OBS_60_CNT_SOCIAL_CIRCLE
numerical[Show Details](#)

Approximate Distinct Count	29	Infinite	0
Approximate Unique (%)	0.0%	Infinite (%)	0.0%
Missing	541	Memory Size	24514
Missing (%)	0.4%	Mean	1.4107
		Minimum	0



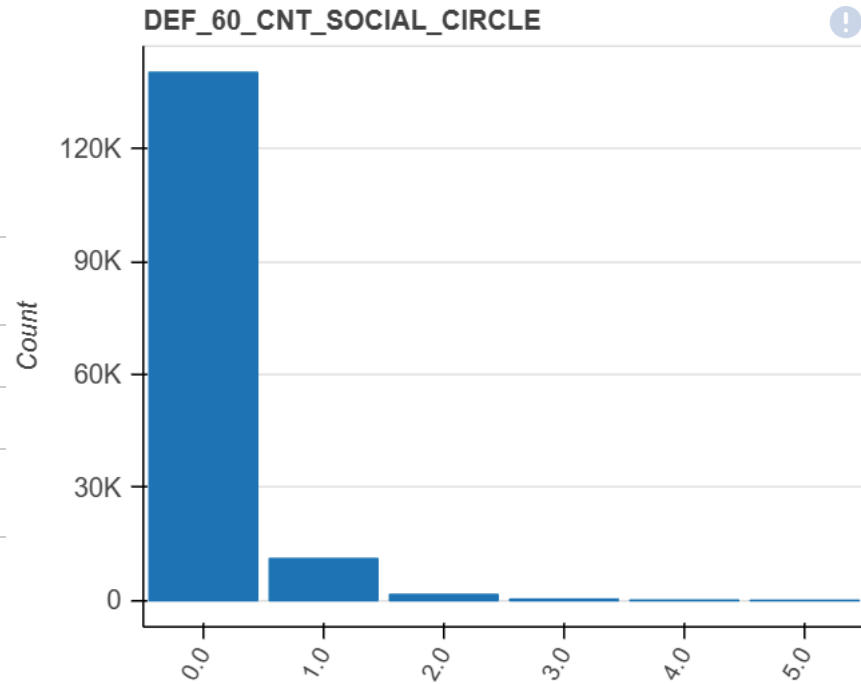
Maximum	47	Negatives	0
Zeros	82221	Negatives (%)	0.0%
Zeros (%)	53.5%		



DEF_60_CNT_SOCIAL_CIRCLE

Show Details

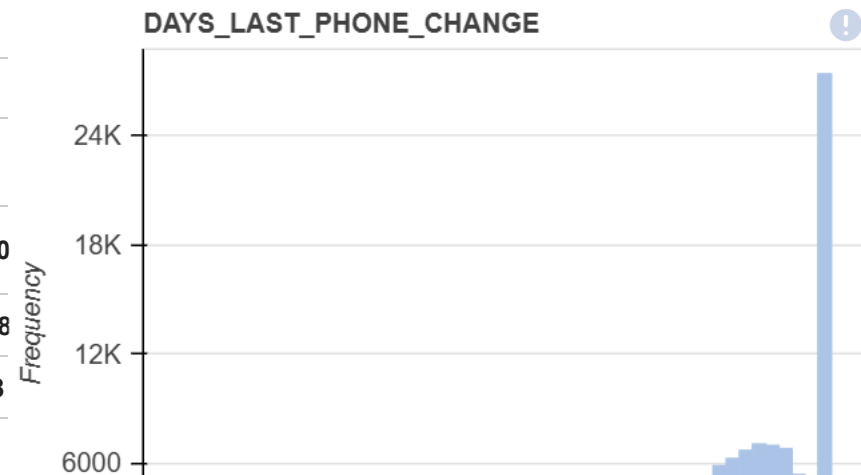
Approximate Distinct Count	6
Approximate Unique (%)	0.0%
Missing	541
Missing (%)	0.4%
Memory Size	10418552

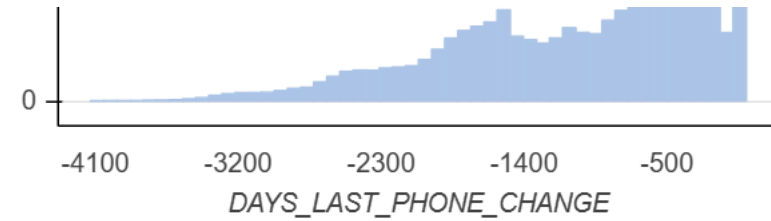


DAYS_LAST_PHONE_CHANGE

Show Details

Approximate Distinct Count	Infinite	0
Approximate Unique (%)	Infinite (%)	0.0%
Memory Size	24600	
Missing	0	Mean -959.8
Missing (%)	0.0%	Minimum -4128
		Maximum 0



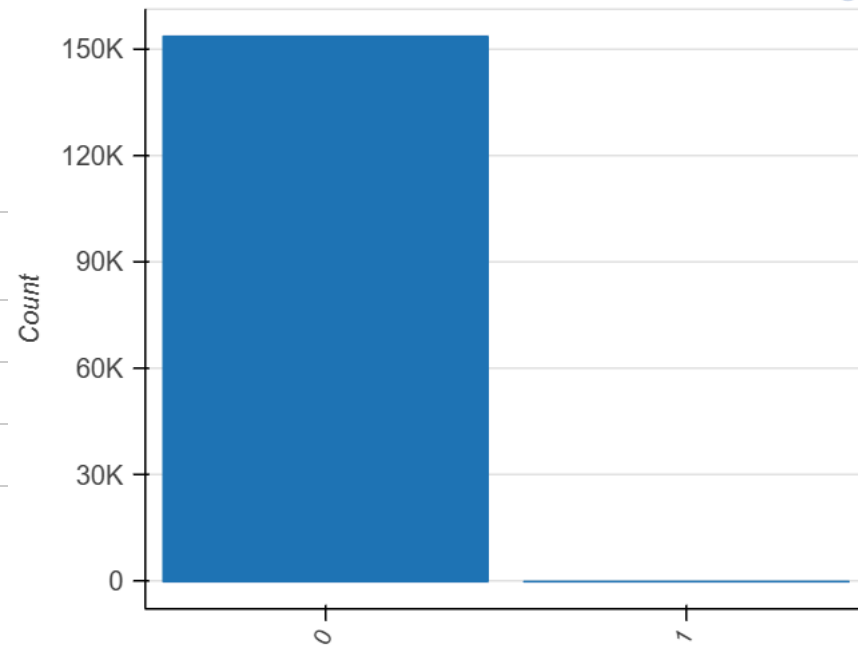
Zeros 18775**Zeros (%) 12.2%****Negatives 18498****Negatives (%) 87.8%**

FLAG_DOCUMENT_2
categorical

Show Details

Approximate Distinct Count	2
Approximate Unique (%)	0.0%
Missing	0
Missing (%)	0.0%
Memory Size	10147830

FLAG_DOCUMENT_2

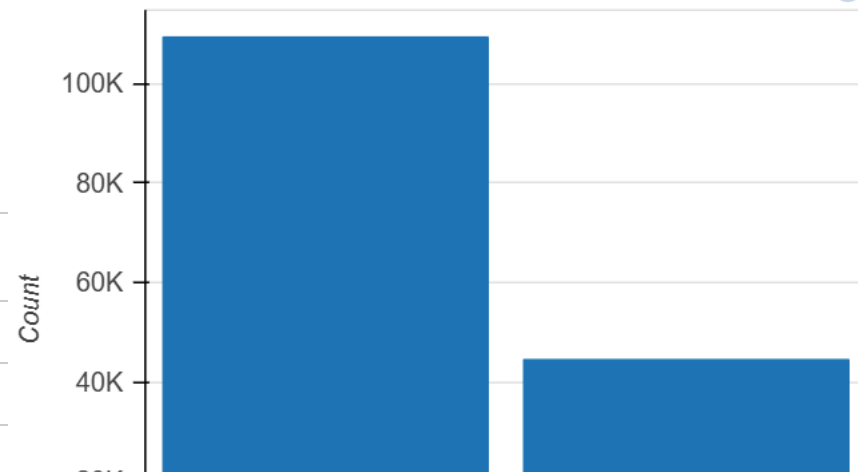


FLAG_DOCUMENT_3
categorical

Show Details

Approximate Distinct Count	2
Approximate Unique (%)	0.0%
Missing	0
Missing (%)	0.0%

FLAG_DOCUMENT_3



Memory Size	10147830
-------------	----------

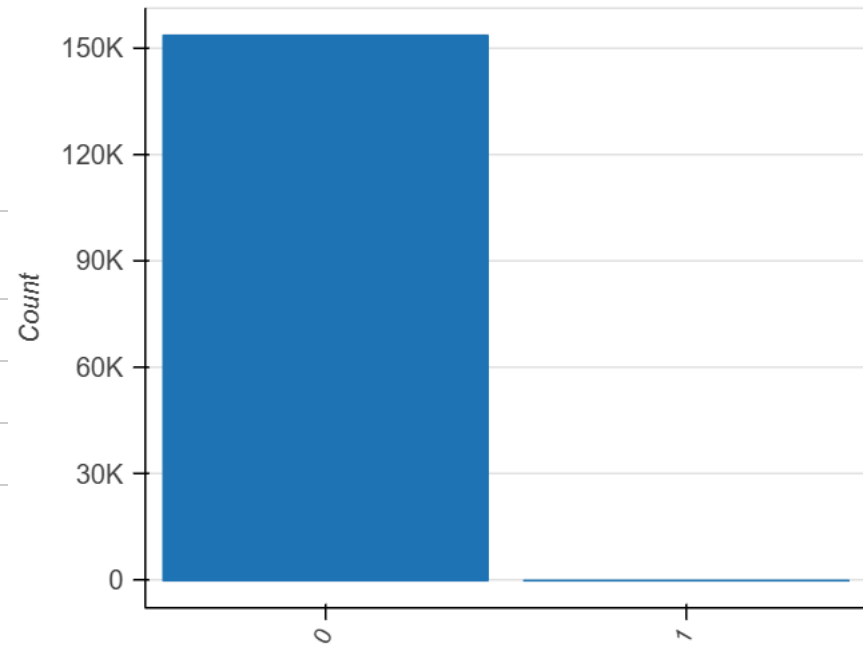


FLAG_DOCUMENT_4
categorical

Show Details

Approximate Distinct Count	2
Approximate Unique (%)	0.0%
Missing	0
Missing (%)	0.0%
Memory Size	10147830

FLAG_DOCUMENT_4

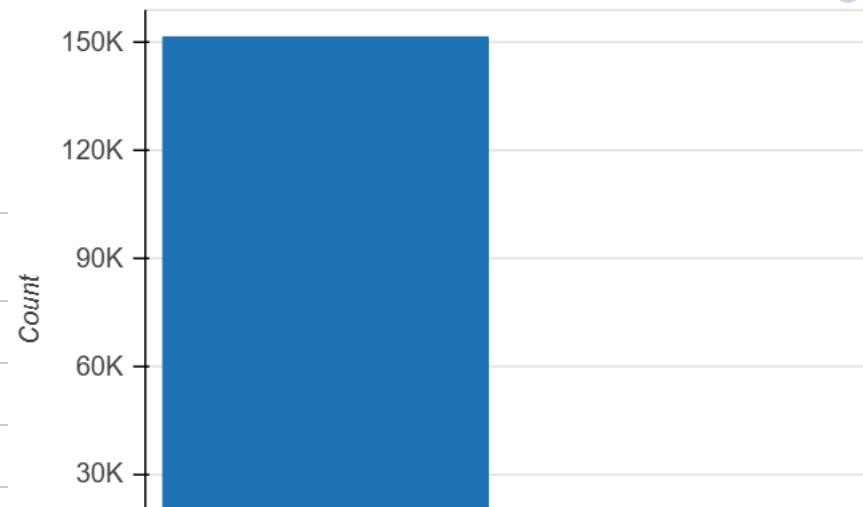


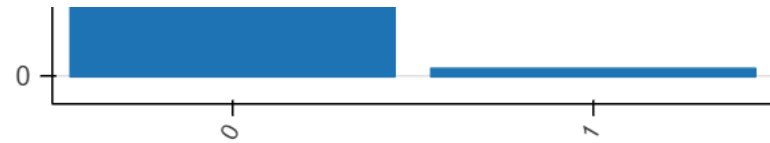
FLAG_DOCUMENT_5
categorical

Show Details

Approximate Distinct Count	2
Approximate Unique (%)	0.0%
Missing	0
Missing (%)	0.0%
Memory Size	10147830

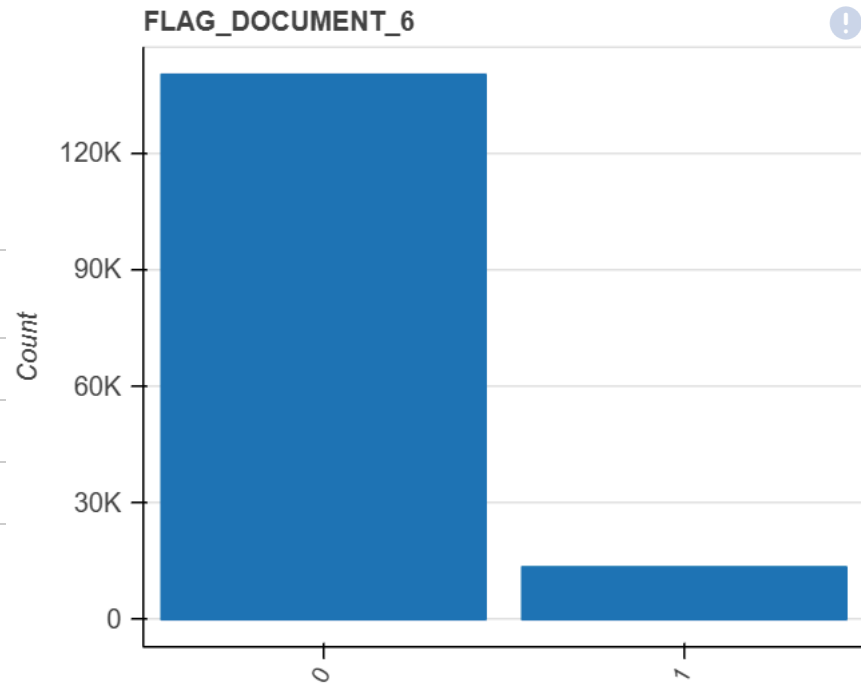
FLAG_DOCUMENT_5





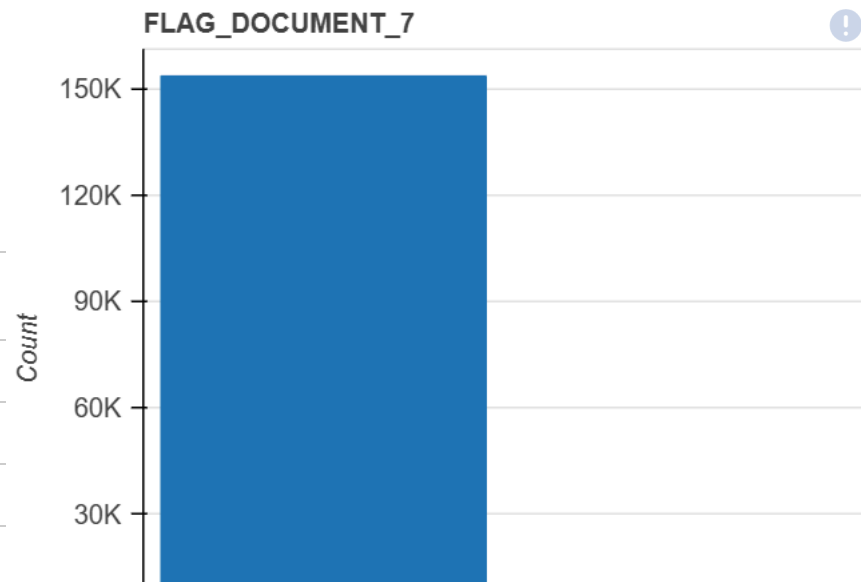
FLAG_DOCUMENT_6 categorical	Approximate Distinct Count	2
	Approximate Unique (%)	0.0%
	Missing	0
	Missing (%)	0.0%
	Memory Size	10147830

Show Details



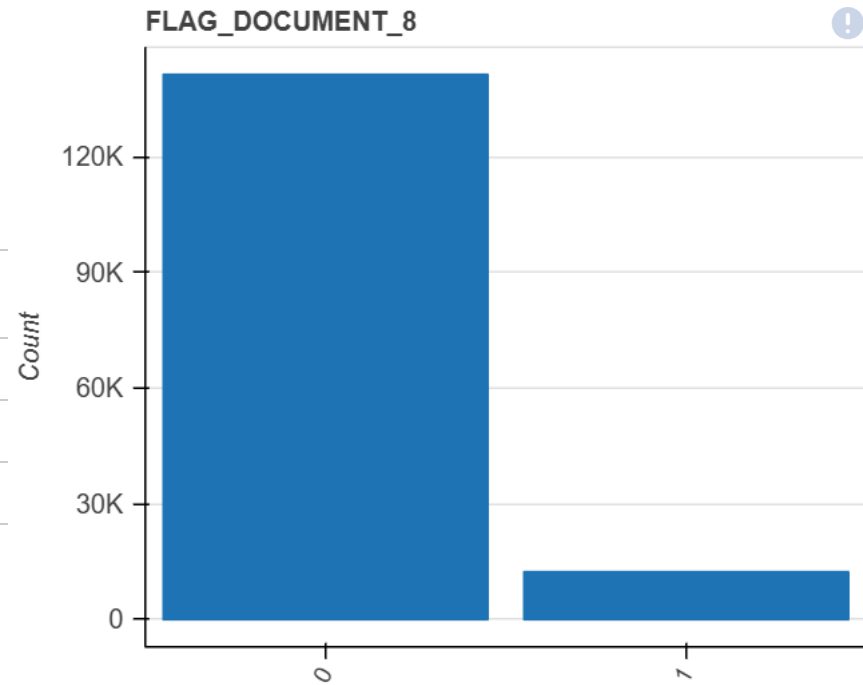
FLAG_DOCUMENT_7 categorical	Approximate Distinct Count	2
	Approximate Unique (%)	0.0%
	Missing	0
	Missing (%)	0.0%
	Memory Size	10147830

Show Details

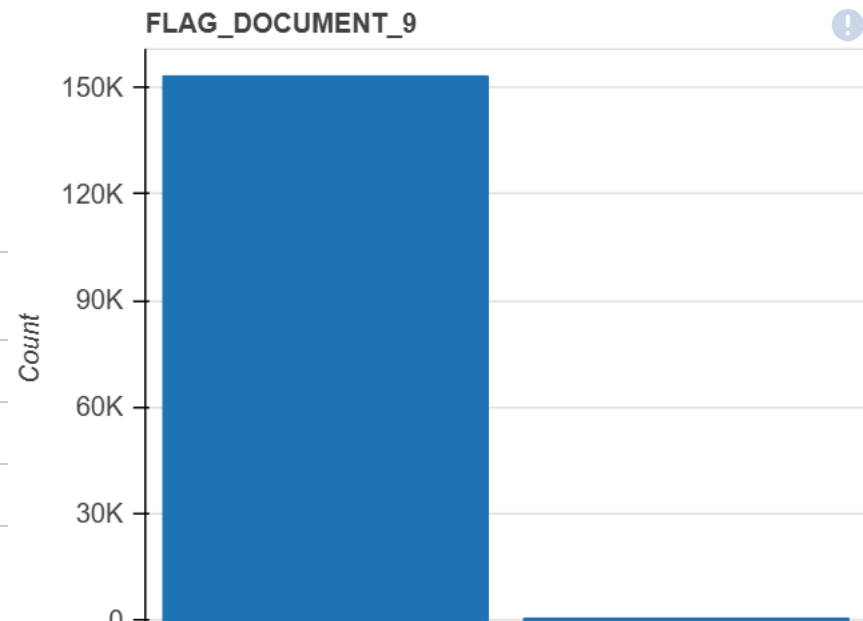




FLAG_DOCUMENT_8 categorical	Approximate Distinct Count	2
	Approximate Unique (%)	0.0%
	Missing	0
	Missing (%)	0.0%
	Memory Size	10147830

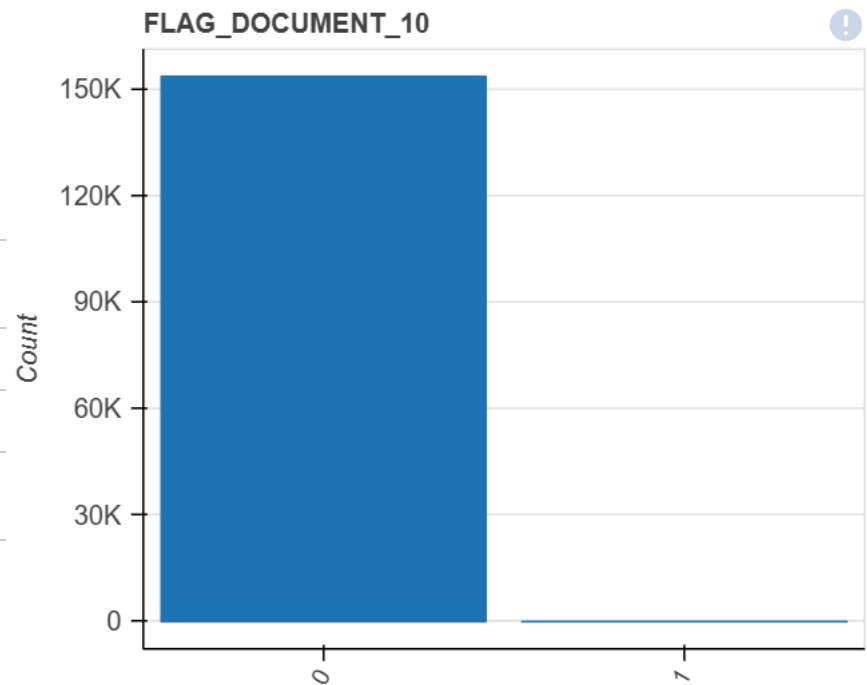
[Show Details](#)

FLAG_DOCUMENT_9 categorical	Approximate Distinct Count	2
	Approximate Unique (%)	0.0%
	Missing	0
	Missing (%)	0.0%
	Memory Size	10147830

[Show Details](#)

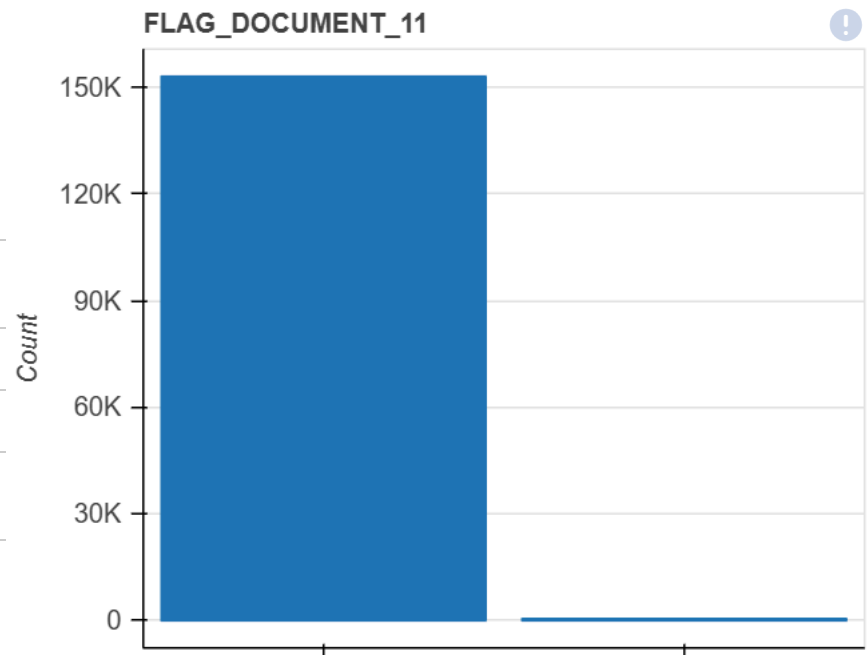
FLAG_DOCUMENT_10 categorical	Approximate Distinct Count	2
	Approximate Unique (%)	0.0%
	Missing	0
	Missing (%)	0.0%
	Memory Size	10147830

Show Details



FLAG_DOCUMENT_11 categorical	Approximate Distinct Count	2
	Approximate Unique (%)	0.0%
	Missing	0
	Missing (%)	0.0%
	Memory Size	10147830

Show Details



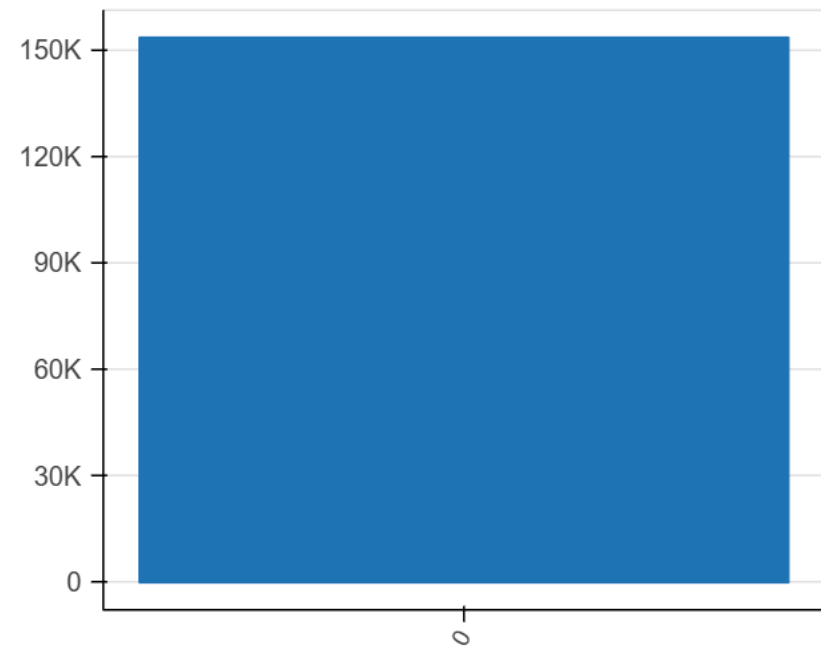
FLAG_DOCUMENT_12
categorical

Show Details

Approximate Distinct Count	1
Approximate Unique (%)	0.0%
Missing	0
Missing (%)	0.0%
Memory Size	10147830

Count

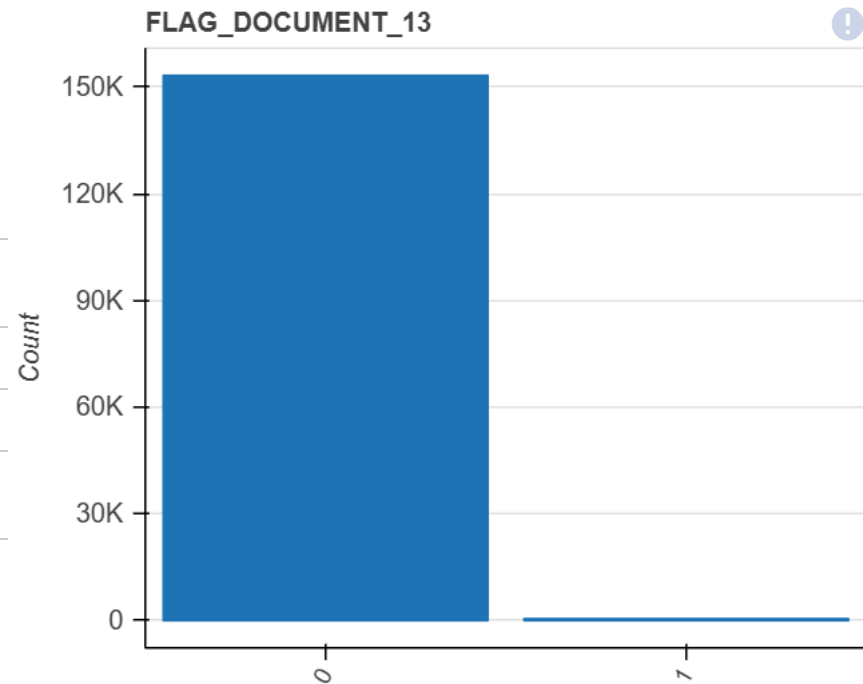
FLAG_DOCUMENT_12



FLAG_DOCUMENT_13
categorical

Show Details

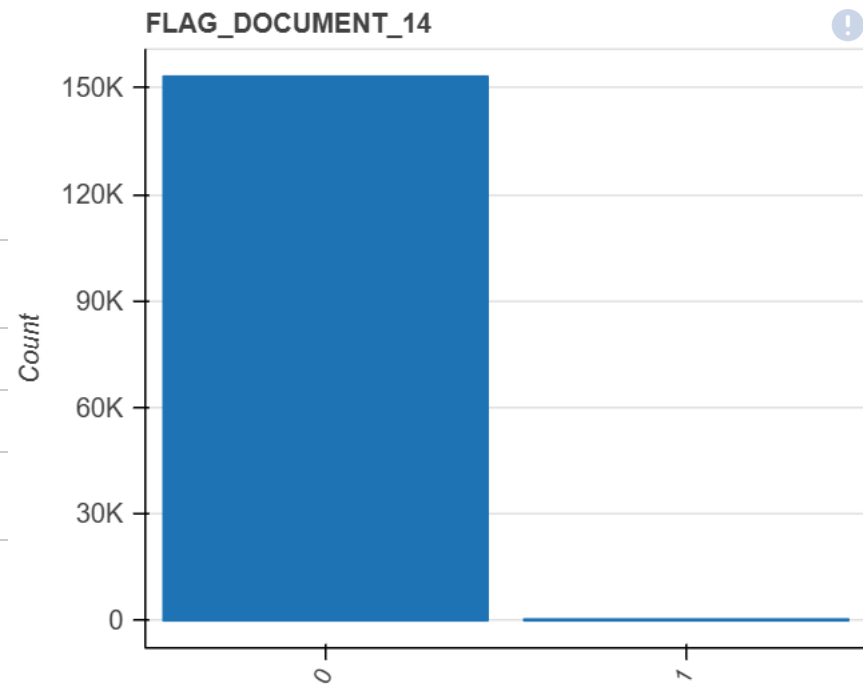
Approximate Distinct Count	2
Approximate Unique (%)	0.0%
Missing	0
Missing (%)	0.0%
Memory Size	10147830



FLAG_DOCUMENT_14
categorical

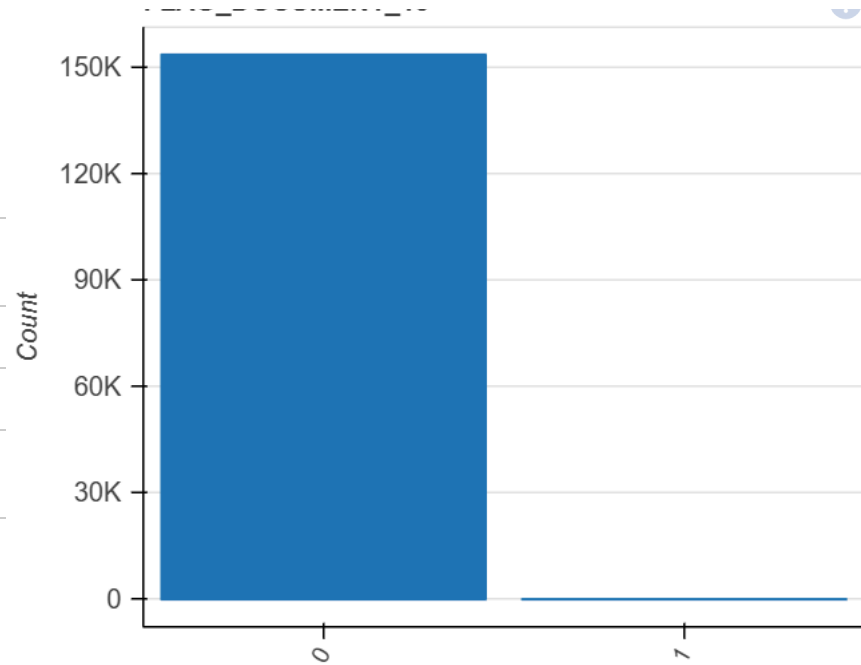
Show Details

Approximate Distinct Count	2
Approximate Unique (%)	0.0%
Missing	0
Missing (%)	0.0%
Memory Size	10147830

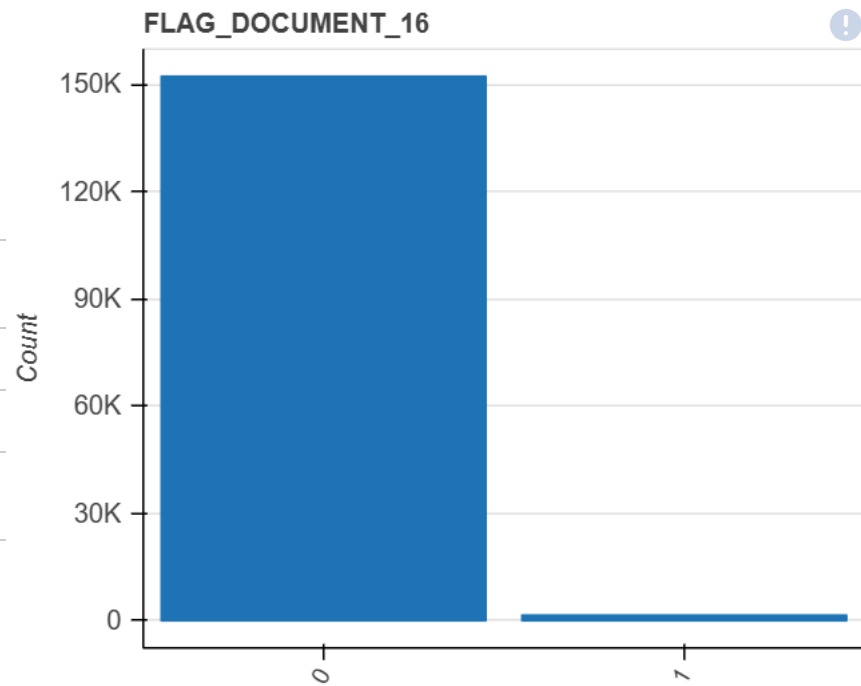


FLAG_DOCUMENT_15

FLAG_DOCUMENT_15 categorical	Approximate Distinct Count	2
	Approximate Unique (%)	0.0%
	Missing	0
	Missing (%)	0.0%
	Memory Size	10147830

[Show Details](#)

FLAG_DOCUMENT_16 categorical	Approximate Distinct Count	2
	Approximate Unique (%)	0.0%
	Missing	0
	Missing (%)	0.0%
	Memory Size	10147830

[Show Details](#)

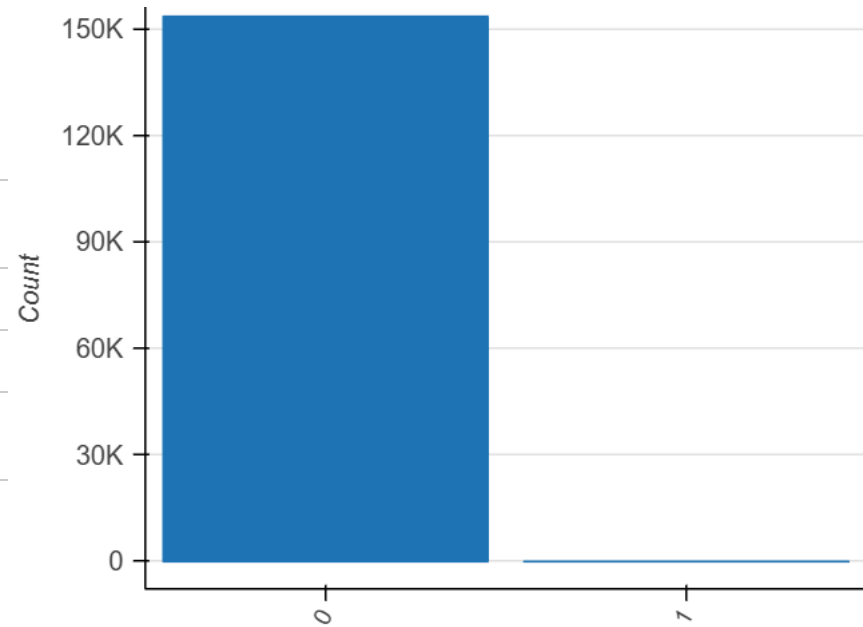
FLAG_DOCUMENT_17



FLAG_DOCUMENT_17
categorical

Show Details

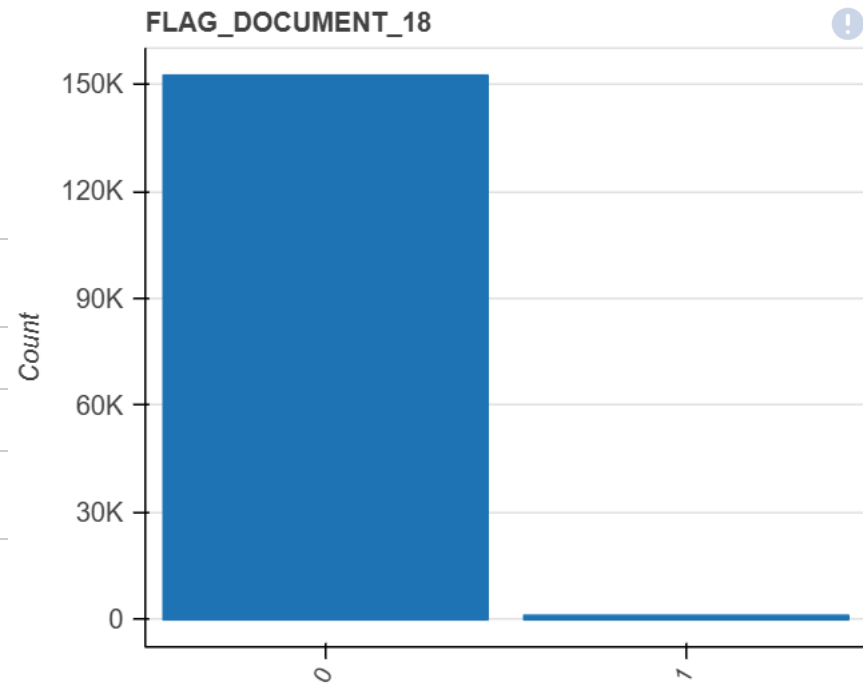
Approximate Distinct Count	2
Approximate Unique (%)	0.0%
Missing	0
Missing (%)	0.0%
Memory Size	10147830



FLAG_DOCUMENT_18
categorical

Show Details

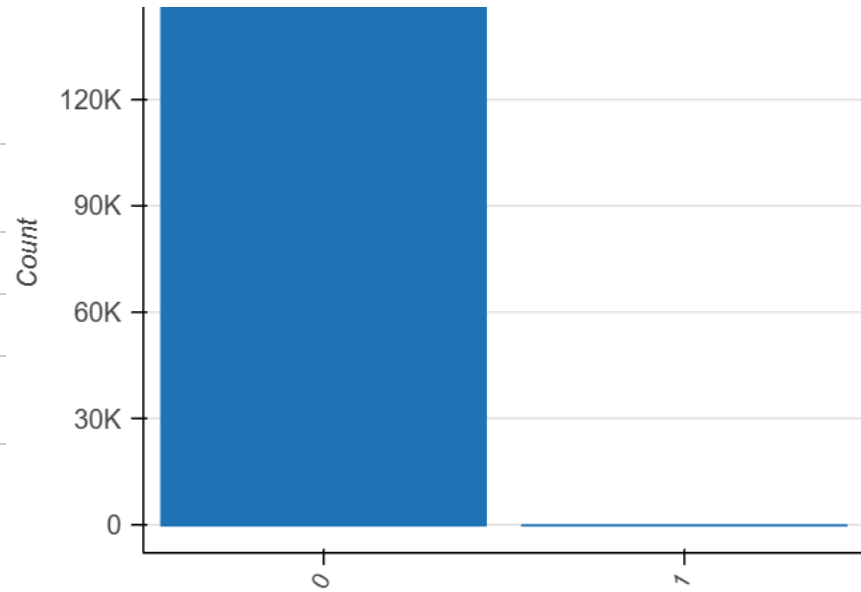
Approximate Distinct Count	2
Approximate Unique (%)	0.0%
Missing	0
Missing (%)	0.0%
Memory Size	10147830



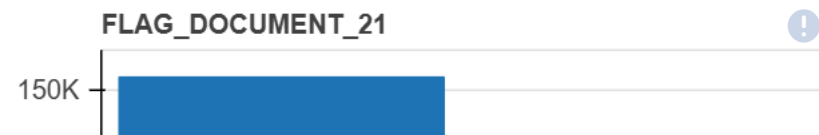
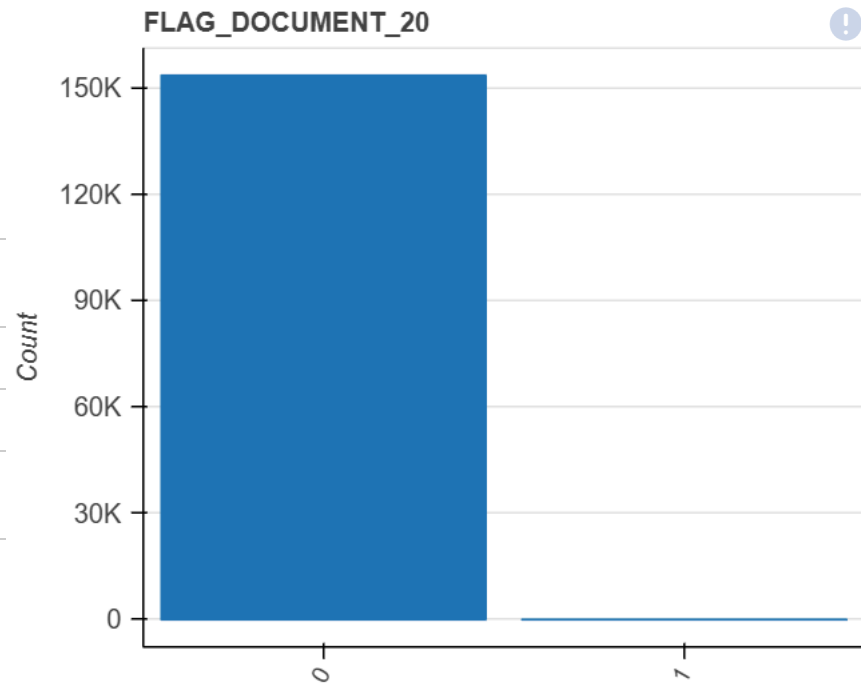
FLAG_DOCUMENT_19



FLAG_DOCUMENT_19 categorical	Approximate Distinct Count	2
	Approximate Unique (%)	0.0%
	Missing	0
	Missing (%)	0.0%
	Memory Size	10147830

[Show Details](#)

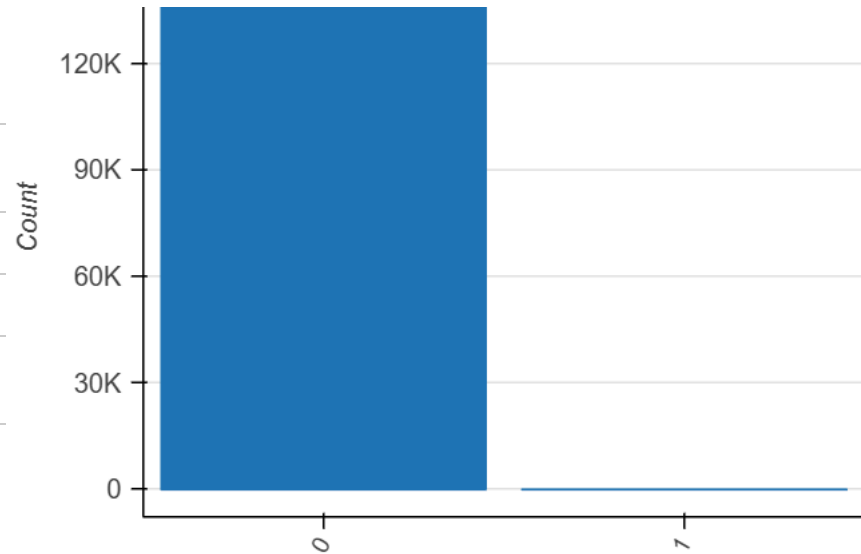
FLAG_DOCUMENT_20 categorical	Approximate Distinct Count	2
	Approximate Unique (%)	0.0%
	Missing	0
	Missing (%)	0.0%
	Memory Size	10147830

[Show Details](#)

FLAG_DOCUMENT_21
categorical

Show Details

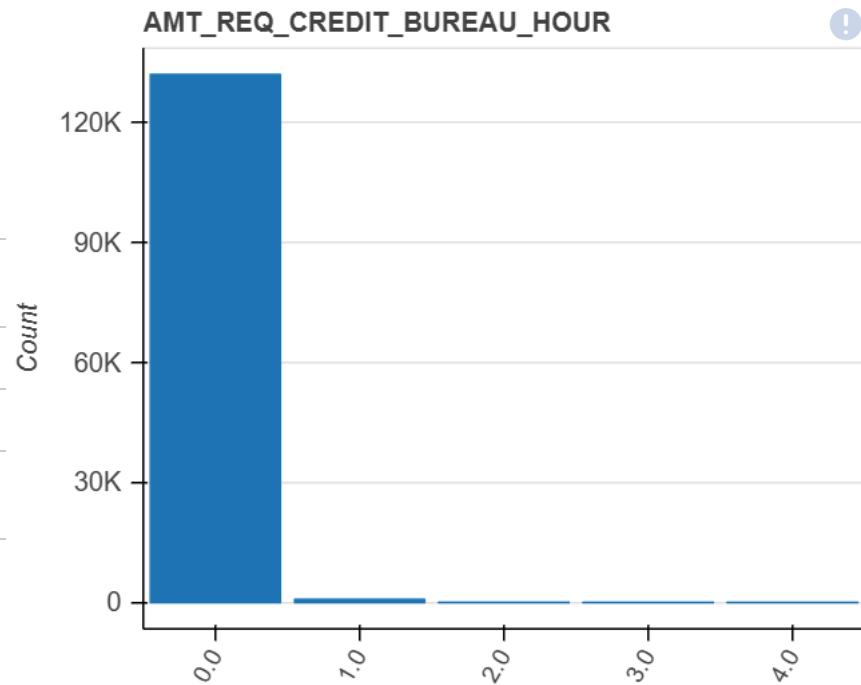
Approximate Distinct Count	2
Approximate Unique (%)	0.0%
Missing	0
Missing (%)	0.0%
Memory Size	10147830



AMT_REQ_CREDIT_BUREAU_HOUR
categorical

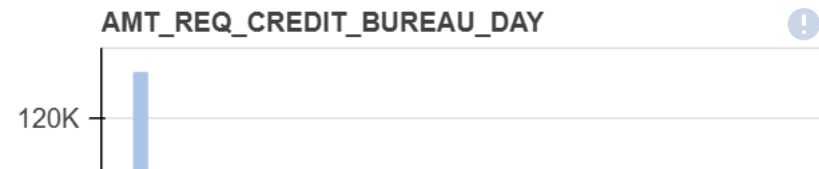
Show Details

Approximate Distinct Count	5
Approximate Unique (%)	0.0%
Missing	20833
Missing (%)	13.6%
Memory Size	9038696



Approximate
Distinct
Count

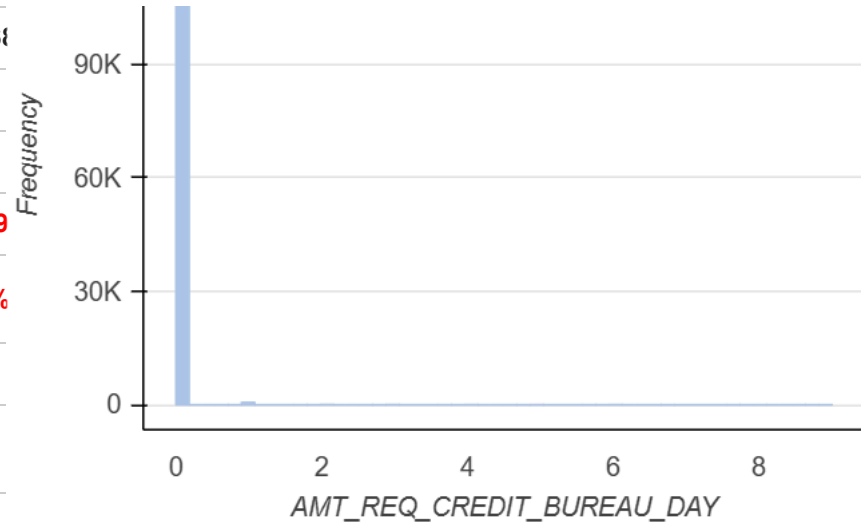
9



AMT_REQ_CREDIT_BUREAU_DAY
numerical

Show Details

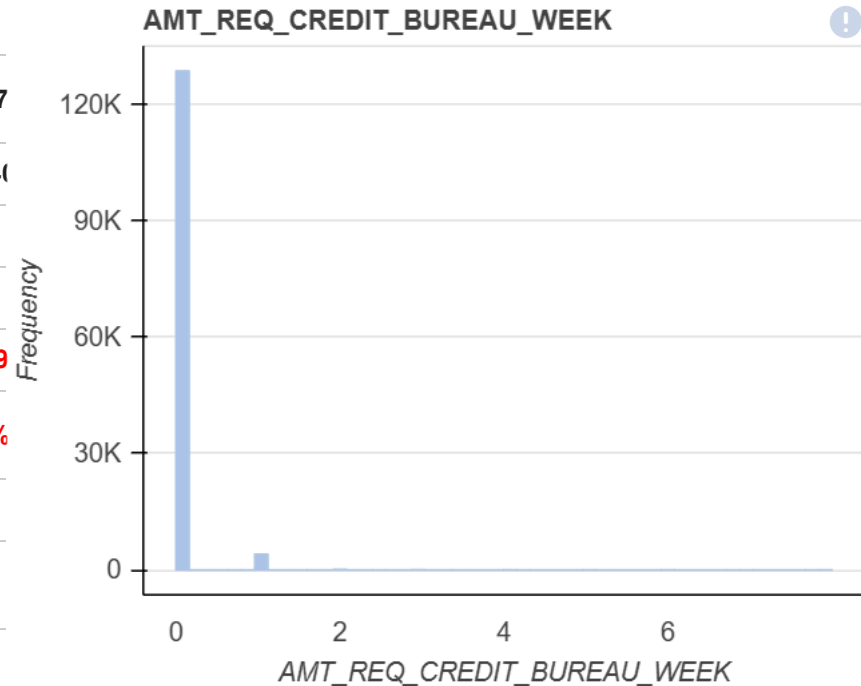
Approximate Unique Count	9	Mean	0.0061
Minimum	0	Maximum	9
Missing (%)	13.6%	Zeros (%)	86.0%
Infinite	0	Negatives	0
Infinite (%)	0.0%	Negatives (%)	0.0%
Memory Size	2126752		



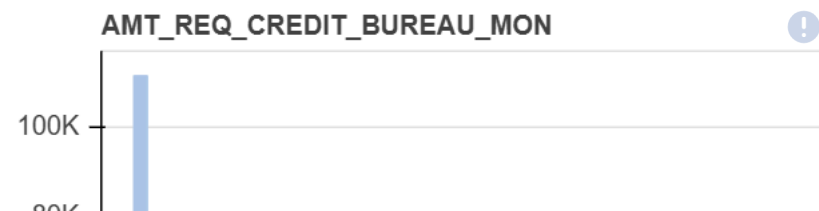
AMT_REQ_CREDIT_BUREAU_WEEK
numerical

Show Details

Approximate Distinct Count	9	Memory Size	21267
Approximate Unique Count	9	Mean	0.0341
Minimum	0	Maximum	8
Missing (%)	13.6%	Zeros (%)	83.7%
Infinite	0	Negatives	0
Infinite (%)	0.0%	Negatives (%)	0.0%



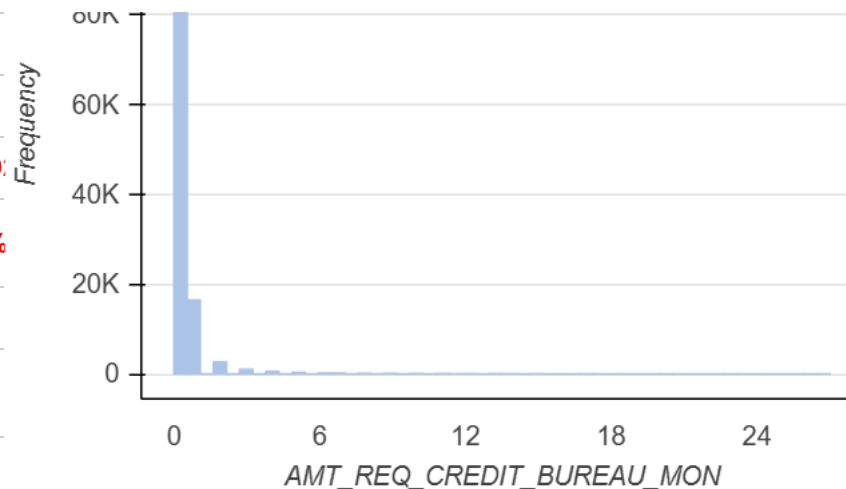
Approximate Distinct Count	22	Approximate Unique Count	22
----------------------------	----	--------------------------	----



AMT_REQ_CREDIT_BUREAU_MON
numerical

Show Details

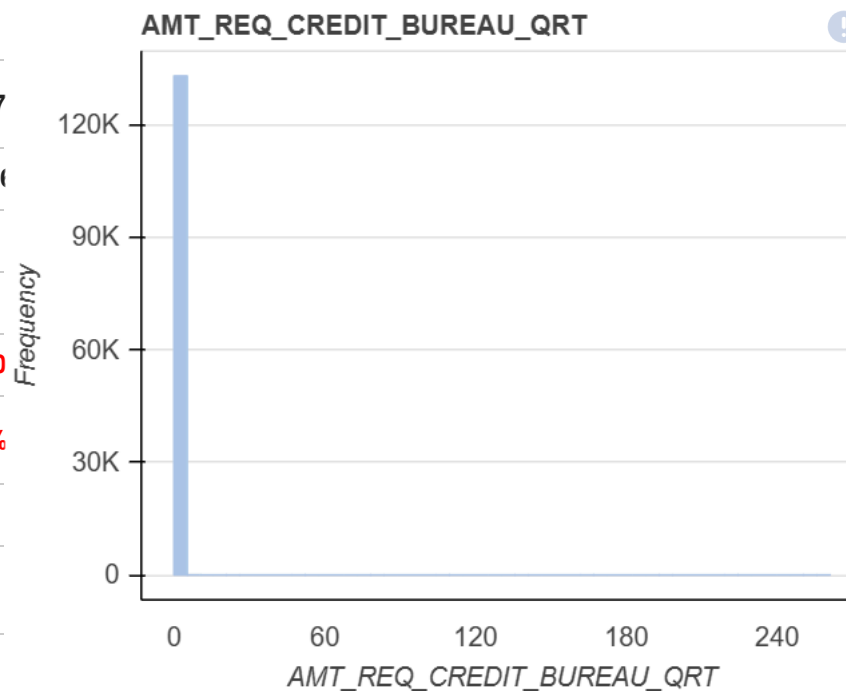
Missing	0833	Minimum	0
Missing (%)	13.6%	Maximum	27
Infinite	0	Zeros	11119
Infinite (%)	0.0%	Zeros (%)	72.3%
Memory Size	2126752	Negatives	0
Mean	0.2655	Negatives (%)	0.0%



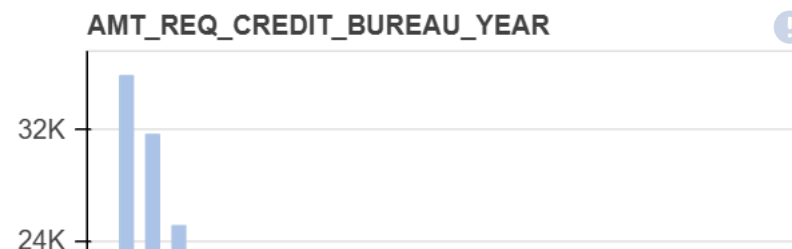
AMT_REQ_CREDIT_BUREAU_QRT
numerical

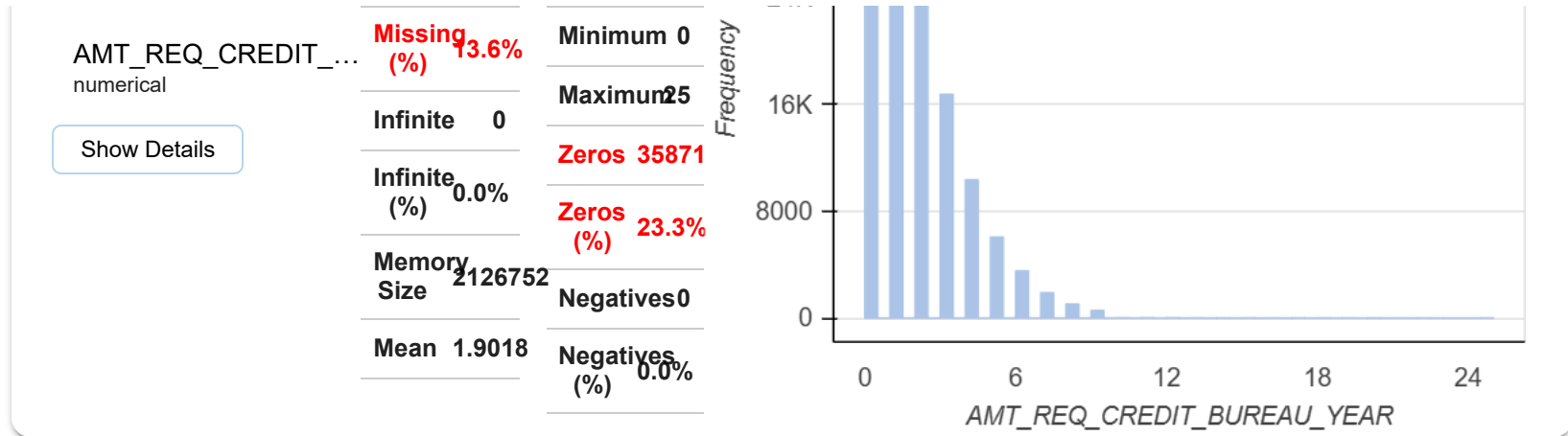
Show Details

Approximate Distinct Count	10	Memory Size	21267
Approximate Unique (%)	0.0%	Mean	0.2671
Missing	0833	Minimum	0
Missing (%)	13.6%	Maximum	261
Infinite	0	Zeros	10750
Infinite (%)	0.0%	Zeros (%)	69.9%
		Negatives	0
		Negatives (%)	0.0%

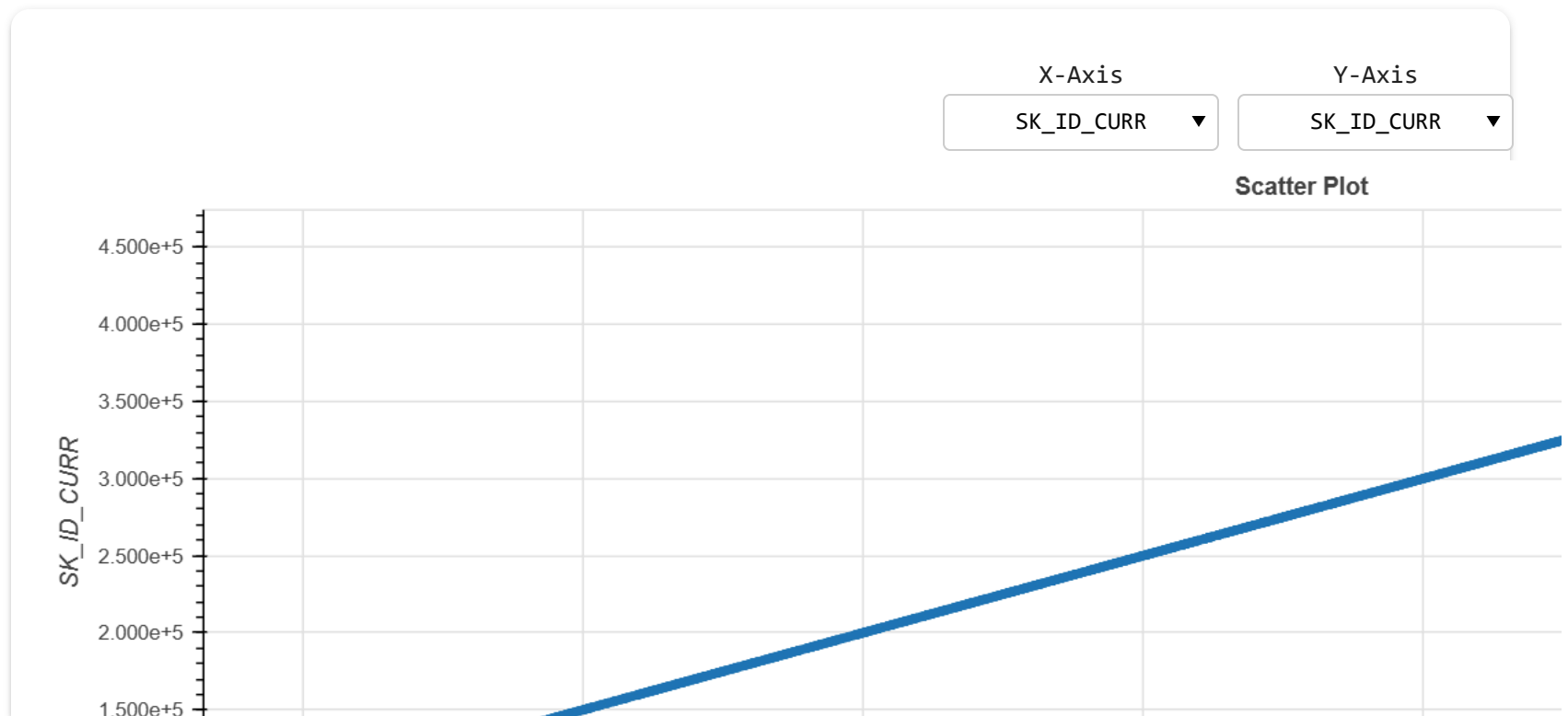


Approximate Distinct Count	24	Approximate Unique (%)	0.0%
		Missing	0833





Interactions





Missing Values

Bar Chart

Spectrum

Heat Map

Dendrogram

Report generated with [DataPrep](#)

Handling Missing values

```
In [131... df2_missing = plot_missing(df2)
df2_missing.save(str(EDA_DIR / 'missing_values.html'))
df2_missing
```

```
100%|#####| 3/3 [00:00<?, ?it/s]
0%|          | 0/1087 [00:00<?, ?it/s]
```

```
E-1019 (DUPLICATE_FACTORS): FactorRange must specify a unique list of categorical factors for an axis: duplicate factors found: 'REG_REGION_NO...ION', 'REG_CITY_NOT...ITY'
```

Out[131...

Stats

Looks like there are columns with lot of missing data. We can remove the columns with more than 50% missing .

```
In [ ]: # MISSING_THRESHOLD = 0.50
# missing_values_counts = df2.isnull().sum()
# missing_values_percent = (missing_values_counts / len(df2)) * 100
# missing_info = pd.DataFrame({
#     'Missing Count': missing_values_counts,
#     'Missing Percent': missing_values_percent
# })
# missing_info = missing_info[missing_info['Missing Count'] > 0].sort_values(by='Missing Percent', ascending=False)
# print("Columns with Missing Values:")
# display(missing_info)
```

Columns with Missing Values:

	Missing Count	Missing Percent
NAME_TYPE_SUITE	654	0.425352
OBS_30_CNT_SOCIAL_CIRCLE	541	0.351858
DEF_30_CNT_SOCIAL_CIRCLE	541	0.351858
OBS_60_CNT_SOCIAL_CIRCLE	541	0.351858
DEF_60_CNT_SOCIAL_CIRCLE	541	0.351858
EXT_SOURCE_2	303	0.197067
AMT_GOODS_PRICE	149	0.096907
AMT_ANNUITY	5	0.003252
CNT_FAM_MEMBERS	1	0.000650

```
In [ ]: # #cols_to_drop = missing_info[missing_info['Missing Percent'] > MISSING_THRESHOLD].index.tolist()
# df2 = df2.drop(columns=cols_to_drop)
# print(f'Columns dropped due to missing values > {MISSING_THRESHOLD*100}%: {cols_to_drop}')
# print(f'New dataset shape after dropping columns: {df2.shape}')
```

Columns dropped due to missing values > 50.0%: []

New dataset shape after dropping columns: (153755, 65)

In [144... `df2.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 153755 entries, 0 to 153754
Columns: 122 entries, SK_ID_CURR to AMT_REQ_CREDIT_BUREAU_YEAR
dtypes: float64(65), int64(41), object(16)
memory usage: 143.1+ MB
```

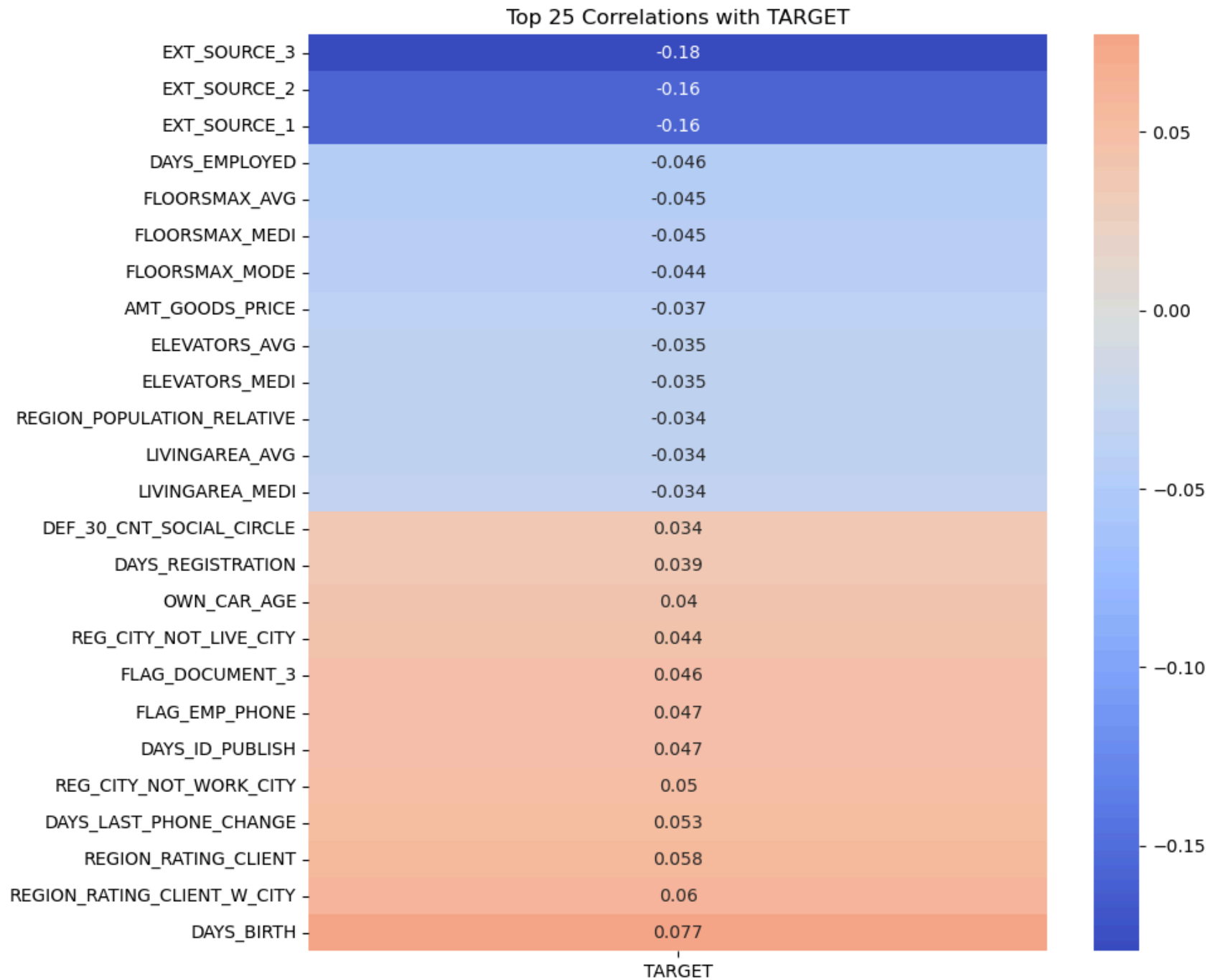
2.3 Correlation plot with TARGET

The correlation plot is overcrowded using dataprep. We can try using seaborn to generate a better correlation plot.

```
In [146... target_col = 'TARGET'
top_n = 25
num_df = df2.select_dtypes(include=[np.number]).copy()
corr = num_df.corr(method="pearson")[target_col].drop(labels=[target_col], errors="ignore")

### Select top +ve and -ve correlations
top_corr = corr.reindex(corr.abs().sort_values(ascending=False).head(top_n).index)
top_corr = top_corr.sort_values()
#lables = top_corr.index

### Plot
plt.figure(figsize=(10,8))
sns.heatmap(top_corr.to_frame(), annot=True, cmap='coolwarm', center=0)
plt.title(f'Top {top_n} Correlations with {target_col}')
plt.tight_layout()
plt.show()
```



NEGATIVE CORRELATED

- EXT_SOURCE 1,2,3 ==> Shows strongest negative correlation with TARGET. ie high external credit score is less like to be default.
- DAYS_EMPLOYED ==> Longer employment history, lower the risk of default.
- AMT_GOODS_PRICE ==> Higher value goods applicants tent to be lower risk.
- REGION_POPULATION_RELATIVE ==> Urban regions lower risk

POSITIVE CORRELATED

- DAYS_BIRTH = 0.77 ==> Younger applicant at higher risk of defaults.
- REGION_RATING_CLIENT_W_CITY ==> Higher region risk rating more defaults.
- DAYS_LAST_PHONE_CHANGE ==> Recent phone change , 50% chances of defaulters.
- REG_CITY_NOT_WORK_CITY ==> if living and working city are different, 50% risk.
- DAYS_ID_PUBLISH ==> Recently issued IDs have higher risk.

2.3 - Feature engineering

Lets select the top related features based on corrleation heatmap

```
In [147... selected_features = [  
  
    # Stronger negative correlation  
    "EXT_SOURCE_1",  
    "EXT_SOURCE_2",  
    "EXT_SOURCE_3",  
    "DAYS_EMPLOYED",  
    "AMT_GOODS_PRICE",  
    "REGION_POPULATION_RELATIVE",  
    "AMT_CREDIT",  
    "AMT_ANNUITY",  
  
    # Positive correlation  
    "CNT_CHILDREN",  
    "CNT_FAM_MEMBERS",  
    "LIVE_CITY_NOT_WORK_CITY",  
    "DEF_60_CNT_SOCIAL_CIRCLE",
```

```
"DEF_30_CNT_SOCIAL_CIRCLE",  
"DAYS_REGISTRATION",  
"REG_CITY_NOT_LIVE_CITY",  
"FLAG_EMP_PHONE",  
"DAYS_ID_PUBLISH",  
"REG_CITY_NOT_WORK_CITY",  
"DAYS_LAST_PHONE_CHANGE",  
"REGION_RATING_CLIENT",  
"REGION_RATING_CLIENT_W_CITY",  
"DAYS_BIRTH",  
"AMT_INCOME_TOTAL",  
  
# Others  
"CODE_GENDER",  
"NAME_INCOME_TYPE",  
"NAME_EDUCATION_TYPE",  
"OCCUPATION_TYPE",  
"ORGANIZATION_TYPE",  
"NAME_FAMILY_STATUS",  
"NAME_HOUSING_TYPE",
```

```
]
```

```
In [148... selected_features = [col for col in selected_features if col in df2.columns]  
print(f'Selected Numerical Features: ')  
for col in selected_features:  
    print(f'{col}')
```

Selected Numerical Features:

EXT_SOURCE_1
EXT_SOURCE_2
EXT_SOURCE_3
DAYS_EMPLOYED
AMT_GOODS_PRICE
REGION_POPULATION_RELATIVE
AMT_CREDIT
AMT_ANNUITY
CNT_CHILDREN
CNT_FAM_MEMBERS
LIVE_CITY_NOT_WORK_CITY
DEF_60_CNT_SOCIAL_CIRCLE
DEF_30_CNT_SOCIAL_CIRCLE
DAYS_REGISTRATION
REG_CITY_NOT_LIVE_CITY
FLAG_EMP_PHONE
DAYS_ID_PUBLISH
REG_CITY_NOT_WORK_CITY
DAYS_LAST_PHONE_CHANGE
REGION_RATING_CLIENT
REGION_RATING_CLIENT_W_CITY
DAYS_BIRTH
AMT_INCOME_TOTAL
CODE_GENDER
NAME_INCOME_TYPE
NAME_EDUCATION_TYPE
OCCUPATION_TYPE
ORGANIZATION_TYPE
NAME_FAMILY_STATUS
NAME_HOUSING_TYPE

In [149...

```
### Remove rows with missing target values
df2 = df2[df2['TARGET'].notnull()].copy()

X = df2[selected_features].copy()
y = df2['TARGET']
X.head()
```

Out[149...

	EXT_SOURCE_1	EXT_SOURCE_2	EXT_SOURCE_3	DAYS_EMPLOYED	AMT_GOODS_PRICE	REGION_POPULATION_RELATIVE	AMT.
0	NaN	0.397762	0.177704	-2037	900000.0	0.010006	9
1	0.697782	0.696312	0.538863	-162	612000.0	0.031329	7
2	NaN	0.716972	0.355639	365243	1795500.0	0.028663	17
3	NaN	0.662210	0.085033	365243	450000.0	0.004960	4
4	NaN	0.774506	0.627991	-886	1575000.0	0.018850	17

5 rows × 30 columns



In [150...

`X.info()`

```

<class 'pandas.core.frame.DataFrame'>
Int64Index: 153755 entries, 0 to 153754
Data columns (total 30 columns):
 #   Column                                  Non-Null Count  Dtype
---  -
 0   EXT_SOURCE_1                          67021 non-null  float64
 1   EXT_SOURCE_2                          153452 non-null  float64
 2   EXT_SOURCE_3                          123226 non-null  float64
 3   DAYS_EMPLOYED                         153755 non-null  int64
 4   AMT_GOODS_PRICE                       153606 non-null  float64
 5   REGION_POPULATION_RELATIVE            153755 non-null  float64
 6   AMT_CREDIT                            153755 non-null  float64
 7   AMT_ANNUITY                           153750 non-null  float64
 8   CNT_CHILDREN                          153755 non-null  int64
 9   CNT_FAM_MEMBERS                       153754 non-null  float64
10   LIVE_CITY_NOT_WORK_CITY               153755 non-null  int64
11   DEF_60_CNT_SOCIAL_CIRCLE              153214 non-null  float64
12   DEF_30_CNT_SOCIAL_CIRCLE              153214 non-null  float64
13   DAYS_REGISTRATION                     153755 non-null  float64
14   REG_CITY_NOT_LIVE_CITY                 153755 non-null  int64
15   FLAG_EMP_PHONE                        153755 non-null  int64
16   DAYS_ID_PUBLISH                       153755 non-null  int64
17   REG_CITY_NOT_WORK_CITY                 153755 non-null  int64
18   DAYS_LAST_PHONE_CHANGE                 153755 non-null  float64
19   REGION_RATING_CLIENT                   153755 non-null  int64
20   REGION_RATING_CLIENT_W_CITY            153755 non-null  int64
21   DAYS_BIRTH                            153755 non-null  int64
22   AMT_INCOME_TOTAL                      153755 non-null  float64
23   CODE_GENDER                           153755 non-null  object
24   NAME_INCOME_TYPE                       153755 non-null  object
25   NAME_EDUCATION_TYPE                    153755 non-null  object
26   OCCUPATION_TYPE                       105665 non-null  object
27   ORGANIZATION_TYPE                     153755 non-null  object
28   NAME_FAMILY_STATUS                     153755 non-null  object
29   NAME_HOUSING_TYPE                      153755 non-null  object
dtypes: float64(13), int64(10), object(7)
memory usage: 36.4+ MB

```

```

In [151...  ### Convert all DAYS columns like DAYS_EMPLOYED, DAYS_BIRTH etc
            ### Negative Days to years.
            days_cols = [col for col in X.columns if "DAYS" in col]

```



```

for col in days_cols:
    X[col] = X[col].abs()

if "DAYS_BIRTH" in X.columns:
    X["AGE_YEAS"] = X["DAYS_BIRTH"] / 365

### Employment ratio = DAYS_EMPLOYED / DAYS_BIRTH
### Higher ratio means stable work experience.
if "DAYS_EMPLOYED" in X.columns and "DAYS_BIRTH" in X.columns:
    X["EMPLOYMENT_RATION"] = X["DAYS_EMPLOYED"] / X["DAYS_BIRTH"]

### Convert Credit and amount annuality to ration
### This ratio can provide insights into the financial burden of the credit compared to the annuity payments.
### A higher ratio may indicate a larger credit amount relative to the annuity, which could be a risk factor for defc
if "AMT_CREDIT" in X.columns and "AMT_ANNUITY" in X.columns:
    X["CREDIT_ANNUITY_RATIO"] = (
        X["AMT_CREDIT"] / (X["AMT_ANNUITY"] + 1)
    )

### Goods product price to credit ratio.
### This ratio can provide insights into the relationship between the price of goods and the credit amount.
if "AMT_GOODS_PRICE" in X.columns and "AMT_CREDIT" in X.columns:
    X["GOODS_PRICE_CREDIT_RATIO"] = (
        X["AMT_GOODS_PRICE"] / (X["AMT_CREDIT"] + 1)
    )

### Children to family members ratio.
### This ratio can provide insights into the family structure and potential financial responsibilities of the applica
if "CNT_CHILDREN" in X.columns and "CNT_FAM_MEMBERS" in X.columns:
    X["CHILDREN_FAMILY_RATIO"] = (
        X["CNT_CHILDREN"] / (X["CNT_FAM_MEMBERS"] + 1)
    )

### replace infinite values with NaN
X = X.replace([np.inf, -np.inf], np.nan)

```

In [152... X.head()

Out[152...

	EXT_SOURCE_1	EXT_SOURCE_2	EXT_SOURCE_3	DAYS_EMPLOYED	AMT_GOODS_PRICE	REGION_POPULATION_RELATIVE	AMT_
0	NaN	0.397762	0.177704	2037	900000.0	0.010006	9
1	0.697782	0.696312	0.538863	162	612000.0	0.031329	7
2	NaN	0.716972	0.355639	365243	1795500.0	0.028663	17
3	NaN	0.662210	0.085033	365243	450000.0	0.004960	4
4	NaN	0.774506	0.627991	886	1575000.0	0.018850	17

5 rows × 35 columns



In [153...

```

categorical_cols = X.select_dtypes(include=["object"]).columns.tolist()
numerical_cols = X.select_dtypes(exclude=["object"]).columns.tolist()
print("\nCategorical Columns:")
print(categorical_cols)

print("\nNumerical Columns:")
print(numerical_cols)

```

Categorical Columns:

```
['CODE_GENDER', 'NAME_INCOME_TYPE', 'NAME_EDUCATION_TYPE', 'OCCUPATION_TYPE', 'ORGANIZATION_TYPE', 'NAME_FAMILY_STATU
S', 'NAME_HOUSING_TYPE']
```

Numerical Columns:

```
['EXT_SOURCE_1', 'EXT_SOURCE_2', 'EXT_SOURCE_3', 'DAYS_EMPLOYED', 'AMT_GOODS_PRICE', 'REGION_POPULATION_RELATIVE', 'A
MT_CREDIT', 'AMT_ANNUITY', 'CNT_CHILDREN', 'CNT_FAM_MEMBERS', 'LIVE_CITY_NOT_WORK_CITY', 'DEF_60_CNT_SOCIAL_CIRCLE',
'DEF_30_CNT_SOCIAL_CIRCLE', 'DAYS_REGISTRATION', 'REG_CITY_NOT_LIVE_CITY', 'FLAG_EMP_PHONE', 'DAYS_ID_PUBLISH', 'REG_
CITY_NOT_WORK_CITY', 'DAYS_LAST_PHONE_CHANGE', 'REGION_RATING_CLIENT', 'REGION_RATING_CLIENT_W_CITY', 'DAYS_BIRTH',
'AMT_INCOME_TOTAL', 'AGE_YEAS', 'EMPLOYMENT_RATION', 'CREDIT_ANNUITY_RATIO', 'GOODS_PRICE_CREDIT_RATIO', 'CHILDREN_FA
MILY_RATIO']
```

2.4 Train / Test Split

In [154...

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)
```

Encode Categorical columns

The HistGradientBoostingClassifier model natively handles NaNs in numerical columns. OrdinalEncoder works with ColumnTransformer Dataframes

```
In [156... num_transformers = Pipeline(steps=[
    ('imputer', SimpleImputer(strategy='median'))
])

cat_transformers = Pipeline(steps=[
    ('imputer', SimpleImputer(strategy='most_frequent')),
    ('encoder', OrdinalEncoder(handle_unknown='use_encoded_value', unknown_value=-1))
])

preprocessor = ColumnTransformer(
    transformers=[
        ('num', num_transformers, numerical_cols),
        ('cat', cat_transformers, categorical_cols)
    ]
)
```

Handle class imbalance using SMOTE Oversampling

```
In [109... from imblearn.over_sampling import SMOTE
from imblearn.pipeline import Pipeline
#as ImbPipeline

# smote = SMOTE(random_state=42, n_jobs=-1)
# pipeline = ImbPipeline([('smote', smote)])
# X_train_resampled, y_train_resampled = pipeline.fit_resample(X_train, y_train)
# print(f'Original training set shape: {X_train.shape}, {y_train.shape}')
# print(f'Resampled training set shape: {X_train_resampled.shape}, {y_train_resampled.shape}')
```

Create pipeline and Train the Model

```
In [157... pipeline = Pipeline(steps=[
    ('preprocessor', preprocessor),
    ("smote", SMOTE(random_state=42, n_jobs=-1)),
    ('model', HistGradientBoostingClassifier(
```

```

learning_rate=0.05,
max_iter=300,
max_depth=6,
min_samples_leaf=50,
l2_regularization=1.0,
random_state=42,
class_weight='balanced')),
])

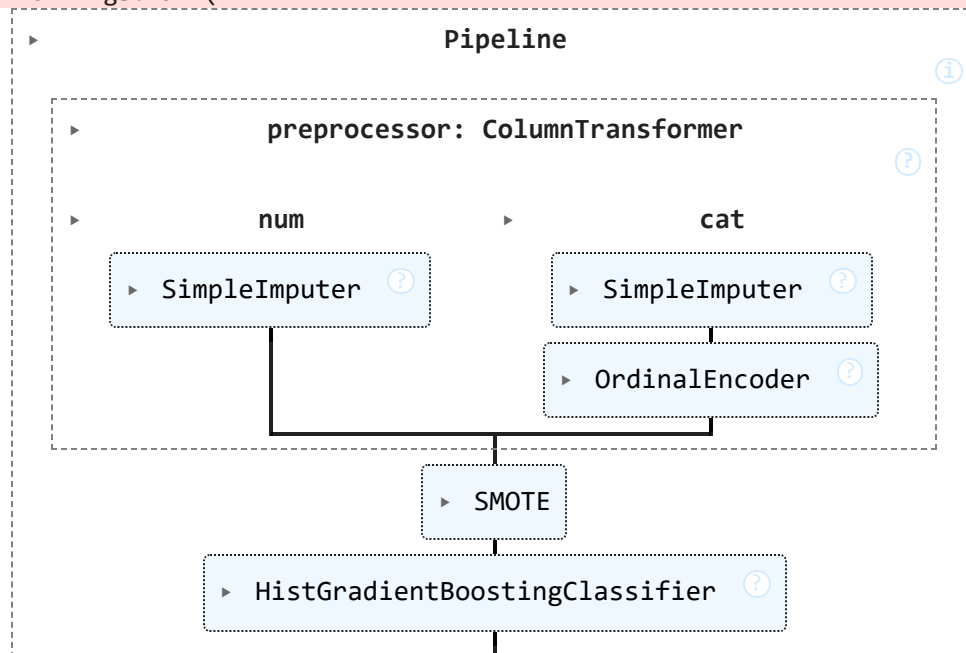
pipeline.fit(X_train, y_train)

```

d:\TECH\AI\envs\aa1-501-m5\lib\site-packages\sklearn\base.py:474: FutureWarning: `BaseEstimator.\_validate\_data` is deprecated in 1.6 and will be removed in 1.7. Use `sklearn.utils.validation.validate\_data` instead. This function becomes public and is part of the scikit-learn developer API.

warnings.warn(
d:\TECH\AI\envs\aa1-501-m5\lib\site-packages\imblearn\over_sampling_smote\base.py:370: FutureWarning: The parameter `n\_jobs` has been deprecated in 0.10 and will be removed in 0.12. You can pass an nearest neighbors estimator where `n\_jobs` is already set instead.
warnings.warn(

Out[157...



Predictions

```
In [158... y_prob = pipeline.predict_proba(X_test)[: , 1]
threshold = 0.3
y_pred = (y_prob >= threshold).astype(int)
```

2.5 Evaluation

```
In [159... print("\n===== MODEL PERFORMANCE =====\n")
print("Accuracy      :", accuracy_score(y_test, y_pred))
print("PR-AUC Score :", average_precision_score(y_test, y_prob))
print("ROC-AUC Score :", roc_auc_score(y_test, y_prob))
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred))
print("\nConfusion Matrix:\n")
print(confusion_matrix(y_test, y_pred))
```

===== MODEL PERFORMANCE =====

Accuracy : 0.9038730447790316
 PR-AUC Score : 0.21879129255882912
 ROC-AUC Score : 0.7391282963040495

Classification Report:

	precision	recall	f1-score	support
0	0.93	0.97	0.95	28269
1	0.32	0.17	0.22	2482
accuracy			0.90	30751
macro avg	0.62	0.57	0.58	30751
weighted avg	0.88	0.90	0.89	30751

Confusion Matrix:

```
[[27385  884]
 [ 2072  410]]
```

- The Metric shows the Model performance has improved after the feature engineering.
- The improvment are specially on the minory class 1.

Metric	Without Feature Engineering	With Feature Engineering	Improvement
Accuracy	0.92	0.90	Slight decrease
ROC-AUC	0.745	0.739	Almost same
Precision (Class 1)	0.49	0.32	Decreased
Recall (Class 1)	0.02	0.17	Huge improvement
F1-score (Class 1)	0.03	0.22	Major improvement
True Positives	~49	410	Big increase

Feature engineering improved the model's ability to detect minority-class samples. Although overall accuracy slightly decreased, recall and F1-score for the positive class improved significantly, indicating better class separability and a more balanced classifier. This demonstrates that feature engineering added predictive signal that the original feature set lacked. The Prediction can be further improved by handling the class weights, More feature engineering, using Stronger model like XGBOost.